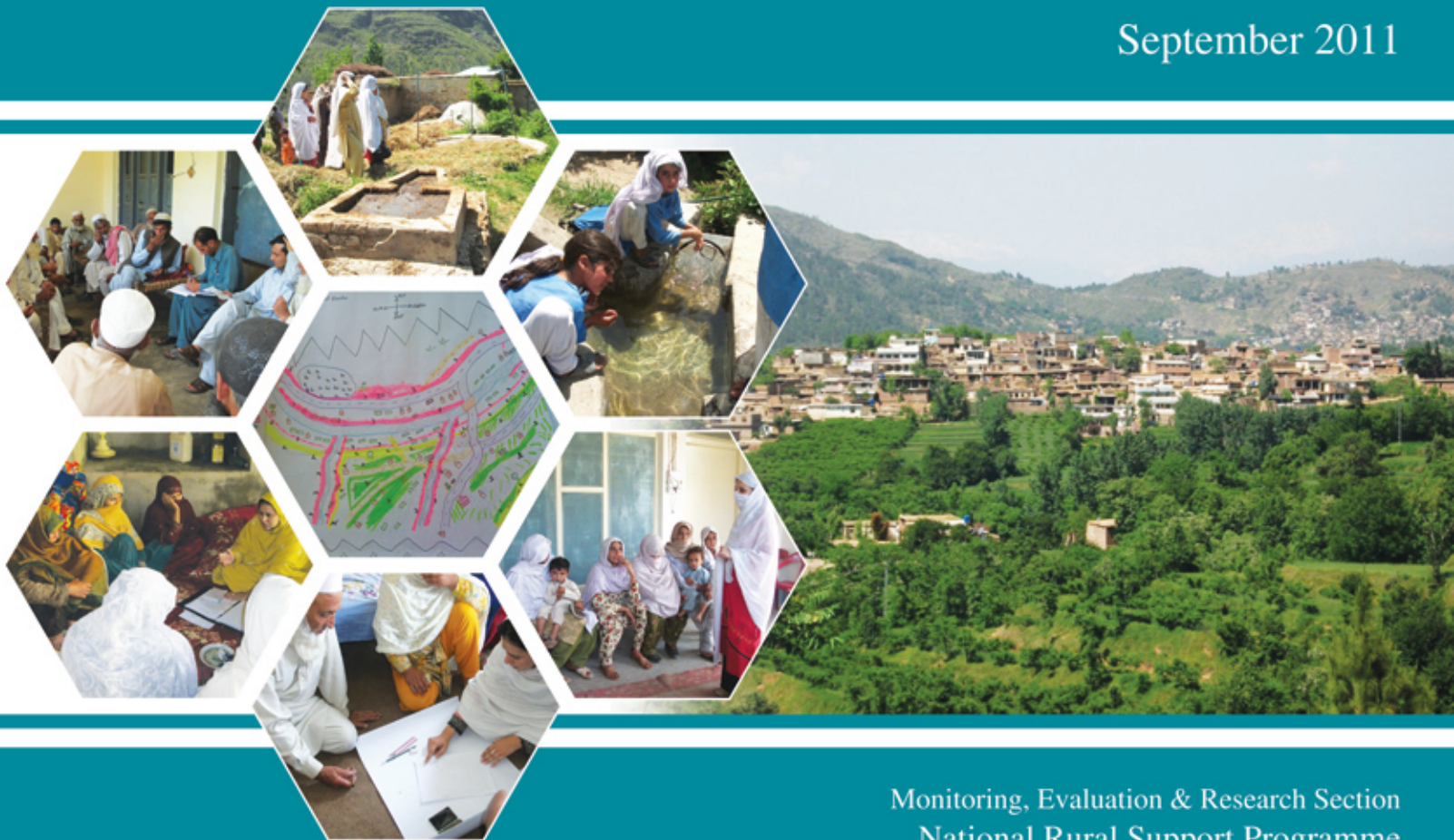


Disaster Risk Mapping

District Swat

September 2011



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Disaster Risk Mapping - District Swat

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Printed by: PanGraphics (Pvt) Ltd., Islamabad

Table of Contents

Introduction	1
Methodology of Risk Assessment	2
Hazard Mapping	2
Methodology for calculating Hazard Score for different Hazards	3
Vulnerability Assessment	4
Matrix for calculating vulnerability assessment	5
Social, political and economic framework conditions	6
In the province	6
In the project area	6
Location of the risk assessment	7
Initial Justification of the Survey	8
Current risk propensity of selected villages	10
Process and the criteria applied for the selection of the proposed households	12
Collection of secondary data (sources and information)	13
Topography of Swat	13
Types of Hazard in Project area and their definition	13
Hazard Analysis	16
Hazard Ranking of villages	16
Methodology notes for Ranking	17
Social Mapping	17
Transect Walk	18
Focus Group Discussions	18
Community insights on Disaster History, their coping mechanisms and superstitions	19
Village wise summary of Focus group Discussions	20
Information gathered using other PRA tools	22
Summary of household vulnerability Analysis	23
Exposure Analysis	23
Fragility Analysis	23
Indicators of Socio economic wellbeing of communities in context of Hazard	24
Lack of resilience	26
Overall Vulnerability Score Range	26
Identified DRR & CCA project interventions	28
Ranking of Communities	29

List of Annexes

Annex 1: Village Report - Murguzar	31
Annex 2: Village Report - Sapal Banday	36
Annex 3: Village Report - Gul Banday	41
Annex 4: Village Report - Kokrai	46
Annex 5: Village Report - Saidu Shagae	51
Annex 6: Village Report - Saidu	55
Annex 7: Village Report - Ghari	59
Annex 8: Village Report - Shagai Shahgram	64
Annex 9: List of Targeted household for Assessment Survey	68
Annex 10: Questionnaire for conducting Vulnerability Survey	82

Acronyms

ACTED	Agency for Technical Cooperation & Development
BHU	Basic Health Unit
BISP	Benazir Income Support Program
CBDRM	Community Based Disaster Risk Management
CFW	Cash for Work
DMC	Disaster Management Committee
FGDs	Focus Group Discussions
HHs	House Holds
HUJRA	Holistic Understanding for Justified Research Action
INGO	International Non Governmental Organization
Lasoon	“Hands” in Pushto Language
NFI	Non-Food Items
NGO	Non-Governmental Organization
NRSP	National Rural Support Programme
PRA	Participatory Rural Appraisal
OXFAM	Oxford Committee for Famine Relief
SPADO	Sustainable Peace & Development Organization
UC	Union Council
WASH	Water, Sanitation & Hygiene

Introduction

In April 2011 NRSP signed an agreement for disaster risk assessment survey in Swat with Diakonie, a German based INGO which has been its strategic partner since Earthquake 2005. The assessment was carried out in 8 villages located in UC Madyan, Tirat, and Islampur & Saidu Sharif in Swat district.

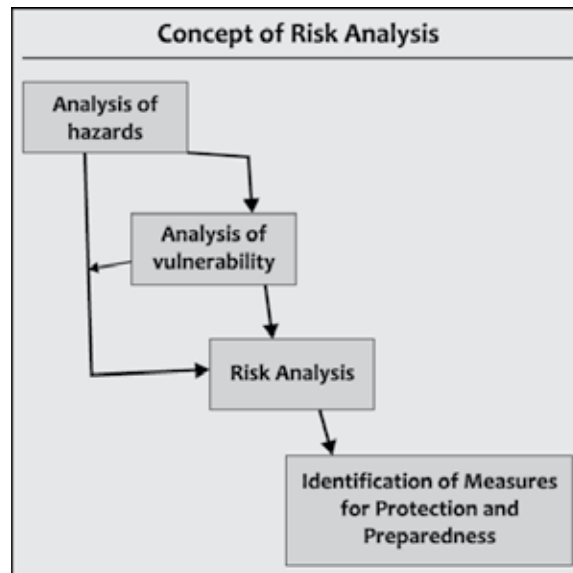
The study was initiated with the orientation of the study team comprising of 15 members including, MER Officer, a data entry officer, 12 field supervisors and 60 enumerators. For data collection, District Swat was divided into two zones (upper and lower Swat) and two teams, each comprising of 6 field supervisors (3 male and 3 female) were deputed. Field supervisors were responsible for village level information collection using PRA tools (FGDs, transect walk, hazard mapping etc.). Each Field Supervisor also supervised a team of 6 enumerators (on an average) for Household survey.

The mentioned project while adopting an innovative approach carried out hazard mapping and analysis within different villages in terms of occurrence and magnitude of seasonal flooding, major flooding and soil erosion. The mapping exercise was supplemented by focus group discussions, village situational analysis, and consultation with different metrological and line departments. Based on hazard mapping about 765 households who were identified as being in high risk zones were targeted for vulnerability assessment survey. The survey, focusing on indicators such as exposure, fragility and lack of resilience of communities in terms of various hazards was able to assimilate proposed interventions with respect to disaster preparedness and mitigation strategies which can be employed to reduce risk level of targeted population (see questionnaire at Annex 10).

The proposed interventions focused upon capacity building initiatives in CBDRM, formation of Disaster Management Committee with formulation of a proper village level emergency, evacuation and preparedness plans. Mitigation measures which were suggested by stakeholders focused upon encouragement of ecological farming, kitchen gardening, terracing and reforestation alongside river banks along with fortification of houses and community buildings with more stable materials.

Methodology for Risk Assessment

While being prone to multitude of natural disasters globally in form of floods, earthquakes, cyclones and land sliding, disaster risk reduction practitioners have endeavored constantly for envisioning of appropriate tools and methodologies whereby risk in a certain hazard prone zone can be identified. Moreover, appropriate and suitable interventions can be planned and implemented both at community level and under fold of policy guidelines keeping into account vulnerability levels of communities. Risk assessment is thus the most appropriate tool in context of which hazard and vulnerability levels are



analyzed and computed quantitatively through measure of certain indicators. Hazard mapping exclusively takes into account hazard risk of a certain area and magnitude and frequency of its occurrence while vulnerability risk computes socio-economic indicators that are affected by hazard such as food insecurity, livestock and land vulnerability, income patterns etc. Based on analyzing both aspects and deriving risk assessment scoring range, disaster mitigation measures can be planned along with disaster preparedness, recovery and rehabilitation initiatives.

Hazard Mapping

Hazard mapping is the process of estimating, for defined areas, the probabilities of the occurrence of potentially damaging phenomenon of given magnitude within a specified period of time. To compute hazard mapping, two fold indicators are taken into account which are as follows;

Probability of occurrence (frequency) describes on average how often a hazard of a specific magnitude, duration, and/or extent typically occurs. For instance, flood frequency analysis uses historical records of peak flows to produce guidance about the expected behavior of future flooding. To be able to analyze the frequency of hazards, the question to ask is “How often do floods occur in the given area?”

Magnitude/Extent describes the strength or force of an event. Only occurrences exceeding some defined level of magnitude are considered extreme, disastrous, or even hazardous. In the case of floods, for example, magnitude is often described as the maximum height of floodwaters above average sea level, flood stage, or simply above ground.

Hazard mapping scoring Matrix

Hazard Analysis		
Probability of Occurrence (Value/Score)	Magnitude/ Extent [Volume, Speed, Force] (Value/Score)	Final Value/ Score
3 = highest	3 = highest	Collection of Hazards:
2 = medium	2 = medium	Examples: $3+2 = 5/2 = 2.5$
1 = lowest	1 = lowest	Hazard value B = 2.5

Formula for risk assessment: [Probability (Frequency) + Magnitude (Extent)] / 2 = Hazard risk value

Construction of Hazard Maps

Construction of hazard mapping adopts a community based participatory approach whereby the mapping exercise is conducted through mutual collaboration of communities in villages which are selected for the exercise;

- During construction of Hazard maps Community member, elders & notables are involved for inputting history of disasters and for constructing hazard maps, initial outlay of the village is drawn with boundaries.
- Through Transect walk, study areas/sub communities in a village is determined by plotting those areas in a village that were directly affected by a disaster.
- Multiple hazards such as floods and soil erosion are plotted along the hazard map.
- Community mapping exercise is then conducted by drawing major land marks such as roads, bridges, community places, school etc.

Methodology for calculating Hazard Scores for different hazards

As there are a multitude of different hazards globally such as floods, cyclones, earthquakes and landslides; hence each has got customized methodology to calculate hazard score. Even in floods there are two categories, major flooding is the one which occurs once in 20 years and seasonal flooding which occurs on a yearly basis. In case of this pilot study flood 2010 was taken as major floods. Following is customized means of calculating each hazard score

Standardised scoring for Seasonal Flooding (Annual flooding)

Magnitude (description)	Magnitude Ranking	Frequency Ranking*	Hazard score
Area with more than 5 feet of water	3	3	3
Area with between 3 and 5 feet of water	2	3	2.5
Area with up to 3 feet of water	1	3	2

*Frequency ranking is three because it is a seasonal flood which occurs every year.

Standardised scoring for Major Flood (e.g. same as 2010 flood in Pakistan)

Magnitude (description)	Magnitude Ranking	Frequency Ranking*	Hazard score
Area with more than 5 feet of water	3	1	2
Area with between 3 and 5 feet of water	2	1	1.5
Area with up to 3 feet of water	1	1	1

*Frequency ranking is 1 because it is a major flood which occurs once in 20 years

Standardised scoring for Riverbank Soil Erosion

Ranking Score:	1	2	3
Frequency definitions	Major flooding	Seasonal (yearly)	On-going all the time
Magnitude definitions	Less than 10% of agricultural land	Between 10% and 20%	More than 20%

Standardised scoring for Major Earth quake (e.g. Earthquake 2005 in Pakistan)

Magnitude (description)	Magnitude Ranking	Frequency Ranking	Hazard score
Area destroyed completely	3	1	2
50% of the area is damaged	2	1	1.5
25% of the area is damaged	1	1	1

* However as it is difficult to determine predictability of earthquake in the same area; more technical input in terms of geological oversight of earth patterns and fault line is needed

Standardised scoring for Land sliding

Magnitude (description)	Magnitude Ranking	Frequency Ranking	Hazard score
Area destroyed completely	3	1	2
50% of the area is damaged	2	1	1.5
25% of the area is damaged	1	1	1

* However as it is difficult to determine predictability of land sliding in the same area; more technical input in terms of geological oversight of earth patterns is needed

Vulnerability Assessment

- The extent to which a person or group is likely to be affected by a hazard (related to their capacity to anticipate, cope, resist and recover from its impact)
- Vulnerability assessment depicts vulnerability at household level while hazard mapping is at community level. Thus risk assessment sums up both together for holistic preparedness, mitigation and prevention mechanisms.
- Vulnerability is the potential to suffer harm or loss, related to the capacity to anticipate

a hazard, cope, resist and recover from its impact. Both vulnerability and its antithesis, resilience, are determined by physical, environmental, social, economic, political, cultural and institutional factors.

- Vulnerability is more than just poverty, but the poor tend to be more vulnerable

Matrix for calculating vulnerability assessment

VULNERABILITY ANALYSIS			
Exposure to Hazard	Fragility	Lack of Resilience	Value of Vulnerability
Value 1-3	Indicators: to be defined locally with population e.g. Source of Income Labour force (Ranking) Assets and physical conditions (Ranking) Community conditions Value 1-3	Indicators: to be defined locally e.g. Educational level (Ranking) Neighbourhood support systems (Ranking) Value 1-3	Example: $3+2+1 = 6/3 = 2$ $P = 2$

Risk Assessment and Mapping Process

During the study the information was collected using secondary and primary sources. Various publications and reports were reviewed for conceptual clarity and better understanding of the area. For primary source information a total of 14 villages were selected in consultation with the other stakeholders. In each of these villages, community level information was collected (May 2012) using various PRA tools (28 FGDs, transect walk, social mapping, hazard mapping etc). At the second stage (June 2012), household level survey was conducted and for this only those households were selected which were directly affected by the floods 2010. The list of these households was prepared during the village level meetings.

In Monitoring of field activities, hazard maps were changed and refined several times considering change in guidelines. Sometimes, during focus group discussions, staff was not able to probe into the communities effectively so questionnaires were re-designed so that results are derived in accordance with DRR perspective. During the vulnerability survey, data collection activities were monitored and it was made sure that staff completes the activities within the work plan that was set. Preparation of Village and final reports took into account surveyed data which was separately analyzed in different categories to derive results.

Social, political and economic framework conditions

In the province

Communities in Swat have been facing man-made and natural disasters from last 5 years i.e. earth quake, Talibanization, terrorist attacks, bombings, militancy and in most recent context heavy monsoon rains and over flowing of swat river that led to severe flash floods in period of late July 2010. The flood spread across 15 UCS of swat creating wide scale damages ranging from devastation of roads, bridges, infrastructure to destruction of houses, shops and cropped fields. The after effects of the flood are still quite visible across different parts of swat with people still being posed with lack of proper shelter, food and health facilities and as a result are struggling to revitalize normal course of activities, with lack of income at their disposal many are reliant on support from humanitarian agencies and relatives to reconstruct their houses and revive their diminished sources of income. As per the current political situation recent killing of one of the major Taliban leaders have sent waves of negativity amongst general public against the armed forces and therefore pressure is being imposed upon them to evacuate area of swat which had been under stringent army control since flood came about.

In the project area

In project areas people are increasingly susceptible to upcoming monsoon rains and the possible damages it can cause. They are gradually reconstructing their damaged houses in the same locations. However due to battered roads, silted water channels and eroded fields, accessibility and socio economic viability is not totally redeemed. Islampur and Saidu Sharif located in lower parts of Swat are more so prone to cloud burst in period of July due to heavy torrential rains, this causes land sliding, opening up of embankments thus leading to destruction of irrigation channels and houses. In upper parts of Swat melting of glaciers and flash floods originating from Kalam river are still a perennial threat to local inhabitants particularly to their agriculture fields because normally people reside at higher zones with their fields at the lower terrain, many villages in upper Swat have been completely washed away hence displaced people have migrated to more safer zones.

Location of the risk assessment

According to Cluster response report UCs of Madyan and Tirat in upper parts of Swat were affected by flash floods with damages ranging to 80.4%, in lower Swat UC Islampur and Saidu Sharif were targeted with damages as far as 35.4%.¹ Lower Swat in which Project UCs of Saidu and Islampur are located is adjacent to Murguzar road which connects the UCs with the main town of Mingora. The area of Islampur is located amid mountainous terrain with water channels and streams penetrating out of Swat River flowing in the midst of plain lands in between Saidu Sharif UCs closely touches



Islampur and is majorly plain area. With a drive of 1 ½ hours from Mingora; terrain of upper Swat starts in which UCs of Tirat and Madyan are adjoined close together in midst of water tributaries gushing out of Kalam river. The area is plain lands with mountain ranges in between which villages are located as well.

During summer season ranging from July-September, areas in Swat witness torrential rains in the range of 750 mm² which causes water level to rise in the river and connected tributaries. Lands close to these over flowing streams are greatly endangered by seasonal floods while areas within steep mountain curves become prone to land sliding which thus disconnects the areas from other parts. The pattern of land sliding and seasonal floods is quite prominent in UC Islampur while in upper Swat seasonal flood is a continuous threat in summers as water gushes to height of over 13 feet to affect land and infrastructure.

Province	District	Area	UC	# of villages
Khyber Pakhtun-Khawa	Swat	Lower Swat	Islampur	4
			Saidu Sharif	2
		Upper Swat	Madyan	4
			Tirat	4

¹ Pak Response website document

² Metrological Department- Swat

Initial Justification of the Survey

The surveyed areas were selected in context of vast scale damage that were witnessed in different areas of Swat in context of Flood 2010. So criterion that were deemed to be important included household damage, destruction of agriculture land, roads and infrastructure. However during selection process, consideration was paid towards extent of vulnerability that was evident in the areas, in terms of future disaster.

Union Council	Justification for Selection
Islampur	Murguzar village is the biggest village of UC Islampur and is located amid steep mountains and patches of plain lands; it generally receives seasonal flooding in the range of 1-3 feet. However in advent of floods 2010 the gushing stream that adjoins the village over flowed from swat river over 5 feet and inflicted damages to buildings, land, infrastructure, Murguzar road and standing crops. Furthermore, due to torrential rains, land sliding often causes blockage of various communities located in mountainous areas. Due to this community became prone to different levels of vulnerabilities in terms of loss of electricity, blockage of pathways and food insecurity. Murguzar village is composed of various sub communities which had damages of variant scale, in Amol Tal community local mosque was destroyed, mud houses and agriculture fields were destroyed, In Zakria community 20 houses were destroyed and heavy torrential rain and cloud burst cut off the area. In Jazwo, community cloud burst created significant damages, walls of many mud houses were destroyed and community was blocked due to ruining of pathways. Other communities including Kadoona and Totramay witnessed loss of agriculture land, damages to mud houses and with land sliding these communities were cut off from rest of the area. Spal Bandai village is closely knitted village in patches of mountains that adjoins it with Murguzar and other village, it also receives seasonal flooding in range of over 1 foot. In advent of flood 2010 the area witnessed significant damage of agriculture land, in various part water channels that connect some of the left over agriculture fields have been severely affected, hence there is a case in which these lands are endangered due to dearth of water supply. Tumbling of rocks due to seasonal flooding also incurred damages to mud houses, walls of which have been battered. Gul Bandai village is closely interconnected with Spal Bandai village. In the area water channels were destroyed and agriculture fields were severely damaged. In various parts of the village due to narrow water channels, water level intensified in minor spaces hence there was penetration of flood water into houses which caused damages particularly to mud houses while also leading to sanitation problems for the communities.

	In 2010 floods water level in Kokrai village was over 5 feet which inflicted damages to the houses and agriculture land in the village. A stream flows adjacent to the village over flowing of which caused water to penetrate into houses which led to sanitation problems. It was also noted that water channels within the village are very narrow hence in case of flood, water is bound to intensify into thin spaces in which houses and fields are located. Therefore these water channels need to be broadened so that flow of water is appropriately channelized and directed.
Saidu Sharif	Saidu village connected to rest of the Swat through Murguzar road, is divided into rural and urban zones In 2005, area was hit by earth quake and still jerks of quake are felt in the area but the losses were minor being the mud houses affected, cracks and hollowness in house walls. In advent of flood 2010, water ranged over 5 feet which affected the rural part of Saidu. Highly affected area being the ones near the river and water streams and water channels not only damaged crops but also the houses near them were severely damaged. Massive destruction has been done to standing crops in fields, land, water channels, bridges and houses. In 2005, earthquake hit the Shagai village and there were minor damages to (kacha) mud houses and shops however during present flood, damages were on bigger scale than the earthquake. Crop fields were badly damaged and have now turned into stony area and needs land restoration, 6 houses were damaged, sanitation being the major problem of the area became worst. During flood the dirty water entered the houses and made conditions unhygienic for the community and numerous diseases spread in the area with diarrhea, dehydration (non availability of clean drinking water) and cholera being the top ones and others were allergies, skin rashes and general illness. Villagers said that due to development of our village we have been neglected in relief projects and hence now no rehabilitation program is being carried out in our area despite being the fact that our village is affected.
Madyan	Ghari Village is adjacent to Madyan (60 km away from Mingora, Swat) with urban and rural mixed area and has 800 households. During flood, water level was 15-20 feet high and the village was cut off as Madyan road was completely washed away and transportation by road was not possible which completely blocked the area. Water level in Cheel stream (Flood) was high and it made damages on high scales. Snow melted from snowy hills and caused flood in Cheel streams (UC Bishigram) which meets River Swat. Jopin and Ingrabad are totally washed away with flood and fields of Ghari Kalay fields damaged, some land is washed away and irrigation channels are damaged. During flood, hospital was damaged which is now shifted to another rented building.

	Kalagay village is located in mid of village Barnavai and Mankyar with a distance of about 2-3 km from Madyan, with a population of about 2000 with 300 Households. In 2005, earth quake hit the area and destroyed houses even earth quake jerks are still felt after every second day. During present flood the major losses in Kalagay were to roads, bridges, as area was disconnected due to Madyan road being washed away there was electricity failure, shortage of food, destruction of telecommunication services and inability to access hospitals.
Tirat	During flood, water level in Shahgram was about 20 feet or above, this caused damages to irrigation channels, water supply, fields and some houses were affected with the flood. Main roads, bridges, communication and electricity were washed away with flood and there was shortage of food however through mutual cooperation with neighbors they shared food. Major area of Shagai which was affected was Drab which is close to the river, presently there are fair chances that river might change course in the future which might hold other communities in close vicinity at risk. In advent of flood 2010, the village Damlay located in mountainous terrain was cut off as Madyan road was blocked away, hence the area faced damages in terms of electricity loss, food insecurity and damaged roads and pathways due to which movability was restricted, secondly village in close vicinity of Damlay namely Aryana was considerably effected, in fact most of it was washed away hence load of communities coming from that village was on it. In comparison Aryana couldn't have been selected because as most of it was damaged; any possible interventions in it were beyond mandate of an NGO, hence this village was selected as it being close to flood areas was under high risk, also as it was also disconnected from rest of the Swat and people in the area faced lot of problems in accessing resources hence its selection was made in that context. Kalagram village in Madyan is located in mountainous terrain, like Damlay. Consequent to flood 2010 this village also suffered damages in terms of devastation of roads and pathways and movability of communities was restricted. On the basis of these criterions it was marked as an effected village by local government authorities.

Note*: Villages marked with asterisk were dropped after review meetings with Diakonie in which they were either not prone to higher risk in the future or intervention demanded there were beyond mandate of an NGOs such as reconstruction of roads, embankments.

Current risk propensity of selected villages

The future risk in these villages were determined through hazard scoring, which was done with respect to river and streams flowing in the areas and their pattern in the past. Many of the areas mentioned below witnessed significant damage during floods and have been prone to seasonal flooding as well. In case of heavy torrential rain which can occur in monsoon

seasons lower Swat is endangered by cloud burst and with areas such as Murguzar, Spal Bandai and Gul Bandai located in vicinity of mountains, damages can be far more significant due to tumbling of rocks and disruption caused along mountain alleys.

Current Risk Levels in Project villages

As per the risk mapping, 117 households in Murguzar are at risk due to damages to houses and agriculture fields caused by flood 2010. Further to that seasonal flooding is a regular occurrence in lower parts of the village which is further accentuated by cloud bursts which causes tumbling off rocks to damage houses and block walking tracks. The area, particularly its lower parts is prone to soil erosion which has eaten away much of the land.

In Spal Bandai, 84 houses are at risk particularly those which are right besides the adjacent stream having no protection mechanism or safety plan. Much of the agriculture land and orchard trees located in the area has been eaten away by flooding. The area apart from major floods also witnesses seasonal flooding due to heavy rainfall and considering past history of rain, here many parts of the area are posed with considerable risk; also soil erosion has degraded agriculture land in the area and in many parts orchards have been destroyed.

In Kokrai, 41 houses have been at risk, the village is located in very narrow curve and in case of seasonal flooding there is less space for water to penetrate hence many houses get effected. During flood 2010, damages were particularly significant to agriculture fields which has caused soil erosion in the area. Now houses located close to river banks and streams are more so at risk in face of seasonal flooding.

In Gul Bandai, 31 houses have been at risk level in face of damages caused by flood 2010 and occurrence of seasonal flooding due to heavy rain fall. Agriculture land too has been damaged significantly which has created soil erosion and many mud dams constructed to channelize flow of water have been destroyed. Many parts of the area residing within mountains have been prone to cloud burst due to heavy torrential rain.

In Shagai village, 96 houses are at risk by nature of them being located close to streams and river banks adjoining the village, this area witnessed damages to many houses during advent of floods and most importantly in the current situation agriculture land which were increasingly effected are being eaten away because of soil erosion.

In Saidu Sharif, about 165 houses are at risk, by nature of them being located close to Murguzar river which is prone to both seasonal and major flooding. The village has several streams flowing in the area from which disruption is caused to houses and land.

Drab community is located right at the brink of Swat river which created significant damages in major and seasonal flooding and has about 150 houses which are at risk. The area is witnessing constant soil erosion.

Join community is located in Gharai village which included few other communities which were completely washed away, adjacent to river many of the houses ranging to about 78 are at high risk in the area.

Process and the criteria applied for the selection of the proposed households

Our definition of a village was guided by government entrenched norms according to which a village outlines segments of Mohalla and different communities closely adjoined together in a homogeneous pattern of landscape, water channels and streams. So in line with that our selection of villages was defined as demarcated by the government. Secondly as mentioned before, villages were selected based on types of hazards such as land sliding and floods which occurred in Islampur in conjunction, other measurable criteria was devastation in terms of houses, land, standing crops, roads and infrastructure, also different levels of vulnerability in terms of local habitat, land holding patterns, income, women/male headed households and food insecurity was also considered while selecting these villages. Once hazard zones in a village were defined through illustration of hazard maps, communities falling under those villages were selected and households within those were selected for survey.

Collection of secondary data (sources and information)

Topography of Swat

Swat valley lies in the north between 34°40' to 35°N latitude and 72 to 74°6' E longitudes at an altitude of 2000 m above sea level. The region is humid having mild summer with average annual rainfall exceeds 1000 mm and mean annual temperature of about 18°C, with increase of up to 30°C in summers. The rainfall patterns in July- August range from 120-134 mm³. The area amid mountainous terrain with intersection of streams flowing from Swat and Kalam Rivers is prone to different natural hazards. Seasonal flooding is a regular occurrence in the period of July-August when water level rises due to torrential rains and melting of glaciers in higher parts of Pakistan. Over the period of time, effect of deforestation has increased in Swat which has made areas more so prone to land sliding and erosion⁴. In context of such situation it is important to have community based tree plantation schemes. In terms of trees that can be planted different varieties are viable in Swat which includes Safaidda variety, it can be used for terracing along boundary walls of the villages and can also be used for candle making considering electricity is a major problem during flooding when infrastructure comes to an immediate halt.

Types of Hazard in Project area and their definition

All of the communities were seriously affected by major floods of 2010, which rattled most parts of Pakistan. The water in project areas ranged up to level of 5 feet, over flowed from Kalam and Swat rivers and skirted away into narrow streams too to effect houses, agriculture fields, roads, walking tracks and infrastructure mechanism. As indicated from the definition below, flooding beyond mean average accentuate within period of 2.33 years.

Flooding is a natural and recurring event for a river or stream. Statistically, streams will equal or exceed the mean annual flood once every 2.33 years.⁵ Flooding is a result of heavy or continuous rainfall exceeding the absorptive capacity of soil and the flow capacity of rivers, streams, and coastal areas. This causes a watercourse to overflow its banks onto adjacent lands. Floodplains are, in general, those lands most subject to recurring floods, situated adjacent to rivers and streams. Floodplains are therefore “flood-prone” and are hazardous to development activities if the vulnerability of those activities exceeds an acceptable level.

It is important to note that many of the low lying areas in project UCs come under category of flood plains which can be looked at from different perspectives. To define a floodplain depends somewhat on the goals in mind. As a topographic category it is quite flat and lies adjacent to a stream; geomorphologic ally, it is a landform composed primarily of unconsolidated depositional material derived from sediments being transported by the related

³ Metrological Department-Swat

⁴ Provincial Disaster Management Authority (PDMA)

⁵ Leopold et al., 1964

stream; hydrologically, it is best defined as a landform subject to periodic flooding by a parent stream. A combination of these [characteristics] perhaps comprises the essential criteria for defining the floodplain” (Schmudde, 1968). Most simply, a flood-plain is defined as “a strip of relatively smooth land bordering a stream and overflowed [sic] at a time of high water.

Flood categories

Seasonal Flood: An overflow of water onto normally dry land. The inundation of a normally dry area caused by rising water in an existing waterway; such as a river, stream, or drainage ditch. Ponding of water at or near the point where the rain fell. Flooding is a longer term event than flash flooding; it may last days or weeks.

Flash flood: A flood caused by heavy or excessive rainfall in a short period of time, generally less than 6 hours. Flash floods are usually characterized by raging torrents after heavy rains that rip through river beds, urban streets, or mountain canyons sweeping everything before them. They can occur within minutes or a few hours of excessive rainfall. They can also occur even if no rain has fallen, for instance after a levee or dam has failed, or after a sudden release of water by a debris or ice jam.⁶

Cloud burst: Many areas of lower Swat particularly Murguzar, Spal Bandai and Gul Bandai have been prone to cloud burst which resulted in damages to houses, infrastructure and walking tracks while also blocking ways and exit routes for the communities. A cloudburst is an extreme amount of precipitation, sometimes with hail and thunder, which normally lasts no longer than a few minutes but is capable of creating flood conditions. Cloudbursts descend from very high clouds, sometimes with tops above 15 kilometers. Meteorologists say the rain from a cloudburst is usually of the shower type with a fall rate equal to or greater than 100mm (3.94 inches) per hour. During a cloudburst, more than 2 cm of rain may fall in a few minutes. When there are instances of cloudbursts, the results can be disastrous.⁷

Soil Erosion: Soil Erosion witnessed in flood plains of project area is the process by which material is removed from a region of the Earth surface. It can occur by weathering and transport of solids (sediment, soil, rock and other particles) in the natural environment, and leads to the deposition of these materials elsewhere. It usually occurs due to transport by wind, water, or ice; by down-slope creep of soil and other material under the force of gravity; or by living organisms, such as burrowing animals, in the case of bioerosion.

Primary data collection of village population

UC	Villages	HH	Population	Source
Islampur	Kokra	424	3,116	BISP
	Sapal Banday	514	4,112	Local Community
	Gul Banday	617	3,288	Local Community

⁶ <http://www.srh.noaa.gov/mrx/hydro/flooddef.php>

⁷ <http://www.islandnet.com/~see/weather/elements/cloudburst.htm>

UC	Villages	HH	Population	Source
	Murguzar	722	9,300	BISP
Saidu Sharif	Saidu	3,383	22,001	BISP
	Saidu Shagai	1,066	60,915	BISP
Madyan	Palam	120	1,800	Local Committees
	Ghari	800	14,400	Local Committees
	Shahgram	1,000	6,000	Local Committees
	Kalagay	300	2,000	Local Committees
Tirat	Kandal	250	2,000	Local Committees
	Shagai	300	1,200	Local Committees
	Damlay	60	480	Local Committees
	Kalagram	200	1,600	Local Committees

Hazard Analysis

As per 8 final villages selected in 4 project UCs of Swat, hazard mapping exercise was conducted in each and accordingly village maps were drawn. While conducting the exercise, team leaders for upper and lower Swat ensured community participation and therefore worked with community resource persons and local villagers to draw hazard maps. Before drawing hazard maps, hazard zones in each village were identified and some preliminary information was also gathered from community members



Hazard mapping exercise

regarding types and history of hazards in the area with effect it had on the community.

Villages of Swat are composed of multi dimensional textures where amid mountains there are plains and river streams as well. So most of the villagers in UC Islampur located in lower Swat cited major flooding, soil erosion, cloud burst and land sliding as major hazards, as a result movability is restricted, food insecurity could prevail and mud houses are continuously under threat of being damaged along with agricultural fields. Often in areas of lower Swat narrow water channels are observed which can smear up water in thin spaces to create vast damage. Based; on community input and discussion with local metrological department particularly in upper Swat rising temperatures causes melting down of glaciers resulting in over flowing of riverine thus leading towards higher probability of flash floods. In upper Swat there is also a tendency of river to change course which endangers areas which might not be at risk previously. Due to seasonal eroding, river becomes more so tied with areas and gradually hazard level increases, a case which was particularly witnessed in Shagai Shahgram village.

Hazard Ranking of Villages

Villages	Flood 2010	Seasonal Floods	Soil Erosion	Cumulative Hazard Score
Gharri	2	3	2.5	2.5
Gul Bandai	2	NA	2.5	2.25
Kokrai	2	NA	2.5	2.25
Murguzar	2	NA	2.5	2.25
Saidu Sharif	2	NA	2.5	2.25
Shagae Shahgram	2	NA	2.5	2.25
Shagai	2	NA	2.5	2.25
Spal Bandai	2	NA	2.5	2.25

Methodology notes for Ranking

Under each village two significant hazards were taken into account and selected in terms of their magnitude and frequency. Barring Gharai village, all other villages were prone to soil erosion and affects of Flood 2010. With regard to Flood 2010, magnitude indicator was selected measured in terms of level of water rise starting from above 5 feet which is denoted by scale of 3.

It was felt that frequency was not a strong indicator in this situation because flood 2010 was of one the worst natural calamities in Pakistan, and if we measure it by frequency intensity of it in terms of hazard position it would be under valued. With regard to seasonal flooding which occurs in most parts of Swat due to over flowing of Swat and Kalam rivers, both magnitude and frequency are emphasized as effective indicators, as it occurs every year in scale of up to 3 feet.

Based on these hazards overall hazard score has been determined. Gharai, located in upper Swat carries the highest score in terms of devastation it has during seasonal as well as 2010 flood. The determined hazard scores have laid basis for conducting vulnerability survey computation of which with hazard scores will give measure of risk assessment of communities.

Social Mapping

To conduct hazard mapping it was imperative to conduct situational analysis of socio economic position and standing of the village, as per which maps drawn depicted that state. Prominent land marks like houses, cropped fields, schools, hospitals, mosque were highlighted in the map with indication of topography and terrain amid which village is positioned. Situational analysis also classified demographic profiling of the village, scale of damage in the area and priority needs

for the villagers. Based on what transpired out of most of the villages, it was understood that in cases of floods or land sliding, areas are completely cut off from the main towns. As a result, people are faced with food insecurity, restricted mobility and lack of access to resources. In many of the locations people having mud houses are more endangered in advent of floods which can disjoin walls of houses and water can come into houses to create sanitation and hygiene problems. In many villages local coping mechanism emphasizes on human resources who at times of floods can safeguard lives of people. However these resources needs to be institutionalized in each village in terms of Disaster Management Committees (DMCs).



Female community based discussion on disaster preparedness

Transect Walk

During the transect walk project teams treaded along the villages along with members of community from different walks of life be it school teachers, farmers, clerics, shop keepers etc. The purpose of these transect walks was to visibly verify the hazard details drawn on the maps while also viewing major damaged areas in terms of destruction of irrigation channels, embankments, destroyed water systems, houses, buildings etc. Based on these walks, some minute details which were missed in different instances during hazard mapping were also notified and incorporated later onto the maps drawn. During these transect walks damaged irrigation land, silted embankments were seen, along with some innovative activities initiated by community which includes biogas systems and water storage techniques. In one of the instance school girls wash their hands from water stored in the water channel, water that comes from mountains is being stored in this concrete water channel which has two open able blockages at both ends. This is a good practice for water storage in village Gul Banday and can be replicated in other mountains. Villagers said that we use this water for washing.



School children using community designed drinking water source



Transect Walk in Kokarai village

Focus Group Discussions

Focus Group Discussion produces data and insights that would be less accessible without interaction found in a group setting—listening to others’ verbalized experiences stimulates memories, ideas, and experiences in participants.⁸ This is also known as the group effect where group members engage in “a kind of ‘chaining’ or ‘cascading’ effect; talk links to, or tumbles out of, the topics and expressions preceding it. Focus group discussion involving segregated groups of men and women were conducted. Each focus group involved 6-12 participants who belonged to all walks of life encompassing farmers, teachers, students, government servants, house wife, tailors, livestock owners, doctors etc. During the focus group discussion different aspects pertaining to disaster risk reduction were probed that covered existing capacities of people, different levels of vulnerability they have

⁸ Lindlof & Taylor, 2002, p. 182

and preparedness and mitigation measures that could be implied at community level. Initially it was felt that community understanding of Disaster risk reduction mechanism was unclear, they instead required relief items in terms of livestock, agri inputs so that they could meet food insecurity. However later on with more probing they started to understand virtues of disaster risk reduction based on which they agreed that certain community based institutions should be formed that could serve as Disaster Management Committees. Within these DMCs, sub school committees, water and health management committees should be formed with each taking up different roles. A disaster management committee shall be required to develop emergency plan at village level to identify evacuation routes and serve as custodian of emergency tool kit. Roles will be ascribed to members in it. As per the current situation it was felt that reliance of traditional and cost effective mitigation measures should be pursued with which constitutes seed banks, food preservation and grain storage methods, water harvesting techniques, other aspects included carving out man made pond and check dams to prevent soil erosion and de-silting of land.



Female group discussion in Murguzar village



Male focus group discussion in Gharai village

Community insights on Disaster History, their coping mechanisms and superstitions

According to FGDs, communities located in Islampur are prone to hazards in form of land sliding occurrence of which is seasonal during spring. Flood occur seasonally as well and water level ranges up to 1-3 feet in areas where damage becomes more so due to narrow water channels. During flood 2010, flood water ranged over 5 feet and communities did not have any coping mechanisms, the only solution they had was to go to houses of their relatives which were located in safe zones. In UC Saidu Sharif areas are segmented according to urban and rural zones, the rural zones receives seasonal flooding of up to 3 feet, while during flood 2010 water level reached over 10 feet which created wide spread damages in the areas particularly those located close to river banks. In upper Swat, floods are a major disaster due to accentuation of flood water from Kalam River due to melting down of glaciers. In flood 2010 water level ranged over 15 feet in some of the villages in which risk assessment was conducted while in seasonal flood water level is up to 3 feet. In these areas communities due to religious beliefs and norms see hazard as consequence of will of God hence they consider themselves

as passive to resort to any measures to counter act its occurrence. Currently they consider themselves in state of risk as they know that climate is changing and floods in range of one which came in 2010 will become more frequent.

Village wise summary of Focus group Discussions

Murguzar: As mentioned before Murguzar is the biggest village of Islampur UC. Within the village there are different sub communities who have been prone to floods and land sliding. Communities in the village mentioned need of a Disaster Management Committees divided by different quarters in the village. Coordination shall be maintained between each in case of eruption of floods in one area which could follow another, as many of the areas of Islampur are located on river banks, community identified plantation along with retaining walls and deflection dams at critical zones, school safety program can be integrated with plantation schemes and at individual level interventions like seed banks and kitchen gardening to meet food insecurity can be incorporated. Other interventions included of energy efficient stoves, mushroom growth through wheat straw techniques and candle making techniques. Places for safe asylum can be identified and selected such as local community mosque or school and evacuation routes and disaster reduction information can be posted there.

Spal Bandai: In Spal Bandai communities required DMC for purpose of awareness building of communities in preventative methods to adopt in case of floods or land sliding and how to evacuate to safe zones. The water channels in the village are very narrow and in case of flooding water can intensify and accumulate in narrow spaces and therefore could come in houses, which restricts movability and creates sanitation problems. So if some deflection dams are placed, flow of water can be channelized. Traditional clean drinking water techniques can be implied in the village by use of crush in pots to clean water, also levels of houses close to river banks can be increased by building of wooden bars at their entrance. Apart from that other interventions like energy efficient stoves and candle making can be implied in the village considering electricity disruption during floods.

Gul Bandai: According to community knowledge 50% of agriculture land in the area has been destroyed, due to narrow water channels, water came into houses and destroyed fields as well. Land sliding is also a major threat in the village which in fact lead to damages on various mud houses, hence tree plantation schemes can be implied here which can be integrated with school safety program. Also if deflection dams are built in the village, flow of water will be effectively channelized into fields.

Kokrai: In Kokrai, communities stated that erosion caused by flood water has damaged land and houses close to narrow water channels. If deflection dams are built in the village flow of water can be appropriately directed into fields without its accentuation. Household level interventions would include energy efficient stoves, candle making techniques and construction of small bars besides houses so that water does not penetrate into houses. Clean water techniques by use of crush can also be incorporated.

Saidu: In Saidu village, rural communities stated that flow of the water was beyond 5 feet in flood 2010. The area also receives seasonal flooding so in various part of the village household level interventions in form of food preservation systems, energy seed banks, efficient stoves and candle making can be implied. At community level disaster management committee can be formed which develops village disaster risk plan and in line with that evacuation routes are identified. Any mitigation measures such as plantation and retaining walls at critical zones can be done through the platform of DMC. The area also faces sanitation and clean water problems so water harvesting techniques can be used.

Gharai: In Gharai village, located in upper Swat, water reached at level of 15 feet during flood 2010 due to which sub communities of Jopin and hospital colony were completely destroyed. In this village disaster management committees divided into quarters can be formed along with school safety programs and asylum fortification. Plantation can be done in critical points and at household level interventions as indicated by communities include energy efficient stoves, candle making techniques, food preservation means and seed banks in areas of agriculture lands can also be implemented.

Shagae Shahgram: Shagae Shagram village, located in upper Swat, is prone to seasonal floods of up to 3 feet. During flood 2010 as per community insights water level accentuated up to 15 feet which greatly affected areas of Drab that was close to river bank. In this community disaster management plan in form of DMC at community level can be formed that develops evacuation plan, ensures liaison with different stakeholders for support during disaster and under platform of which mitigation measures in form of plantation, retaining walls and school safety programs are planned. Household level interventions in the area could include food preservation and clean drinking water techniques, energy efficient stoves and candle making techniques.

UC	Village	# of FGDs	Participants of Male FGD	Participants of Female FGD
Islampur	Kokrai	2	Farmers, teachers, students, shopkeepers, labors, committee member, Government servant	Farmers, barber, retired manager, driver, teachers, students, labors, masons
	Sapal Banday	2	Farmers, barber, retired manager, driver, teachers, students, labors, masons	House wives, livestock keepers, widow
	Gul Banday	2	Farmers, security guards, labors, students	House wives, maid, students, livestock keeper
	Murguzar	2	Farmers, business men/ shopkeepers, labors, driver, hotel cook, students	House wives, handicrafts workers, widow, livestock keepers, teacher
Saidu Sharif	Saidu	2	Community member, labor, farmer, shop keeper, student, mason, bus conductor	House wives, tailors, teachers, livestock keeper
	Saidu Shagae	2	Community chairman, labor, farmer, shop keeper, student, mason, bus conductor	House wife, tailors, teachers, livestock keeper, student

UC	Village	# of FGDs	Participants of Male FGD	Participants of Female FGD
Madyan	Palam	2	Labor, student, tailor, government servant, farmers, medicine store keeper	House wife, student, tailor, farmer
	Ghari	2	Shop keeper, social worker, government teacher, land lord, student, electrician, transporter, retired manager, social activists, doctor	House wife, tailor, tuck shop keeper, student
	Shahgram	2	Shop keeper, farmer, tailor, driver	House wife, tailor
	Kalagay	2	Student, farmer, government servant	House wife, tailor
Tirat	Kandal	2	Teacher, business man, exporter, police man, shop keeper, labor, students	House wife, student
	Shagai	2	Shop keeper, farmer, teachers, student	House wife, student, livestock keeper
	Damlay	2	Cleric, land lord, school teacher, driver, shop keeper	Government primary teacher, house wife, tailor
	Kalagram	2	Shop keeper, farmer, teachers, student	House wife, student, livestock keeper

Information gathered using other PRA tools

As per hazard mapping conducted, it become also important to conduct situational analysis of the village which encapsulates demographic profile of the community along social mechanism that in place in the village while also emphasizing on scale of damage and priority need of the villagers. At different level participatory rural appraisal of community was undertaken in which their information regarding their socio economic standing and capacity to with stand the disaster was also materialized. This information is separately mentioned in separate village reports.

Summary of household vulnerability analysis

In order to set framework for household vulnerability assessment and derive coherent results; there sub indicators including **1) exposure** **2) fragility** and **3) lack of resilience** were separately analyzed in view of both vulnerability ranking and later by categorization through context of Hazard analysis.

Exposure Analysis

Exposure analysis signifies property that is exposed to risk, as out of 765 households surveyed 633 stated that their vulnerability in that respect is in rank of 3 which is the highest rating. Possibly it can be reflective of their perception regarding flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent hence their property would be continuously under risk.

Exposure Analysis in context of Hazard scores

As per computation of households that are exposed in context of hazard, about 687 of houses fall under hazard score of 2.25 out of which 88.2% are exposed to exposure score of 3 which goes to show that even though area is only at medium hazard level but still communities feel that they are still exposed at the highest level considering events of flood 2010. More over it again emphasize the point of vulnerability assessment which is from the perspective of household while hazard is more a trend or factual measurement.

Fragility Analysis

Fragility covers in itself different levels of vulnerabilities; be it land holding, livestock, income, food insecurity, each in itself has a certain measurement because these factor underline communities potential counter action against a disaster. According to fragility indicator, 415 out of 765 households' surveyed fall under range of 2.2-2.5 which is almost touching the

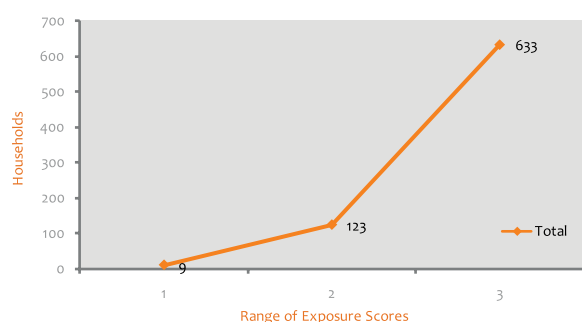


Figure 1: Households under Average Exposure Ranges

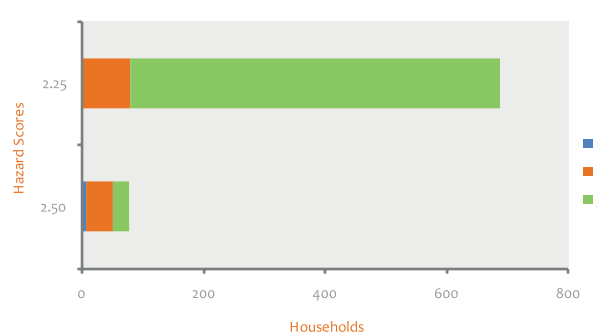


Figure 2: Categorization of Household in context of Exposure

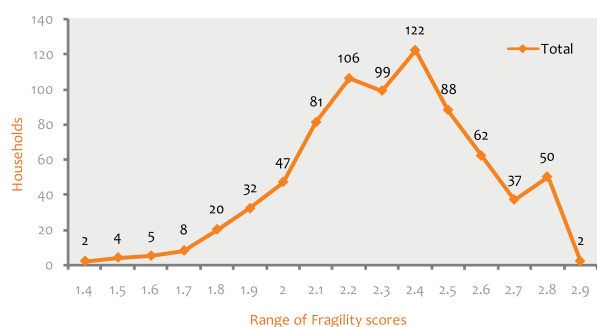


Figure 3: Households under Average Fragility

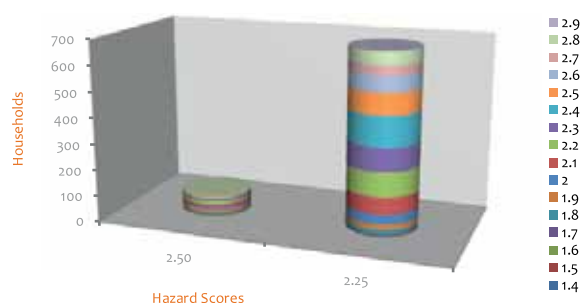


Figure 4: Categorization of Hazard Prone Communities in Context of Fragility Indicator

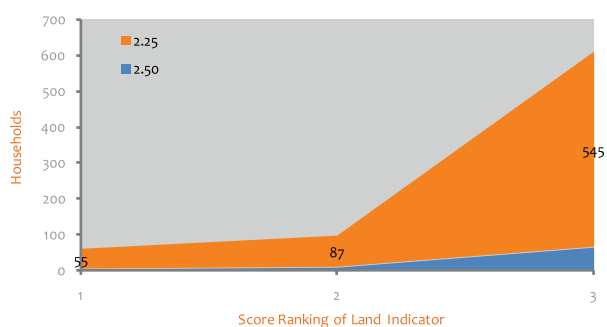


Figure 5: Land Holding Vulnerability

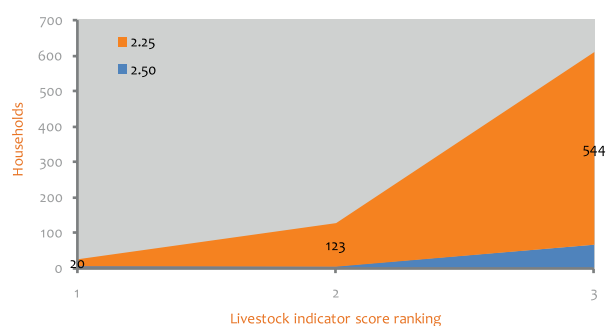


Figure 6: Livestock Vulnerability

higher end. Further analysis would show that under fragility sub indicators basic amenities constituting land, livestock and income which signify monetary wellbeing fall under the rank of 3.

Fragility Analysis in context of Hazard

In comparison of fragility analysis in the context of hazard scores, it was indicated that 90% of the households fall under hazard score of 2.25 out of which highest proportions fall under 2.2-2.6 which is almost touching higher end as indicated by second bar in the figure 4.

Indicators of Socio economic wellbeing of communities in context to Hazard

Land Holding

About 687 of households surveyed under risk zones fall within hazard score of 2.25 out of which 545 have a land holding vulnerability of 3. In total 610 households out of total survey of 765 fall under land holding rank of 3 irrespective of their hazard score which equates to 80% of the total survey size. (see figure 5)

Livestock

About 687 households surveyed have fall under hazard score of 2.25 which if compared with specific fragility indicator constituting vulnerability in terms of livestock, it is signified that 544 households out of 765 fall under the rank of 3 for livestock. Out of 765 household surveyed, 611 fall under the vulnerability rank of 3 for livestock.

Income

Comparison of specific vulnerability indicator in terms of income with hazard score signifies that about 610 households fall under rank of 3 with hazard score of 2.25. Out of 765 households surveyed; 649 falls under rank of 3 for income indicator irrespective of their hazard score; a figure which is 85% of the surveyed households.

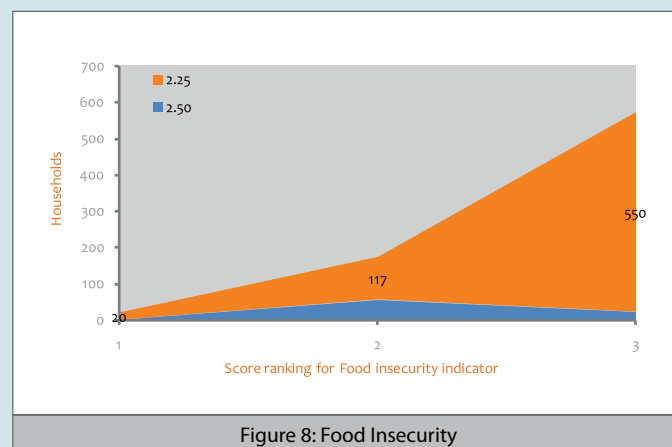
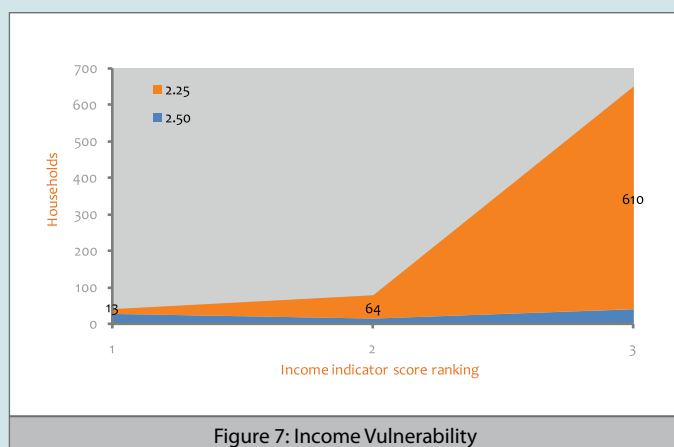
Food Insecurity

Comparison of specific vulnerability indicator constituting food insecurity with hazard score unravels that about 550 households with food insecurity rank of 3 falls under hazard score of 2.25. Out of total household surveyed of 765 about 572 falls under insecurity rank of 3 irrespective of hazard score which equates to 75% of the total households surveyed.

Summary of socio economic indicators

The four indicators including income, food insecurity, livestock and land mentioned above are reflective of socio economic viability of a household. Addressing them in context of hazard allows one to ascertain accentuated nature of vulnerability pattern of houses. In the table below, land, livestock and income being primary monetary sources for households have strong correlation as communities become prone to higher degree of hazard; overall vulnerability in terms of these indicators increases by 5.3%. It was noted that when households become prone to greater hazard range income vulnerability increases to 3.9% above the mean average along with Food insecurity indicator which increases by 5.3%. However there are no overall significant deviations in land holding and livestock indicators.

Indicator	Indicator Analysis		Hazard analysis of Indicators		
	Houses within rank of 3	Proportion of survey	Houses under highest hazard score of 2.50	Houses under highest hazard score with rank of 3	Proportion of households in hazard score of 2.50 with indicator rank of 3
Land	610	79.70%	414	336	81.15%
Livestock	611	79.80%	414	357	86.20%



Indicator	Indicator Analysis		Hazard analysis of Indicators		
	Houses within rank of 3	Proportion of survey	Houses under highest hazard score of 2.50	Houses under highest hazard score with rank of 3	Proportion of households in hazard score of 2.50 with indicator rank of 3
Food Insecurity	572	74.70%	414	310	74.80%
Income	649	84.80%	414	363	87.60%

Lack of resilience

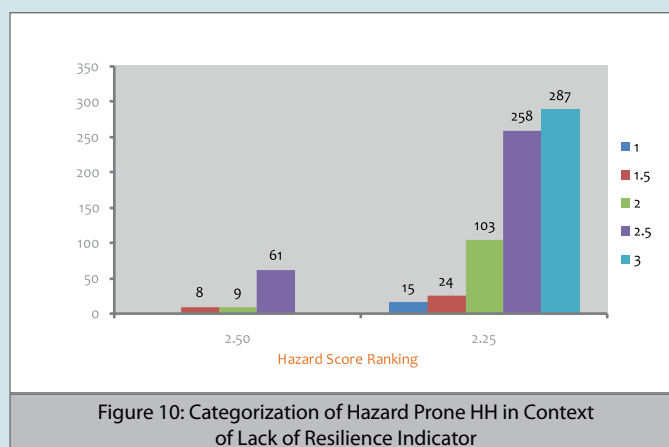
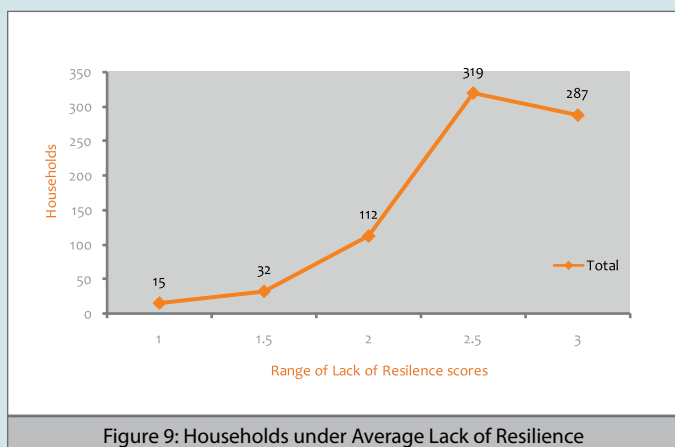
Lack of resilience indicator is measured through education and neighborhood patterns in a village, under it 79% of houses fall within scores of 2-2.5 which almost touches higher boundaries. For further analysis 83% of houses have no member who goes to school so literacy trends to respond to hazard is low. However, for 50% households neighborhood support patterns are ranked at 2, while for 41% ranking is 3. So it can be concluded here that literacy level is greater reflector of vulnerability of households in context of any disaster mainly because of lack of awareness amongst communities. Furthermore, it is felt that neighborhood patterns are not directly correlated with hazards.

Lack of resilience in context of Hazard

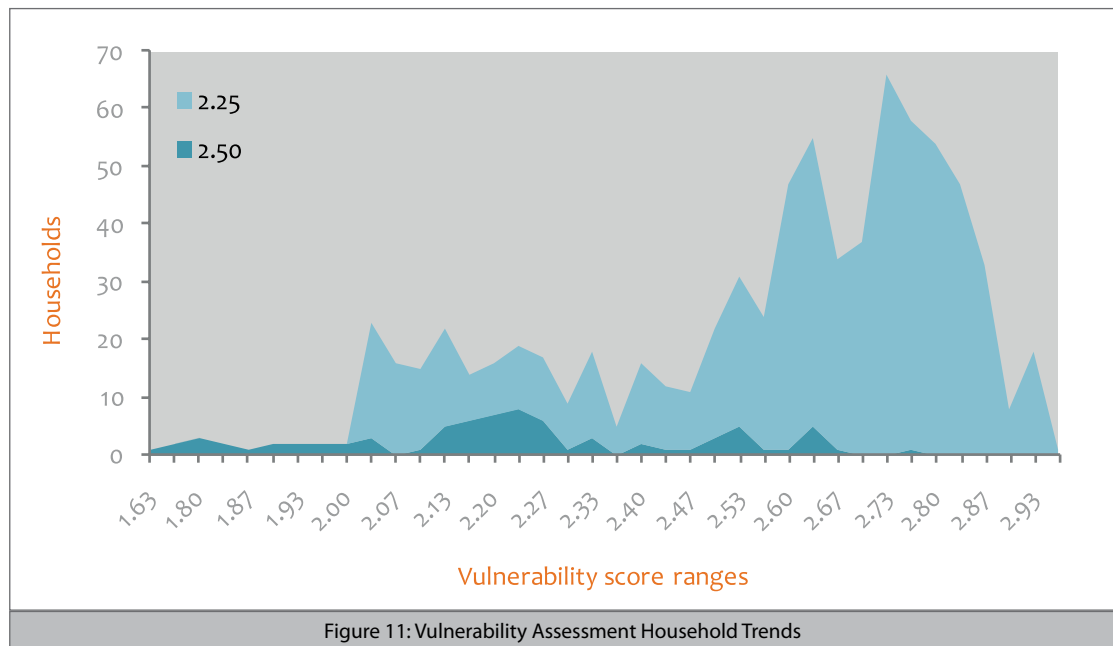
In comparison of hazard scores with lack of resilience indicator, 687 houses fall under hazard score of 2.25 out of which 545 household highlighted in figure 10 fall under zones of 2.5-3, which again show that these indicators despite being indicative of socio economic standing of communities also advocate whether they have coping mechanism to with stand disaster. With this, it can be derived that majority of the households lack literacy levels and social cohesion as integrated elements in context of hazard risk.

Overall Vulnerability Score Range

The depiction as per the figure 11 shows that irrespective of hazard scale vulnerability indicator is on the higher side, 70% of the household fall between ranges of 2.5 and above so it can be determined that already vulnerable household which translate their set of



vulnerability in context of a disaster, when measured with the hazard the results show the same depiction most of the households ranging up to 414 fall under hazard score of 2.50.



Identified DRR & CCA project interventions

Components	Interventions
Disaster Preparedness	Formation of Disaster Management Committees at village level under which roles will be ascribed to members in terms of disaster awareness at community level, coordination with different stakeholder, custodianship of emergency tool kit that is provided. Once a Disaster comes this community will enact as an emergency unit under fold of which they shall be separate search and rescue, communication, first aid team. These DMCs as per community based knowledge will be divided into different village quarters and will develop a network of regular coordination between each other.
	Inclusion of sub school, water management and health committees with defined roles
	Training in CBDRM Model for roles taken up by different committees
	Development of Village level Hazard planning (identification of evacuation routes, fortification of community buildings like schools as safe asylums, mapping of safe and hazard prone areas)
	Establishment of disaster management funds for different mitigation measures at community level.
	Initiating school safety programs (Mock drills)
	Linkage strengthening with local municipalities and line authorities
Mitigation Measures	Plantation of trees alongside river bank, using shrubs and rocks to mitigate risks. Construction of small scale retaining walls.
	Kitchen gardening for food conservation. Encourage use of traditional food preservation methods eg. grain storage, meat preservation. Considering that many of the villages particularly those in lower Swat are cut off from the main area due to land sliding and floods which blocks pathways thus creating inaccessibility towards resources. In lower Swat many people located in mountainous terrain have barren lands as well on which Mushroom plantation can be initialized considering food insecurity in the area.
	Homemade fuel efficient energy stoves, biogas systems and cost effective candle making techniques considering electricity problems that can be created.
	Using humus and organic matter to prevent soil erosion and degeneration
	Creation of seed Banks
	Water harvesting and preservation of rain fed water mechanisms such as roof water harvesting, clean drinking water techniques by use of crush

Ranking of Communities

Communities have been ranked by computation of their hazard and vulnerability scores. Ghari village located in upper Swat; and Shagai in lower Swat ascribes the highest scores whereas mean value of risk assessment is 6.9. So overall risk assessment score is within medium level considering that communities that were selected witnessed damages, but these were not completely washed away. However disaster risk reduction project needs to be initialized in these so that they can minimize effect of future risk.

Total households in risk zones surveyed numbered to 765. The larger survey data had to be churned down considering focus that had to be maintained only on risk prone households. As indicated by figure 12 & 13 households are plotted according to defined range criteria, 400 houses which is the greatest proportion that fall within range of 2 to 4 whereas 356 households fall within range of 6 to 8.

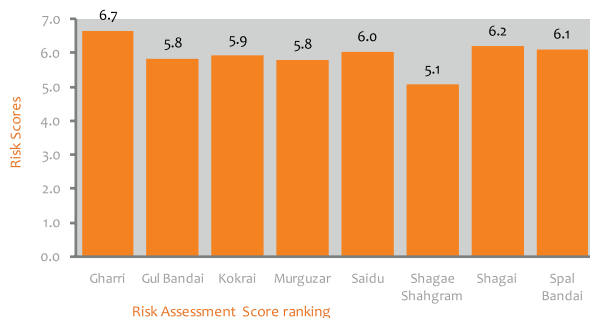


Figure 12: Assessment Categorization of Households

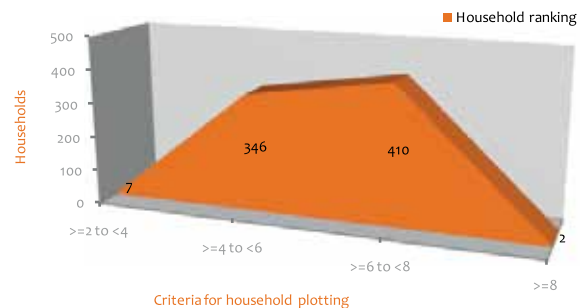


Figure 13: Household Ranking

Annexes

Annex 1: Village Report - Murguzar

Tehsil: Babuzai	Union Council: Islampur	HH Size: 12	Hazard Score: 2.25
Hazards in the village: Seasonal flooding, major flooding & soil erosion			
Surveyed household prone to hazards: 117			
NGOs/ Agencies working in the village			
1	NRSP: Cash for work, NFIs and Livelihoods		
2	Lasoonia: Cash for work		

Profile of the Village

Village Murguzar is basically a mountainous area with a population of about 9,300 and 722 Households (BISP Survey). It is about 21 km away from Mingora and by drive it takes about 30 minutes to reach Murguzar village from Mingora City, Swat. Murguzar Village is situated on both sides of road and some portion of population also lives on upper side of hills which are hilly and track able. Village Murguzar is known for its handicrafts made by women in their houses i.e. Shawls, embroidery, purses etc. The areas near river have the capacity for fishing and fish hotels can be established.



Hazard Assessment

Murguzar village is the biggest village of UC Islampur and is located amid steep mountains and patches of plain lands; it generally receives seasonal flooding in range of 1-3 feet however in advent of floods 2010 the gushing stream that adjoins the village over flowed from Swat river over 5 feet and inflicted damages to buildings, land, infrastructure, Murguzar road and standing crops. House construction designs are not proper as there was no terracing which allowed destruction and damages at large scale. The walking tracts were damaged with cloud burst, houses and mosques near river side were washed away with flood water. Two Steps of Hotel White Palace (Tourist Hotel) are also damaged. Villagers in Murguzar are of the view that there is needs for a disaster management committee which



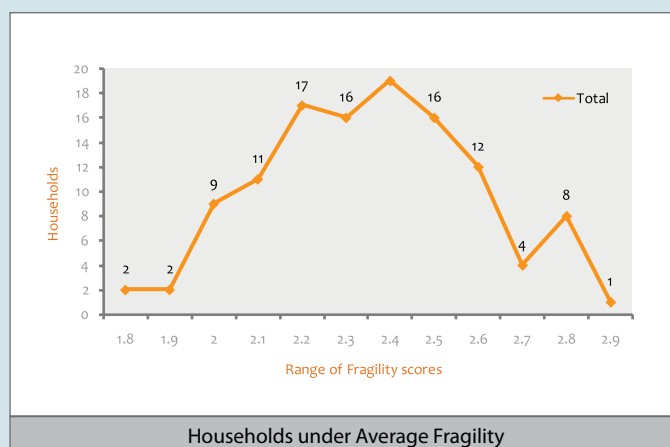
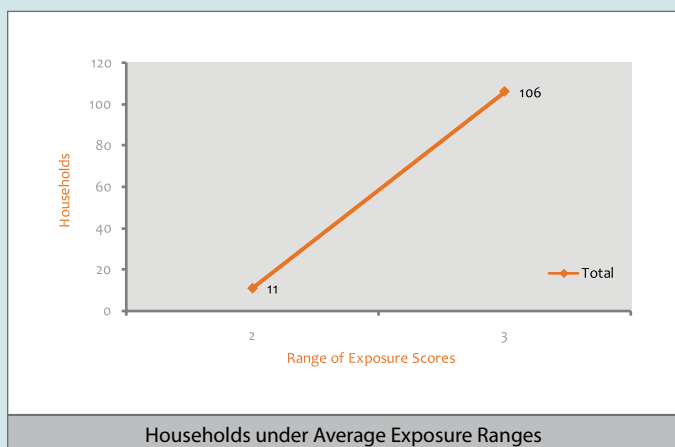
develops an emergency plan and creates awareness among committee. They also need terracing and plantations that can be integrated with school safety program. Committee member of Murguzar Village told that they are working in Cash for work schemes with NRSP and Lasoona Organization but this work is only temporary. They said “we have stones and crush as natural resource which we get from river if we are only provided with cement we can rehabilitate our roads.” This will provide a permanent solution for the walking tracts rehabilitation as well as land sliding will not destroy the walking tract again. Establishment of small dispensaries or first aid trainings and water quality management trainings to villagers will help minimize the effects of disaster.



Vulnerability Assessment

Summary of household vulnerability Analysis: In order to set framework for household vulnerability assessment and derive coherent results; there sub indicators including **1) exposure 2) fragility** and **3) lack of resilience** were separately analyzed in view of both vulnerability ranking. However, as hazard scores for the villages is cumulatively signified, vulnerability ranking would also signify hazard level of 2 which is evident in the village.

Exposure Analysis: Exposure analysis signifies property that is exposed to risk, as per the overall results 106 out of 117 HHs stated that their vulnerability in that respect is in rank of 3 which is the highest rating. Possibly it can be reflective of their perception regarding flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent hence their property would be under continuous risk.



Fragility Analysis: Fragility covers in itself different levels of vulnerabilities; be it land holding, livestock, income, food insecurity. Each in itself has a certain measurement because these factor underline communities potential counter action against a disaster. According to fragility indicator, 80 out of 117 households surveyed fall under range of 2.2-2.6 which is almost touching the higher end. Further analysis would show that under fragility sub indicators basic amenities constituting land, livestock and income which signify monetary wellbeing fall under rank of 3.

Indicators of Socio economic wellbeing of communities in context to Hazard

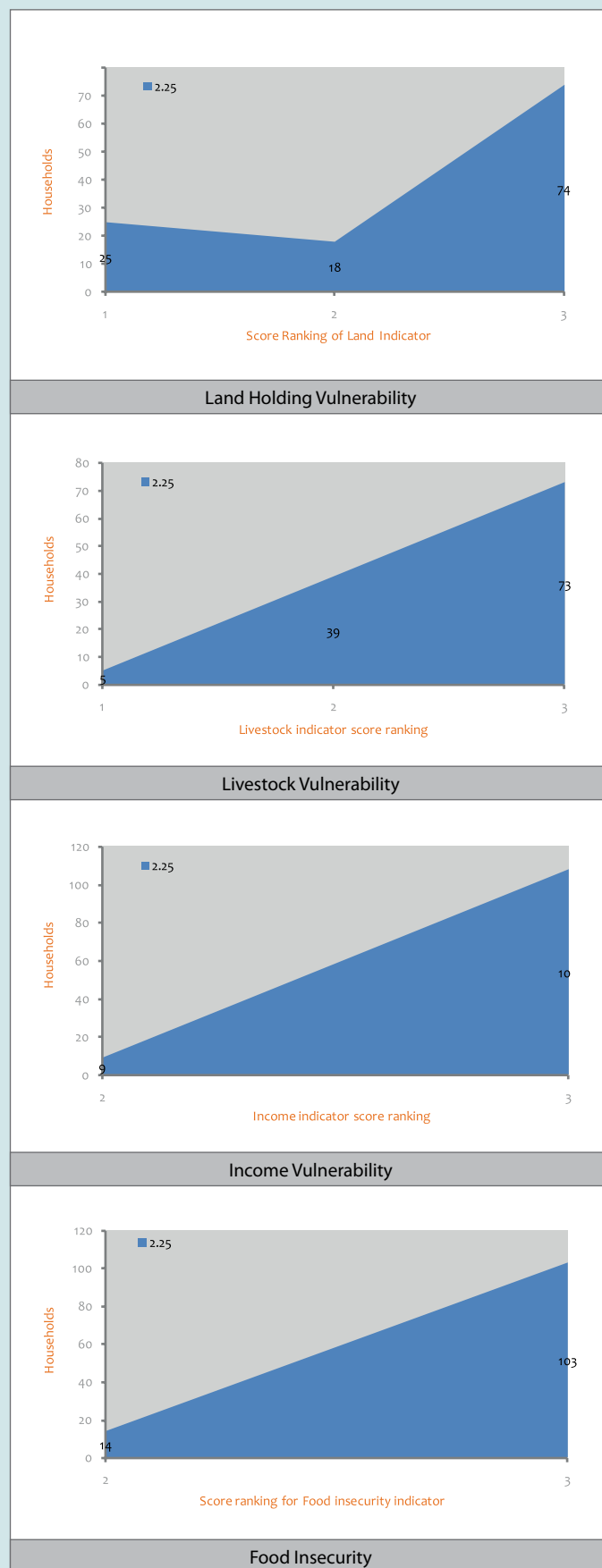
Land Holding: The village has hazard score of 2.25 and 74 out of 117 HHs fall under rank of 3 for land indicator which is 63.2% of total survey size in the village.

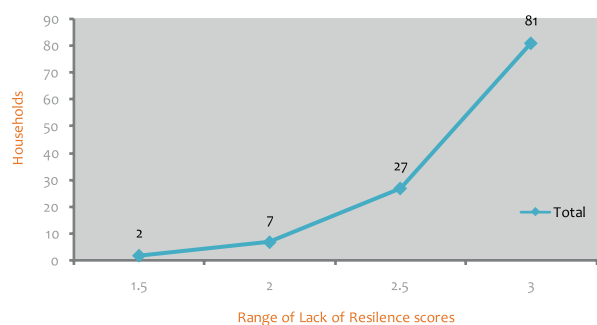
Livestock: The village has hazard score of 2.25 and 73 out of 117 HHs fall under rank of 3 for livestock indicator which is 62% of total survey size in the village.

Income: The village has hazard score of 2.25 and 108 out of 117 HHs fall under rank of 3 for income indicator which is 92% of total survey size in the village.

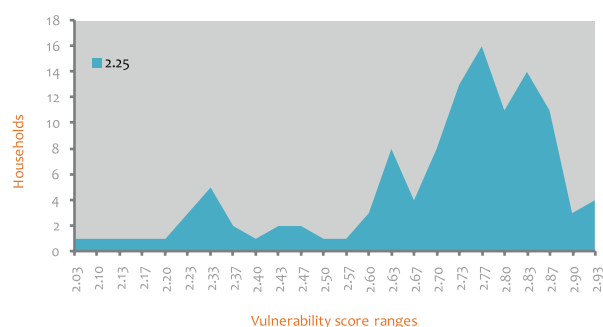
Food Insecurity: The village has hazard score of 2.25 and 103 out of 117 HHs fall under rank of 3 for food insecurity indicator which is 88% of total survey size in the village.

Summary of socio economic indicators: In the village there is strong correlation between income and

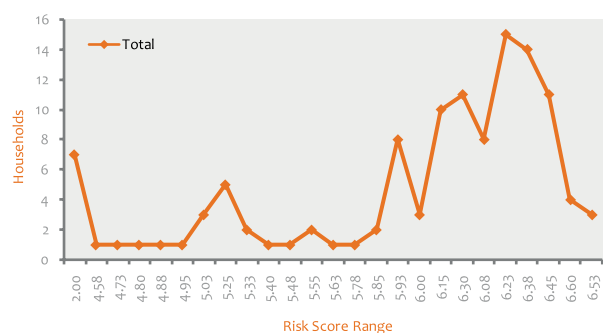




Households under average Lack of Resilience



Vulnerability Assessment Household Trends



Household Count of Risk Assessment



Murguzar Risk Map

food insecurity indicators with both falling under same proportion under rank of 3 and therefore household sizes are comparatively greater in number than for similar correlation seen between land and livestock. So in the village, income and food insecurity are greater drivers of vulnerability for majority of hazard prone households in contrary to land or livestock.

Lack of resilience

Lack of resilience indicator is measured through education and neighborhood patterns in a village, under it 92% of houses fall within scores of 2-2.5 which almost touches higher boundaries. For further analysis 90% of households have no member who goes to school so literacy trends to respond to hazard is low; lack of neighborhood support despite being in lower proportion to education level has proportion of 75% households in rank of 3. Hence it can be concluded that education level is greater driver of vulnerability in the village.

Overall Vulnerability Score Range

The depiction as per the graph shows that irrespective of hazard scale, vulnerability indicator is on the higher side, 97 out of 117 surveyed households fall between ranges of 2.5 and above.

Risk Assessment

In calculating risk assessment in Murguzar Village it was revealed that about 74% of households equating to 97 falls under risk score of over 5.33 with mean average of around 6.33.

Accessibility

Average altitude of the village: Low as compared to Upper Swat

Average time that villages are cut off during the winter time: Due to cloud burst and heavy rain falls in winter (December and January) and summers (July and August) cause land sliding and Chitor road (Road turning from Murguzar Road to Kokrai) is blocked

Availability of public and other transport: No public transport is available in the village and villagers have to walk to Murguzar Road to get the public transport.

Demographic Profile

Name of Village	No. of Total HH	No. of widows & Single Women*	No. of Child-headed HHs*	No. of elderly as head of HHs*	No. of disabled as head of HHs*	Priority Needs	Village Infrastructure
Murguzar	722	50	20-30	150	5	DMCs and Sub DMCs, Plantation, Mushroom Cultivation, food preservation, Kitchen gardening, Candle Making, School Safety Programs (6), Seed Banks, Energy Efficient Stoves, Retaining Wall (200 Feet), safe zones. Note: *School students will be motivated to participate in plantation (i.e. sowing and safety of plants and trees) and hazard planning activities (i.e. drawings and coloring)	8 Mosques 4 educational institutions i.e. 1 Government Girls schools (Primary) and 3 Government Boys school (Primary, Middle and High), shops, 1 Hotel and bus stop. All the village houses do have electricity and telecommunication facility.

**Estimated figures based on local community knowledge*

Annex 2: Village Report - Sapal Banday

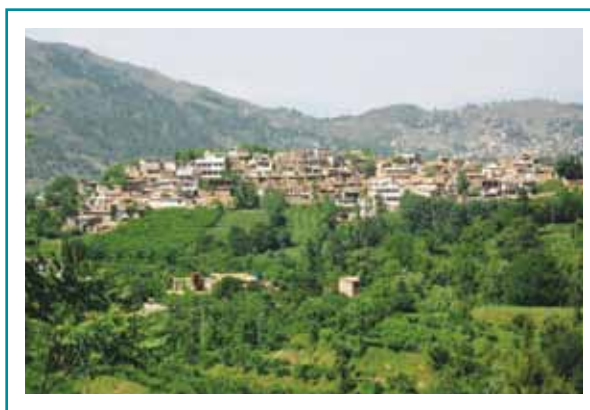
Tehsil: Babuzai	Union Council: Islampur	HH Size: 8	Hazard Score: 2.25
Hazards in the village: Seasonal flooding, major flooding & soil erosion			
Surveyed household prone to hazards: 84			
NGOs/ Agencies working in the village			
1	NRSP: Livelihood		
2	Lasoon: Livelihood and Cash for work		
3	Spado: Child Protection Centers		
4	Red Cross: Assessment for Livelihoods		

Profile of the Village

Sapal Baday is 15 km (driving 20 – 30 minutes) away from Mingora, Swat by going through Nishat Chowk, Mingora to Murguzar Road and then turning to the left hand you will find Sapal Banday Village. Basically it is located on a hill with a population of approximately 4,112 and 514 households. About 70 % of the inhabitants are connected with Mia Gaans caste, and they are land owners while rest of the villagers belongs to farming communities working on yearly payment from the landlords and some share crop production. Women are hardworking they do house chores and as well as rear livestock and assist in farming in the fields while young girls and boys attend primary and middle school in the village. River Murguzar flows along the side of Murguzar road and irrigates the land of Spal Banday and other villages located on its sides. There are small streams of Shera Tarab (springs) flowing from top of the mountains to irrigate agricultural lands of the village Sapal Banday. These streams are now damaged by flood and therefore there is no proper irrigation system for crop fields.

Hazard Assessment

Major occupation of Sapal Banday is farming however flood damaged their irrigation channels, cultivated land and disconnected the village from other areas because roads were blocked. The major flood came from River Murguzar and flood water also came from small irrigation channels from Shera Tarab Stream; water level ranged from 8 to 12 feet. During flood agricultural lands and two bridges were destroyed, lack of transport, electricity (more than one month there was no electricity) and telecommunication made these villagers most vulnerable. Villagers reported that during relief items distribution there was no verification system in practice which led to uneven distribution of relief items.

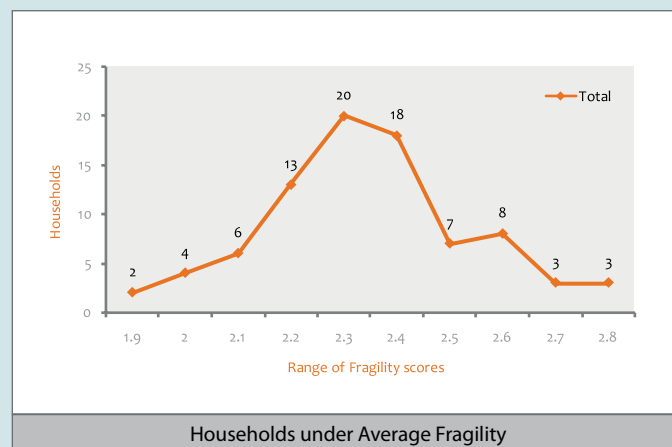
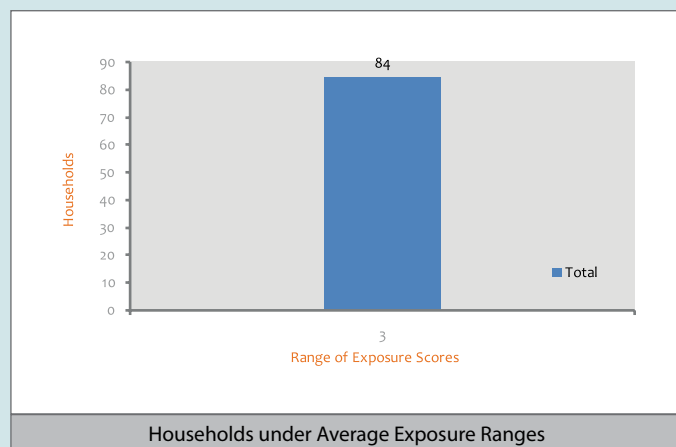


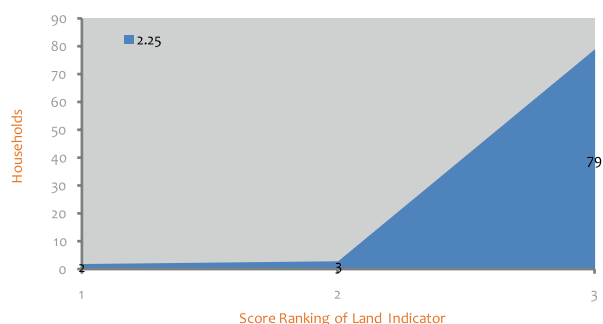
Both males and females groups expressed their willingness to actively participate in any disaster management activity if they get different trainings such as first aid, civil defense, construction of houses and concrete roads. Communities and fields near river and stream (chashma) are still in high risk for future disaster as they are now reconstructing their houses in the same location. Villagers said that cash for work and machinery work is being carried out by NGOs to remove debris from walking tracts and roads, however projects are not focusing on building concrete roads and paved walking tracts. Now when we are only cleaning roads and it is so that on any rainy day the road is again damaged with water and land sliding. Villagers' priorities for prevention and mitigation are according to their occupation (farming) i.e. rehabilitation of water channels, embankments and retaining walls on sides of road. They also suggested that plantation, embankment, plantation and environment friendly practices (proper disposal of wastes, hygiene practices) should be incorporated in the preparedness programs which will in larger contribute to climate change adaptation strategies. Villagers said, "We are united and we do help each other in need and we do have good social relationships with our village and neighboring villages. If we are organized in a committee, we will be proactively working if provided skills and tools which are helpful in time of disaster. We can work in the construction schemes which will ensure good material usage and quality work in the construction and community members will also get paid as the labors in the program which will give them employment opportunity".

Vulnerability Assessment

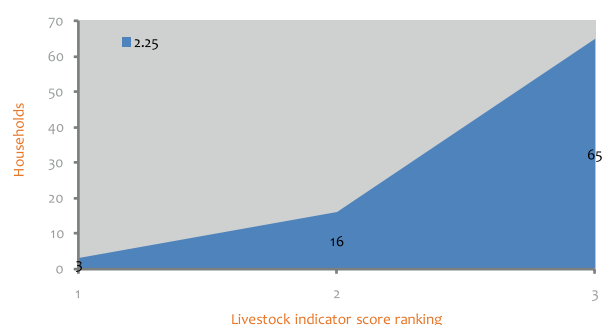
Summary of household vulnerability Analysis: In order set framework for household vulnerability assessment and derive coherent results; there sub indicators including **1) exposure 2) fragility** and **3) lack of resilience** were separately analyzed in view of both vulnerability ranking. However as hazard scores for the villages is cumulatively signified vulnerability ranking would also signify hazard level of 2.5 which is evident in the village.

Exposure Analysis: Exposure analysis signifies property that is exposed to risk, as in the overall results all 84 households stated their vulnerability is in rank of 3 which is the highest rating. Possibly it can be reflective of mud houses in the area and their perception regarding

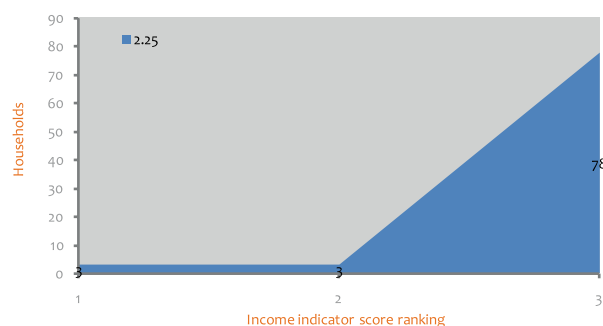




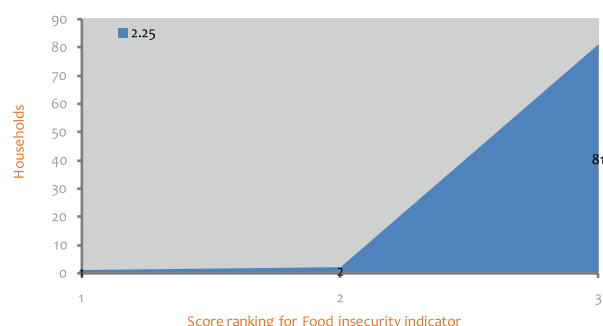
Land Holding Vulnerability



Livestock Vulnerability



Income Vulnerability



Food Insecurity

flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent hence their property would be under risk.

Fragility Analysis: Fragility covers in itself different levels of vulnerabilities; be it land holding, livestock, income, food insecurity, each in itself has a certain measurement because these factor underline communities' potential counter action against a disaster. According to fragility indicator, 66 out of 84 households surveyed, fall under range of 2.2-2.6 which is almost touching the higher end. Further analysis would show that under fragility sub indicators majority of basic amenities constituting land, livestock and income which signify monetary wellbeing, fall under rank of 3.

Indicators of Socio economic wellbeing of communities in context to Hazard

Land Holding: The village has hazard score of 2.25 and 79 out of 84 HHs fall under rank of 3 for land indicator which is 94% of total survey size in the village. As only risk prone households are chosen, this indicator would bound to have a higher proportion.

Livestock: The village has hazard score of 2.25 and 65 out of 84 HHs fall under rank of 3 for land indicator which is 77% of total survey size in the village. Please note that it is comparatively lower for land indicator hence it can be determined the people in the village do not consider livestock on same parity as land holding in context of drivers of vulnerability.

Income: The village has hazard score of 2.25 and 78 out of 84 HHs surveyed fall under rank of 3 for income indicator which is 92% of total survey size in the village.

Food Insecurity: The village has hazard score of 2.25 and 81 out of 84 HHs fall under rank of 3 for food insecurity indicator which is 96% of total survey size in the village.

Summary of socio economic indicators:

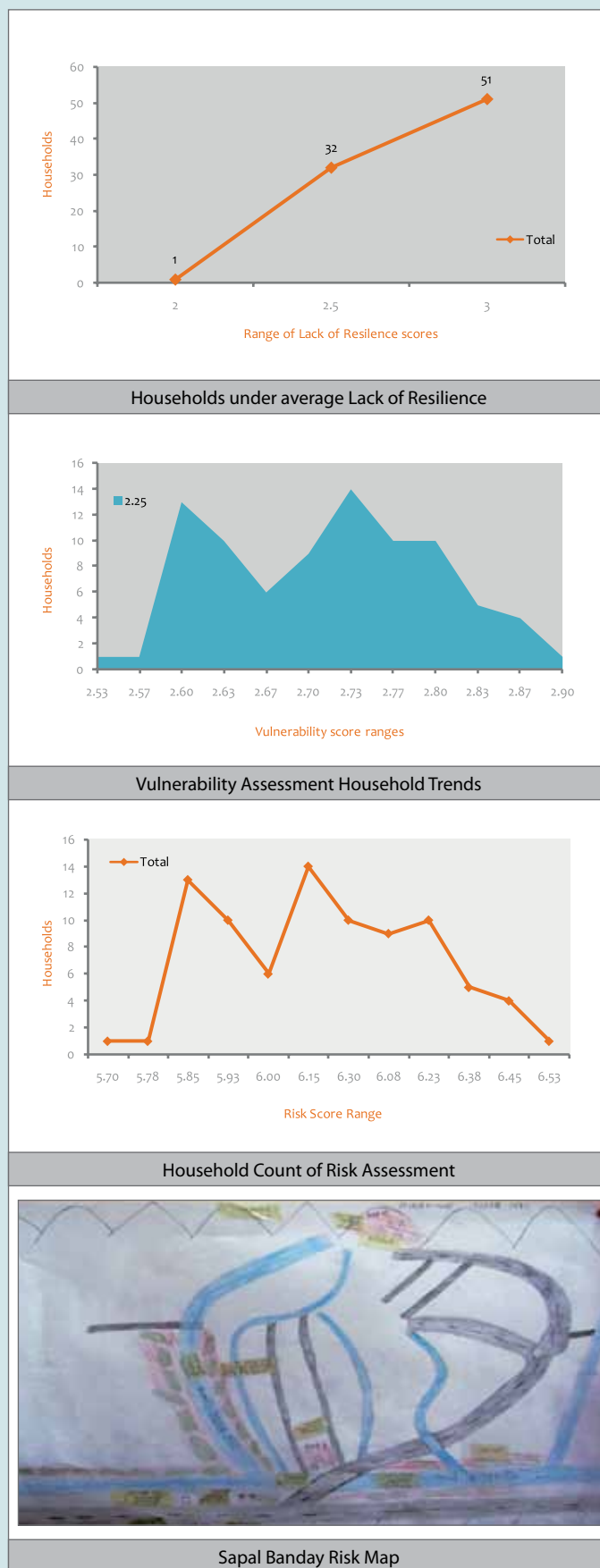
In the village there is strong correlation between income, and food insecurity indicators with all falling under same proportion under rank of 3. However livestock and land is lesser a driver of vulnerability in comparison to these three monetary sources of economic wellbeing in the village.

Lack of resilience

Lack of resilience indicator is measured through education and neighborhood patterns in a village. Under it 83 HHs (out of 84 surveyed) fall within scores of 2-2.5 which almost touches higher boundaries. Further analysis shows 98% of HHs have no member who goes to school so literacy trends to respond to hazard is low. Lack of neighborhood support despite being in lower proportion to education level has proportion of 61% households in rank of 3. Hence it can be concluded that education level is greater driver of vulnerability in the village.

Overall Vulnerability Score Range

The depiction as per the graph shows that irrespective of hazard scale vulnerability indicator is on the higher side. Out of 84 households surveyed 72 fall between ranges of 2.60 and 2.80.



Risk Assessment

In calculating risk assessment number of households in Spal Bandai Village, it was revealed that about 86% of households equating to 72 falls under risk score of 5.85-6.23 with greatest number of 14 households at score of 6.14 with mean average of 6.07.

Crops Harvest Timing

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
										Maize	
								Rice			
					Peach, Pears, Apple, Japanese Persimmon						
						Wheat					

Accessibility

Average altitude of the village: Low as compared to Upper Swat

Average time that villages are cut off during the winter time: Due to heavy rain falls in winter (December and January) and summers (July and August) cause land sliding and roads are blocked

Availability of public and other transport: local transport is available

Demographic Profile

Name of Village	No. of Total HH	No. of widows & Single Women*	No. of Child-headed HHs*	No. of elderly as head of HHs*	No. of disabled as head of HHs*	Priority Needs	Village Infrastructure
Sapal Banday	514	100-120	100	100	10-15	DMCs, Plantation integrated with school safety programs, safe asylum, check and deflection dams, housing safety measures (using plastic sheets and tins as tops), candle making, food preservation techniques, energy efficient stoves, retaining wall of 300 meters.	5 Mosques 4 educational institutions i.e. 2 Government Girls schools (Primary and Middle) and 2 Government Boys school (Primary and Middle). All the village houses do have electricity facility and do have Telecommunication facility. Villagers need a hospital or trained medical practitioner to deal with emergencies and provide first aid and midwives to deal with pregnant women. NRSP, Lasoona, Red cross and Spado are doing rehabilitation work i.e. Cash For Work Schemes and Livelihood Programs.

**Estimated figures based on local community knowledge*

Annex 3: Village Report - Gul Banday

Tehsil: Babuzai	Union Council: Islampur	HH Size: 5.3	Hazard Score: 2.25
Hazards in the village: Seasonal flooding, major flooding & soil erosion			
Surveyed household prone to hazards: 31			
NGOs/ Agencies working in the village			
1	NRSP: Livelihoods		
2	Lasoonia: Wash		

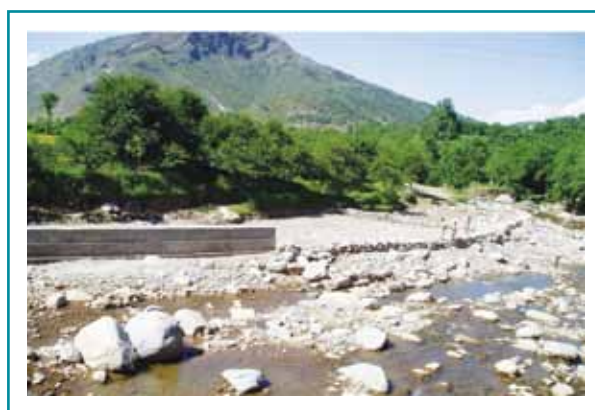
Profile of the Village

Gul Banday Banday is known as Topae Gatt (Stone at top) because of the stone at top of the village. From Mingora, Swat by going through Nishat Chowk, Mingora to Murguzar Road, the distance to the village is 17km (30 minutes) It has a population of 3,288 and 617 Households. The Murguzar River flows along the Murguzar Road irrigating the land of Gul Banday. Tangae Streams (now damaged due to flooding) coming from mountains and irrigate the village land.



Hazard Assessment

Village Gul Banday is prone to flood and land sliding, during the 2010 flood level was about 8 ft. high and land sliding disconnected village from other villages as the main bridge was damaged. Fields and houses near river stream were also destroyed. Irrigation channels coming from Tangae Streams are damaged and still not rehabilitated. There is need to cement plaster the front of houses adjoining the land.



Both men and women of Gul Banday are highly motivated to form village committees and work on water management and receive trainings on first aid and civil defense. They said “we do not have hospital and lack of medicines and first aid facility is the top most problem for our community, moreover during floods and land sliding our area is cut off from other villages

and we face hardship in getting first aid. If our community is trained in such trainings they will be proactively working on providing first aid in normal routine and as well as in disaster situation. Secondly, embankments and retaining walls near river will help in protection of our cultivated land and crops from damages and lastly we need clean drinking water tools and manual procedures.”

Vulnerability Assessment

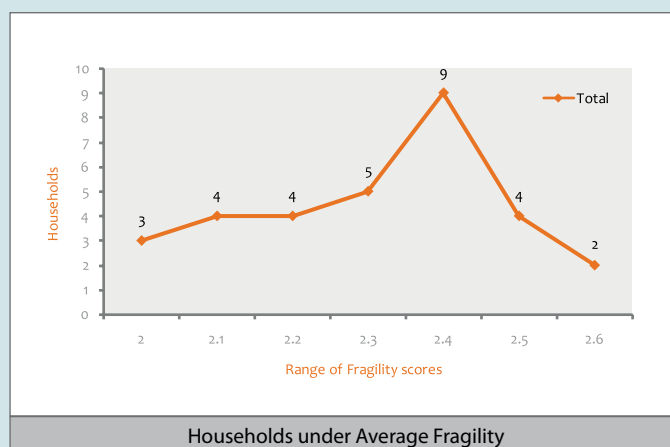
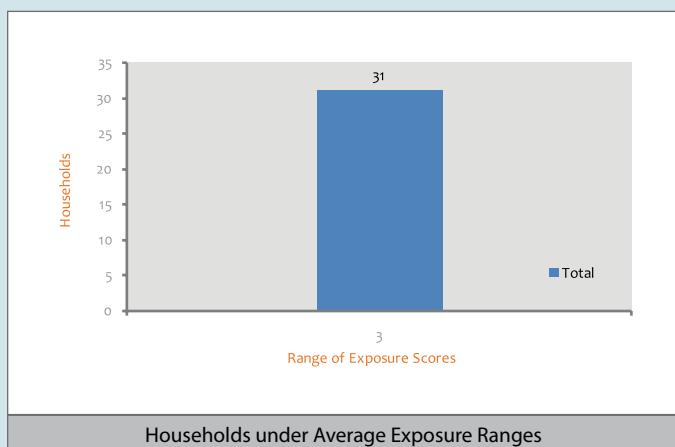
Summary of household vulnerability

Analysis: In order set framework for household vulnerability assessment and derive coherent results; there sub indicators including **1) exposure 2) fragility** and **3) lack of resilience** were separately analyzed in view of both vulnerability ranking. However as hazard scores for the villages is cumulatively signified vulnerability ranking would also signify hazard level of 2.0 which is evident in the village.



Exposure Analysis: Exposure analysis signifies property that is exposed to risk, as in the overall results all 31 households stated their vulnerability is in rank of 3 which is the highest rating. Possibly it can be reflective of mud houses in the area and their perception regarding flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent.

Fragility Analysis: Fragility covers in itself different levels of vulnerabilities; be it land holding, livestock, income, food insecurity. Each in itself has a certain measurement because these factor underline communities potential counter action against a disaster. According to fragility indicator, 18 out of 31 households surveyed fall under range of 2.2-2.4 which



is almost touching the higher end. Further analysis would show that under fragility sub indicators majority of basic amenities constituting land, livestock and income which signify monetary wellbeing fall under rank of 3.

Indicative of Socio economic wellbeing of communities in context to Hazard

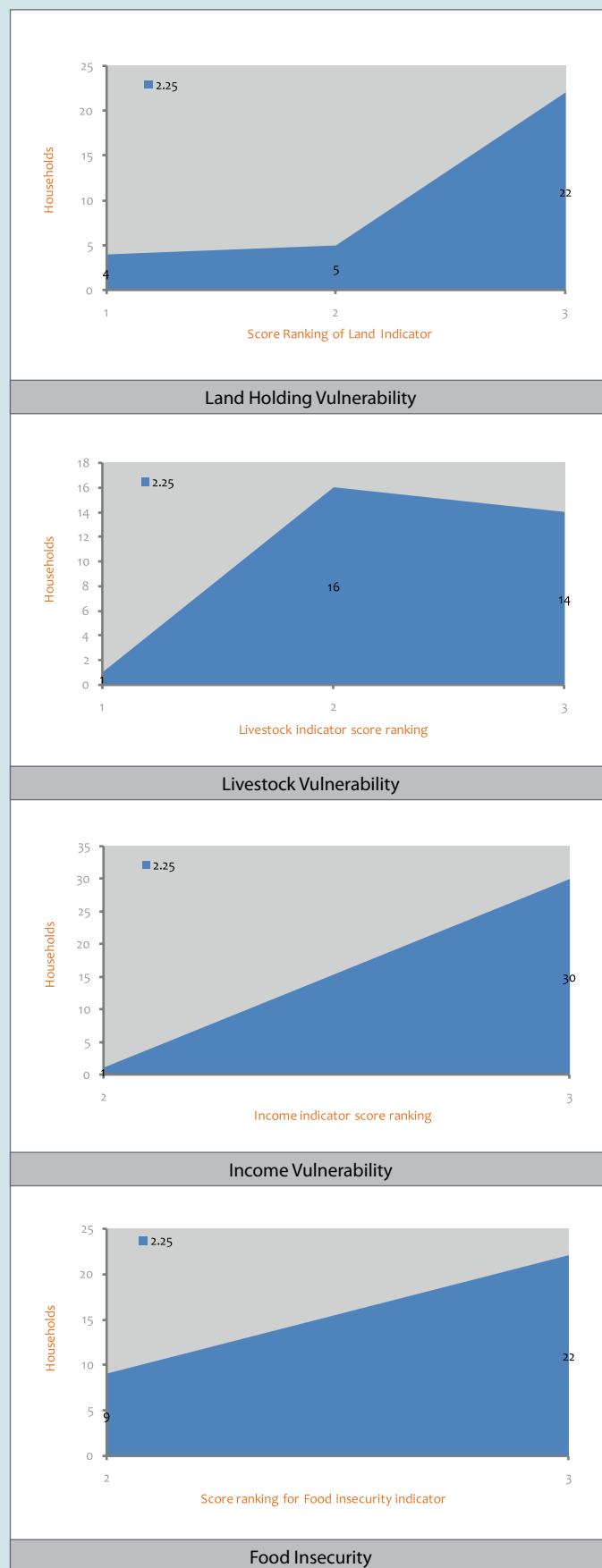
Land Holding: The village has hazard score of 2.25 and 22 out of 31 HHs fall under rank of 3 for land indicator which is 71% of total survey size in the village.

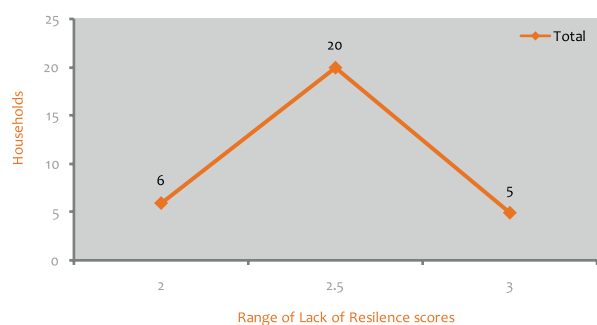
Livestock: The village has hazard score of 2.25 and 16 out of 31 HHs fall under rank of 2 for livestock indicator which is 52% of total survey size in the village. Please note that livestock vulnerability seems a lower driver of vulnerability in the village.

Income: The village has hazard score of 2.25 and 30 out of 31 HHs surveyed fall under rank of 3 for income indicator which shows that almost all of the HHs surveyed have meager source of income hence it seems a very strong driver of vulnerability in the village.

Food Insecurity: The village has hazard score of 2.25 and 22 out of 31 HHs fall under rank of 3 for food insecurity indicator which is 71% of total survey size in the village.

Summary of socio economic indicators: In the village there is strong correlation between land, income and food insecurity indicators with all





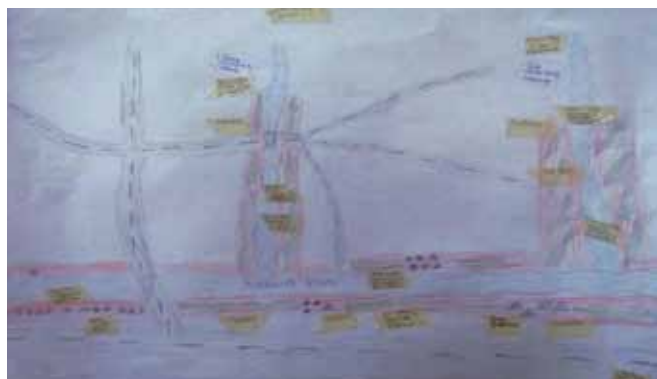
Households under average Lack of Resilience



Vulnerability Assessment Household Trends



Household Count of Risk Assessment



Gul Bandy Risk Map

falling under same proportion under rank of 3. However, livestock is lesser a driver of vulnerability in comparison to these three monetary sources of economic wellbeing in the village.

Lack of resilience

Lack of resilience indicator is measured through education and neighborhood patterns in a village, under it 20 out of 31 houses surveyed fall within scores of 2.5 which almost touches higher boundaries. Further analysis shows 90% HHs have no member who goes to school so literacy trends to respond to hazard is low; lack of neighborhood support in the village is a lower driver of vulnerability with 21 households falling in rank of 2.

Overall Vulnerability Score Range

The depiction as per the graph shows that majority of household surveyed fall between ranges of 2.60 and are more vulnerable in context of a disaster.

Risk Assessment

In calculating risk assessment in Gul Bandy Village it was revealed that about 80% of households equating to 25 falls under risk score of 5.5-6 and above with mean average of 5.8.

Crops Harvest Timing

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
										Maize	
								Rice			
					Japanese Persimmon						
						Wheat					

Accessibility

Average altitude of the village: Low as compared to Upper Swat

Average time that villages are cut off during the winter time: Due to heavy rain falls in winter (December and January) and summers (July and August) cause land sliding and roads are blocked

Availability of public and other transport: local transport not available inside village however can be accessed from Murguzar Road

Demographic Profile

Name of Village	No. of Total HH	Priority Needs	Village Infrastructure
Gul Banday	514	Check and deflection dams, DMCs, Safe asylum, school safety programs, using rocks and shrubs to mitigate risks	1 Mosque 2 educational institutions i.e. 1 Government Girls schools (Primary) and 1 Government Boys school (Primary). All the village houses do have electricity and telecommunication facility. There is one Biogas Plant made by community but is not functional. (See 3.1 picture below) Community has Good practices (See 3.2 picture Below) for water storage, they store water coming from stream and use for washing hands and clothes). Villagers need a hospital or trained medical practitioner to deal with emergencies and provide first aid and midwives to deal with pregnant women. NRSP and Oxfam are doing rehabilitation work i.e. WASH and Livelihood Programs.

**Estimated figures based on local community knowledge*

Water Storage

In this picture, school girls washing their hands from water stored in the water channel, water is coming from mountains and is being stored in this concrete water channel which has two open able blockages at both ends. This is a good practice for water storage in village Gul Banday and can be replicated in other villages that have small streams coming from mountains. Villagers said that we use this water for washing our body and clothes.



Annex 4: Village Report - Kokrai

Tehsil: Babuzai	Union Council: Islampur	HH Size: 7	Hazard Score: 2.25
Hazards in the village: Seasonal flooding, major flooding & soil erosion			
Surveyed household prone to hazards: 41			
NGOs/ Agencies working in the village			
1	NRSP: Livelihood		
2	Spado: Child Protection Centers		
3	Lasoon: Livelihood		
4	Red Cross: Livelihoods		
5	ACTED: Wash and CFW (rehabilitation of water courses)		

Profile of the Village

Kokrai is a hilly village located 8 to 10 km away from Mingora (Swat). It takes about 10 minutes drive from main Mingora and on left and right side of Murguzar road you will find Village Kokrai with a population of 3,116 and 424 HHs.

Major sources for income are farming, labor and remittance while women make handicrafts, rear livestock and stitch dresses. Basically there are four prominent communities in the villages out of which

Mia Gaans (20 HHs) hold the first rank being the most well to do, 100 households are steel workers, 100 households are Masons, 60 households are Gujar rearing and milking livestock while rest are labors working on daily wages. River Murguzar flows side by side on Murguzar Road and irrigates land of Kokrai, a stream (Charoona Kanda/Dara Kokrai) flows from hills and irrigates land across the village.



Hazard Assessment (these insights derived through FGDs transect walk and mapping)

Villagers informed that during flood there was about 8 to 9 feet water level in the village. The water from stream damaged girls primary school and Village Bridge was also damaged. Murguzar road was blocked due to land sliding. Village has its hospital and people received medicines and treatment on time. There was no loss of lives, only partial damages occurred to bridge, the houses and one school building. Their top most need is clean drinking water and agricultural land restoration by building safety walls. Moreover, roof tops of the houses are also not safe as the roof allows the water entry and can be easily damaged as construction design is not durable. Villagers indicated that safe zones on left side of the Murguzar road exist and people can be gathered here if informed through early warning system in the village.

They said that they need clean drinking water and water storage facility. They said we can form our committees at school and community level to cope with disaster and the committees now formed will be involved in the livelihood rehabilitation projects. Female group showed resilience to participate in the water management committees. However they can contribute in livelihood rehabilitation for their families at household level by enterprise development programs, livestock rearing and poultry keeping.

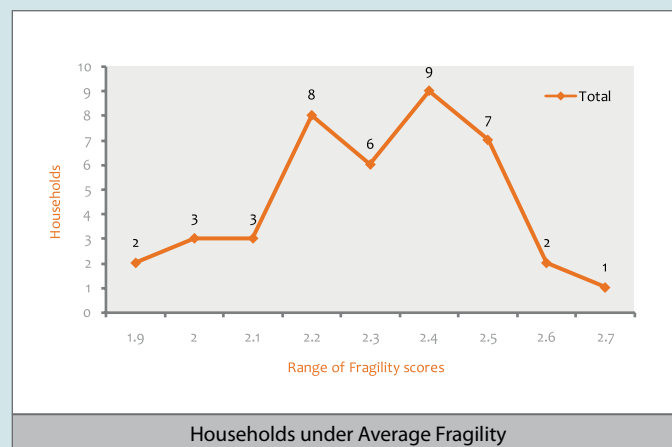
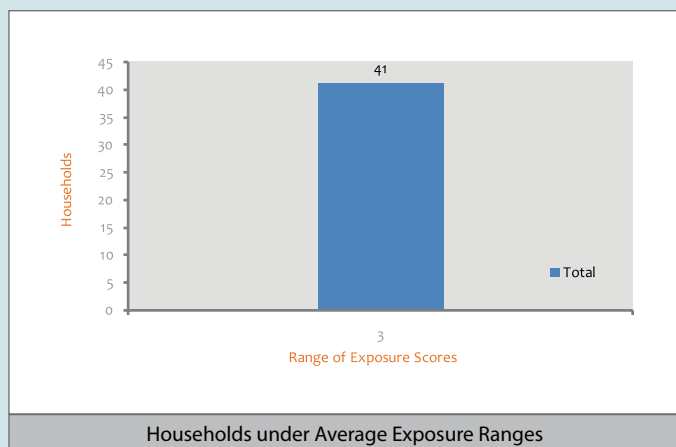


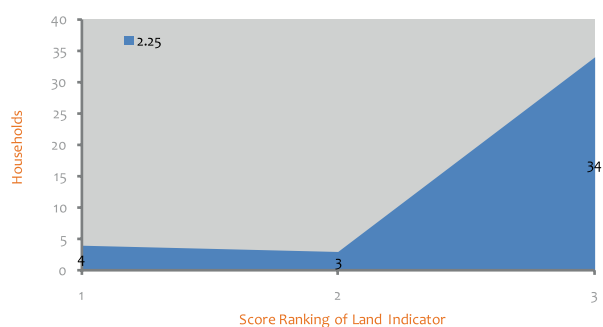
Vulnerability Assessment

Summary of household vulnerability Analysis: In order to set framework for household vulnerability assessment and derive coherent results; there sub indicators including **1) exposure 2) fragility** and **3) lack of resilience** were separately analyzed in view of both vulnerability ranking; however as hazard scores for the villages is cumulatively signified vulnerability ranking would also signify hazard level of 2.0 which is evident in the village.

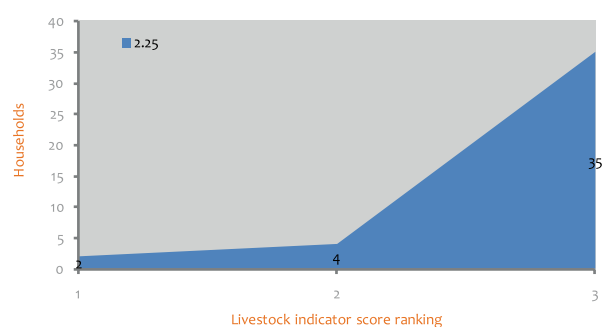
Exposure Analysis: Exposure analysis signifies property that is exposed to risk, as the overall results shows that all 41 households stated their vulnerability is in rank of 3 which is the highest rating. Possibly it can be reflective of mud houses in the area and their perception regarding flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent hence their property would be under risk.

Fragility Analysis: Fragility covers in itself different levels of vulnerabilities; be it land holding, livestock, income, food insecurity, each in itself has a certain measurement because these factor underline communities potential counter action against a disaster. According

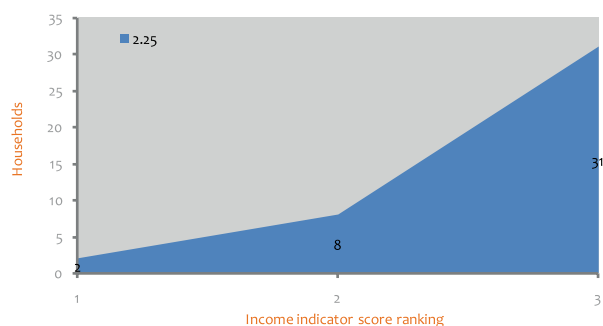




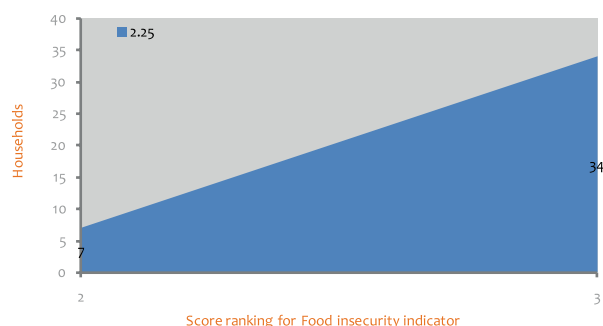
Land Holding Vulnerability



Livestock Vulnerability



Income Vulnerability



Food Insecurity

to fragility indicator, 23 out of 41 households surveyed fall under range of 2.2-2.4 which is almost touching the higher end. Further analysis would show that under fragility sub indicators majority of basic amenities constituting land, livestock and income which signify monetary wellbeing, fall under rank of 3.

Indicative of Socio economic wellbeing of communities in context to Hazard

Land Holding: The village has hazard score of 2.25 and 34 out of 41 HHs fall under rank of 3 for land indicator which is 83% of total survey size in the village. As only risk prone households are chosen, this indicator would bound to have a higher proportion.

Livestock: The village has hazard score of 2.25 and 35 out of 41 HHs fall under rank of 3 for livestock indicator which is 85% of total survey size in the village. Please note that livestock vulnerability in this village seems a strong driver of vulnerability of households.

Income: The village has hazard score of 2.25 and 31 out of 41 HHs surveyed fall under rank of 3 for income indicator which shows that majority of the houses have meager source of income hence it seems a very strong driver of vulnerability in the village.

Food Insecurity: The village has hazard score of 2.25 and 34 out of 41 HHs fall under rank of 3 for food insecurity indicator which is 83% of total survey size in the village.

Summary of socio economic indicators:

In the village there is strong correlation between all four indicators with all falling under same proportion under rank of 3.

Lack of resilience

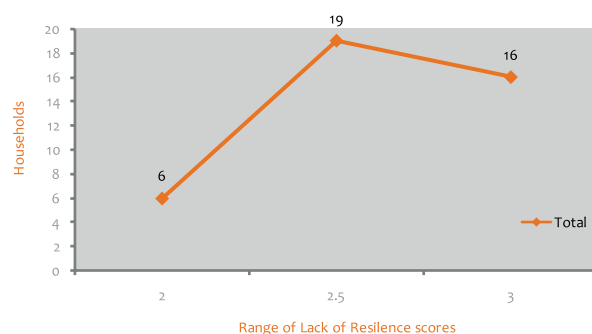
Lack of resilience indicator is measured through education and neighborhood patterns in a village, under it 35 out of 41 HHs surveyed fall within scores of 2.5-3 which almost touches higher boundaries. For further analysis 83% of HHs have no member who goes to school so literacy trends to respond to hazard is low; lack of neighborhood support in the village is a lower driver of vulnerability in comparison with 21 households falling in rank of 3. Hence it can be conducted that education level is greater driver of vulnerability in the village

Overall Vulnerability Score Range

The depiction as per the graph shows that majority of households surveyed fall between ranges of 2.50-2.67 and above, equating to 24, so it can be determined that these households are more vulnerable in context of a disaster.

Risk Assessment

In calculating risk assessment in Kokrai Village it was revealed that about 68% of households equating to 28 falls under Risk score of 5.63-6 with mean average of 5.93.



Households under average Lack of Resilience



Vulnerability Assessment Household Trends



Household Count of Risk Assessment



Kokrai Risk Map

Crops Harvest Timing

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
										Maize	
						Wheat					

Accessibility

Average altitude of the village: Low as compared to Upper Swat

Average time that villages are cut off during the winter time: Due to heavy rain falls in winter (December and January) and summers (July and August) cause land sliding and Chitor road (Road turning from Murguzar Road to Kokrai) is blocked

Availability of public and other transport: No public transport is available in the village and villagers have to walk to Murguzar Road to get the public transport.

Demographic Profile

Name of Village	No. of Total HH	No. of widows & Single Women*	No. of Child-headed HHs*	No. of elderly as head of HHs*	No. of disabled as head of HHs*	Priority Needs	Village Infrastructure
Kokrai	424	25**	20*	150*	1*	School safety program integrated with Plantation , sanitation ,water harvesting, energy efficient stoves, food preservation, mushrooms growth (wheat straw can be used to cultivate mushroom) safe asylum, strengthening pillars of connecting bridges	4 Mosques, 2 educational institutions, 1 BHU, 1 Play ground, 1 Agriculture office and 1 Police Chowky. All the village houses do have electricity and telecommunication facility. NRSP, Lasoona, Red cross, ACTED and Spado are doing rehabilitation work i.e. Cash For Work Schemes and Livelihood Programs. Villagers need clean drinking water and land restoration projects

** Community Estimates

* Total disables in Kokrai are 45 while 1 is the head of the household

Annex 5: Village Report - Saidu Shagee

Tehsil: Babuzai	Union Council: Saidu Sharif	HH Size: 8-10	Hazard Score: 2.25
Hazards in the village: Seasonal flooding, major flooding & soil erosion			
Surveyed household prone to hazards: 99			
NGOs/ Agencies working in the village			
None			

Profile of the Village

Saidu Shagee is a hilly and plain mixed area with rural-urban population and about 2-3 km away from Mingora with 700 HHs.

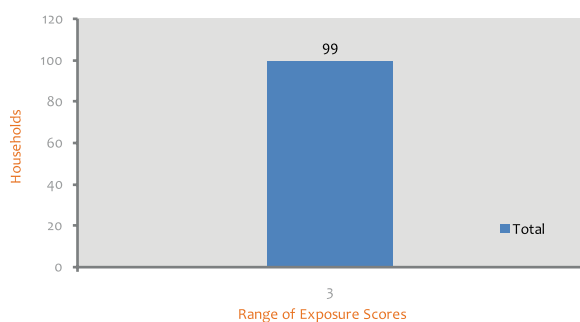


Hazard Assessment

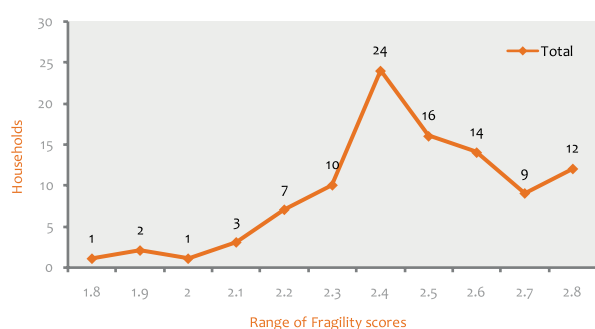
In 2005 earthquake hit the area and there were minor damages to (kacha) mud houses and shops. However, during present flood, damages were on bigger scale than the earthquake. Crop fields were badly damaged and have now turned into stony area and needs land restoration, 6 houses were also damaged. Sanitation being the major problem of the area became worst condition during flood as water entered the houses and made conditions unhygienic for the community and numerous diseases spread in the area with diarrhea, dehydration (non availability of clean drinking water) and cholera being the top ones and others were allergies, skin rashes and general illness. Villagers said that due to development of our village we have been neglected in relief projects and no rehabilitation program is being carried out in our area. The top most priority is the need for culverts, proper sanitation systems and water supply schemes. Villagers suggested that seed banks be established at village level so that farmers can withstand the disasters i.e. if crops are damaged they can replace with new seeds stored in the seed bank. Further more land restoration of agricultural land is needed through embankments on both sides of the river and cleaning of land. Cash grants for skilled people will help them reestablish their businesses. Community has established one committee at village level and want to work on disaster management. Progressive goals of the committee are establishment of school, BHU, Agricultural improvement methods and Pest Control Medicines and Techniques.

Vulnerability Assessment

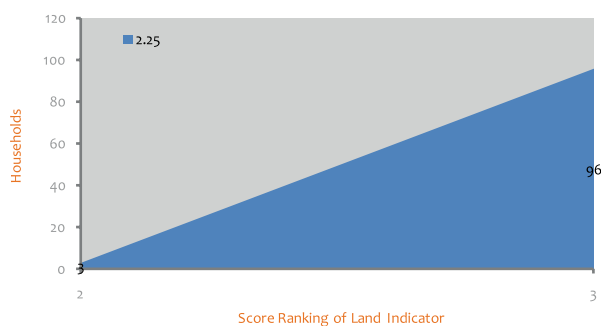
Summary of household vulnerability Analysis: For household vulnerability assessment; there sub indicators including **1) exposure** **2) fragility** and **3) lack of resilience** were separately analyzed in view of both vulnerability ranking. However as hazard scores for the



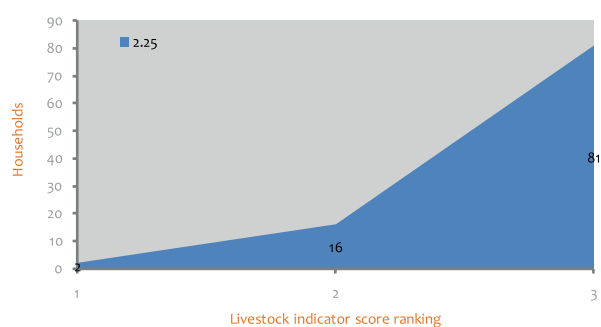
Households under Average Exposure Ranges



Households under Average Fragility



Land Holding Vulnerability



Livestock Vulnerability

villages is cumulatively signified, vulnerability ranking would also signify hazard level of 2.5 which is evident in the village.

Exposure Analysis: Exposure analysis signifies property that is exposed to risk. In the overall results, all 99 households stated their vulnerability in rank of 3 which is the highest rating. Possibly it can be reflective of mud houses in the area and their perception regarding flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent hence their property would be under risk.

Fragility Analysis: Fragility covers in itself different levels of vulnerabilities; be it land holding, livestock, income, food insecurity, each in itself has a certain measurement because these factor underline communities' potential counter action against a disaster. According to fragility indicator, 57 out of 99, households fall under range of 2.2-2.5 which is almost touching the higher end. Further analysis would show that under fragility sub indicators majority of basic amenities constituting land, livestock and income which signify monetary wellbeing fall under rank of 3.

Indicative of Socio economic wellbeing of communities in context to Hazard

Land Holding: The village has hazard score of 2.25 and 96 out of 99 Hhs fall under rank of 3 for land indicator which is 97% of total survey size in the village. As only risk prone households are chosen this indicator would bound to have a higher proportion considering already vulnerable land holding patterns in the village.

Livestock: The village has hazard score of 2.25 and 81 out of 99 HHs fall under rank of 3 for livestock indicator which is 82% of total survey size in the village. Please note that livestock vulnerability in this village seems a strong driver of vulnerability of households.

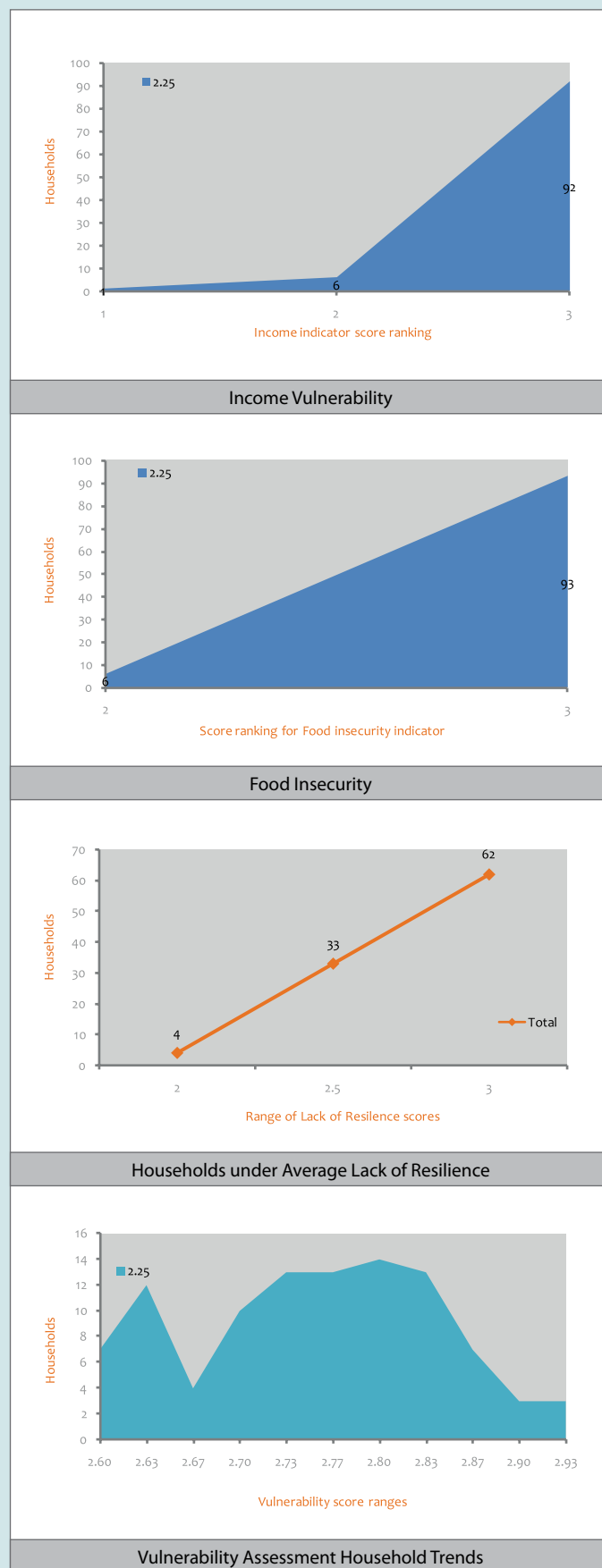
Income: The village has hazard score of 2.25 and 92 out of 99 HHs surveyed fall under rank of 3 for income indicator which shows that majority of the houses have meager source of income hence it seems a very strong driver of vulnerability in the village.

Food Insecurity: The village has hazard score of 2.25 and 93 out of 99 HHs fall under rank of 3 for food insecurity indicator.

Summary of socio economic indicators: In the village there is a strong correlation between all four indicators with all falling under same proportion under the rank of 3. So with these four vulnerability drivers integrated in close correlation, the vulnerability of houses at risk in the village is more so accentuated.

Lack of resilience

Lack of resilience indicator is measured through education and neighborhood patterns in a village. Under it 62 out of 99 surveyed houses fall within scores of 2.5-3 which almost touches higher boundaries. 99.9% of houses have no member who goes to school so literacy trends to respond to hazard is significantly low. Lack of neighborhood support in the village is a lower driver of vulnerability as 62 households falling in rank of 3.



Overall Vulnerability Score Range

The depiction as per the graph shows that majority of households (63) fall between ranges of 2.7-2.83

Risk Assessment

In calculating risk assessment in Shagai Village it was revealed that about 67% households fall under risk score of 5.85-6.08 with mean average of 6.9.

Accessibility

Average altitude of the village: Low as compared to Upper Swat

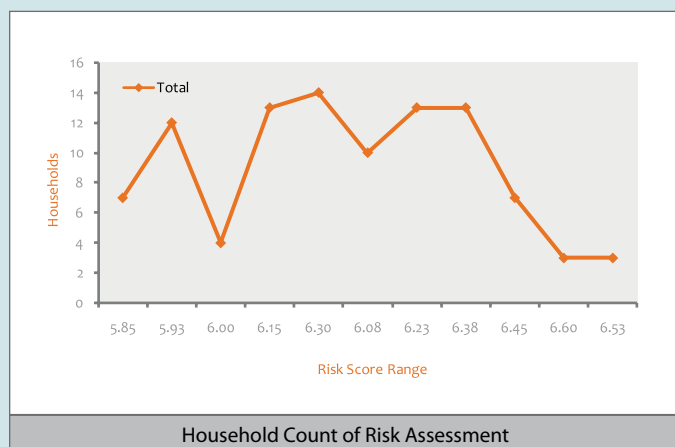
Average time that villages are cut off during the winter time: Due to cloud burst and heavy rain falls in winter (December and January) and summers (July and August) cause land sliding and Chitor road (Road turning from Murguzar Road to Kokrai) is blocked

Availability of public and other transport: No public transport is available in the village and villagers have to walk to Murguzar Road to get the public transport.

Demographic Profile

Name of Village	No. of Total HH	No. of widows & Single Women*	No. of Child-headed HHs*	No. of elderly as head of HHs*	No. of disabled as head of HHs*	Priority Needs	Village Infrastructure
Shagae	700	20	10	5	2	School Safety Programs (2), DMC, Retaining Wall (300 Feet and 500 Feet), Plantation, Sanitation (widening of narrow water channels), Safe Asylum,	Schools (4), Mosques (7), Murguzar Road cuts the village into two parts (one on left side and other on right side). Communities on right side are more rural and flood has damaged this side of the village while only one part of left side is affected during seasonal flooding due to sanitation problems.

**Estimated figures based on local community knowledge*



Annex 6: Village Report - Saidu

Tehsil: Babuzai	Union Council: Saidu Sharif	HH Size: 8-10	Hazard Score: 1.5
Hazards in the village: Seasonal flooding, major flooding & soil erosion			
Surveyed household prone to hazards: 99			
NGOs/ Agencies working in the village			
None			

Profile of the Village

Saidu is a village of Saidu Sharif Union Council located at about 2 km from main Mingora on Saidu Road. It is divided into two parts. On one side of the road is the rural community and on the other side is the urban community. Rural part of Saidu is agricultural and is irrigated by Murguzar River coming from Islampur Union Council and small water channels coming from mountains while the urban part is residential with houses, offices, mosques, small markets and educational institutions.

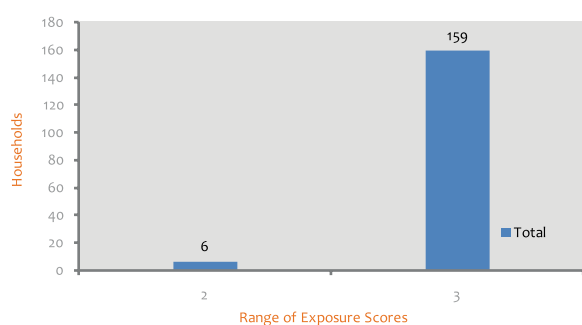


Hazard Assessment

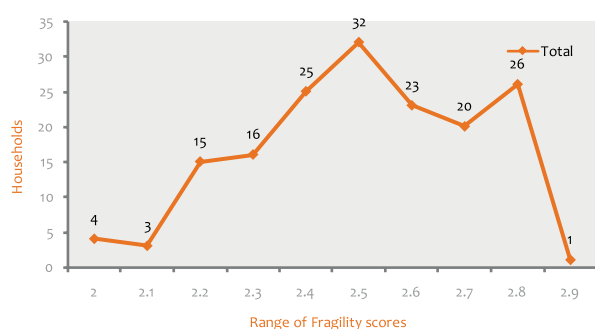
In 2005, area was hit by earth quake and still jerks of quake are felt in the area but the losses were minor being the mud houses affected, cracks and hollowness in house walls. Present floods affected the rural part of Saidu highly affected area being the ones near the river and streams not only damaging crops but also the houses near them. Massive destruction has been done to standing crops in fields, land siltation, water channels, bridges and houses. Villagers said that we are in dire need for restoration of agricultural land, reconstruction of concrete irrigation channels & bridges, plantation, early warning systems, improved/ advanced techniques for agricultural production and reconstruction material for damaged houses. Villagers on road side of Saidu told that they now that urban Saidu is safe zone therefore they can migrate to that area if flood comes again and the villagers on the other side of the river told that they will come to low risk area if alarmed before the disaster. They told that our area is on high risk with disasters i.e. floods and earth quake and therefore we need civil defense trainings and children should be educated in schools for response/ coping strategies for disasters.

Vulnerability Assessment

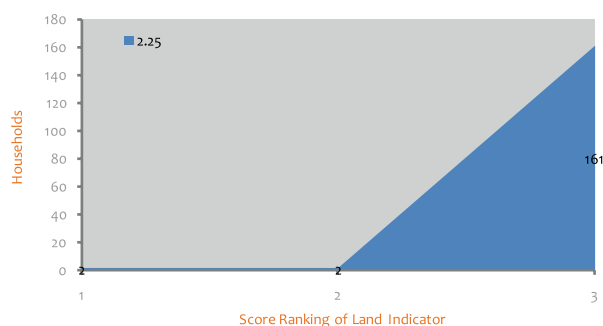
Summary of household vulnerability Analysis: In order to set framework for household



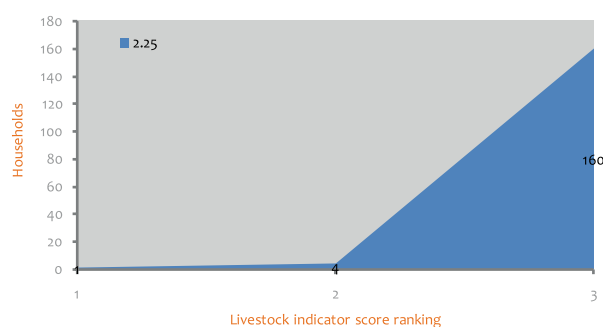
Households under Average Exposure Ranges



Households under Average Fragility



Land Holding Vulnerability



Livestock Vulnerability

vulnerability assessment and derive coherent results; there sub indicators including **1) exposure 2) fragility and 3) lack of resilience** were separately analyzed in view of both vulnerability ranking; however as hazard scores for the villages is cumulatively signified vulnerability ranking would also signify hazard level of 1.5 which is evident in the village.

Exposure Analysis: Exposure analysis signifies property that is exposed to risk. In the overall results all 159 out of 165 HHs stated their vulnerability is in rank of 3 which is the highest rating. It can be reflective of mud houses in the area and villagers perception regarding flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent hence their property would be under continuous risk.

Fragility Analysis: Fragility covers in itself different levels of vulnerabilities; be it land holding, livestock, income, food insecurity, each in itself has a certain measurement because these factor underline communities' potential counter action against a disaster. According to fragility indicator, 126 out of 165 households' surveyed fall under range of 2.4-2.8 which is almost touching the higher end further analysis would show that under fragility sub indicators majority of basic amenities constituting land, livestock and income which signify monetary wellbeing fall under rank of 3.

Indicators of Socio economic wellbeing of communities in context to Hazard

Land Holding: The village has hazard score of 2.25 and 161 out of 165 HHs

fall under rank of 3 for land indicator. As only risk prone households are chosen this indicator would bound to have a higher proportion considering already vulnerable land holding patterns in the village.

Livestock: The village has hazard score of 2.25 and 160 out of 165 HHs fall under rank of 3 for livestock indicator.

Income: The village has hazard score of 2.25 and 148 out of 165 HHs surveyed fall under rank of 3 for income indicator which shows that majority of the HHs have meager source of income hence it seems a very strong driver of vulnerability in the village.

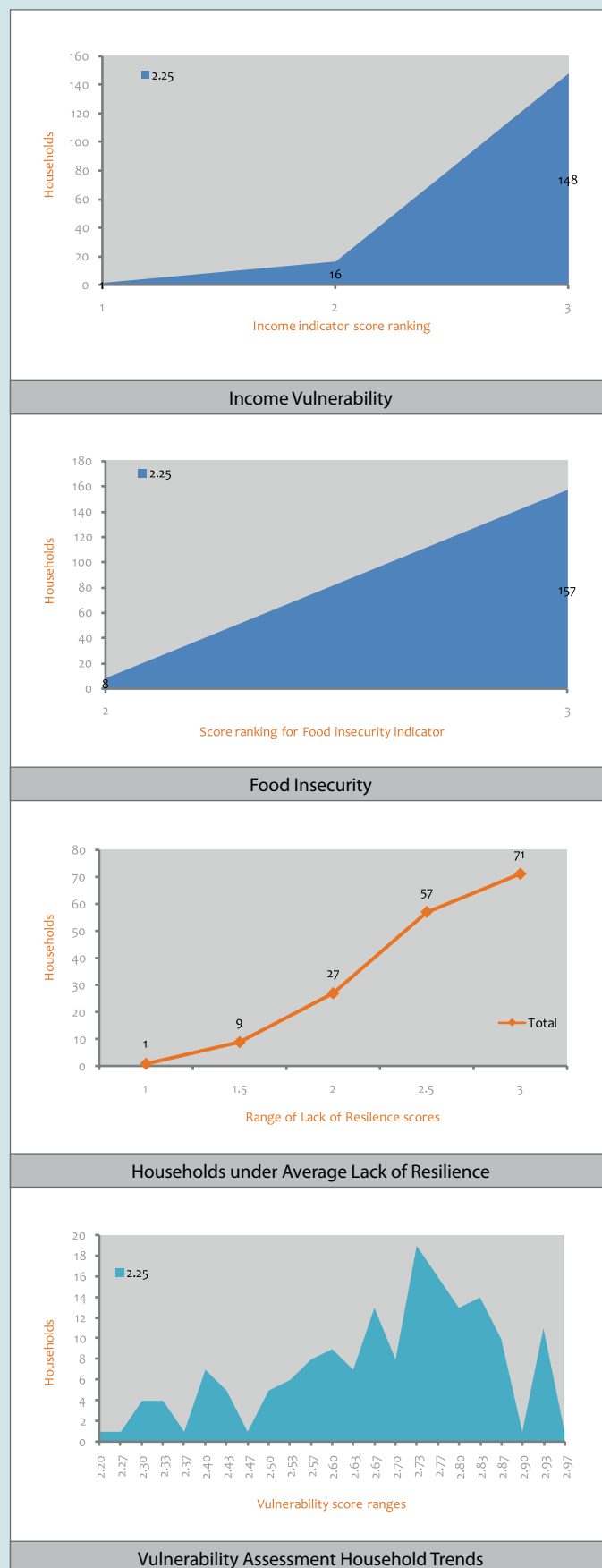
Food Insecurity: The village has hazard score of 2.25 and 157 out of 165 HHs fall under rank of 3 for food insecurity indicator.

Summary of socio economic indicators:

In the village there is strong correlation between all four indicators with all falling under same proportion under rank of 3. So with these four vulnerability drivers integrated in close correlation vulnerability of houses at risk in the village is more so accentuated.

Lack of resilience

Lack of resilience indicator is measured through education and neighborhood patterns in a village. Under it 128 out of 165 houses surveyed fall within scores of 2.5-3 which almost touches higher boundaries. Further analysis shows 81% of houses have no member who goes to school so literacy trends to respond to hazard is significantly low; lack of neighborhood support in the village is a lower driver of vulnerability in



comparison with 83 households falling in rank of 3. Hence it can be conducted that education level is greater driver of vulnerability in the village.

Overall Vulnerability Score Range

The depiction as per the graph shows that majority of households (93) surveyed fall between ranges of 2.67-2.87 and above.

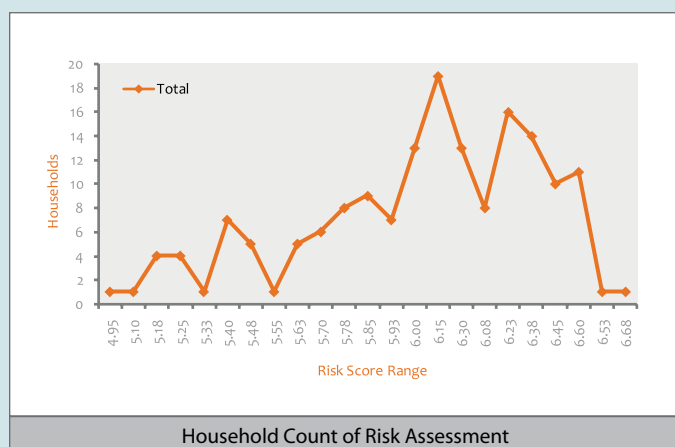
Risk Assessment

In calculating risk assessment number of households in Saidu Village, it was revealed that about 79% households falls under risk score of 5.4-6.38 and above with mean average of 6.15.

Demographic Profile

Name of Village	No. of Total HH	No. of widows & Single Women*	No. of Child-headed HHs*	No. of elderly as head of HHs*	No. of disabled as head of HHs*	Priority Needs	Village Infrastructure
Saidu	950	9	3	33	2	Plantation, sanitation to widen narrow water channels, Disaster management to be created at quarter level, committee, School safety, food kitchen gardening seed banks, retaining wall of 200 feet	1 hospital, 4 schools, 2 mosques, 1 commissioner office, shops, agriculture fields amid plain land areas

**Estimated figures based on local community knowledge*



Annex 7: Village Report - Ghari

Tehsil: Behrain	Union Council: Madyan	HH Size: 8.8	Hazard Score: 2.5
Hazards in the village: Seasonal flooding & soil erosion			
Surveyed household prone to hazards: 78			
NGOs/ Agencies working in the village			
1	NRSP: Cash for work on irrigation channels		
2	Hujra: Cash for work		

Profile of the Village

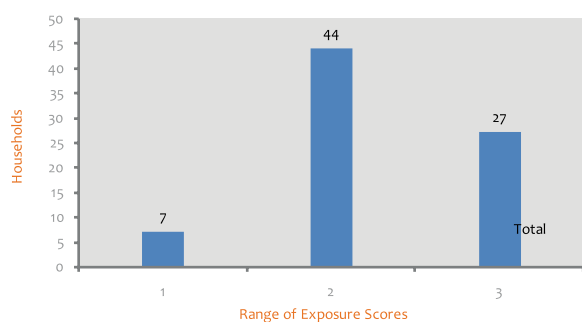
Ghari Village is adjacent to Madyan (60 km away from Mingora, Swat) with urban and rural mixed area and has 800 households. People of Ghari are hard working and are proactively involved in doing their businesses i.e. farming, small businesses, labors, doctors, tailors, engineers, plumber, mason, carpenter, photographer and hair dresser. Most of the population is poor and have good social relationships. Different communities exist in Ghari which can be segregated on the basis of localities: Jopin, Ingrabad, Proper Ghari Kalay. Electricity and mobile phones are available but there is no gas. People collect water from streams (Chashma).

Hazard Assessment

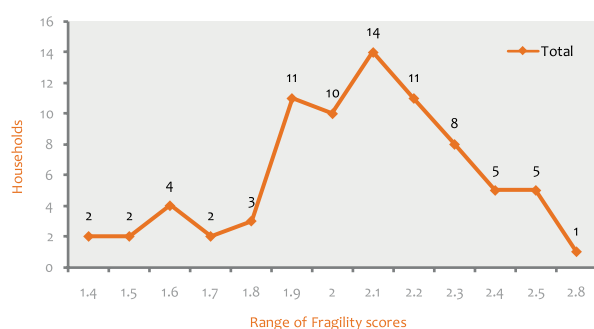
During flood 2010, water level was 15-20 feet high and the village was cut off as Madyan road was completely washed away and transportation by road was not possible. Water level in Cheel stream was high and it made damages on high scales. Snow melted from snowy hills and caused flood in Cheel streams (UC Bishigram) which meets River Swat. Jopin and Ingrabad were totally washed away with flood and fields of Ghari Kalay



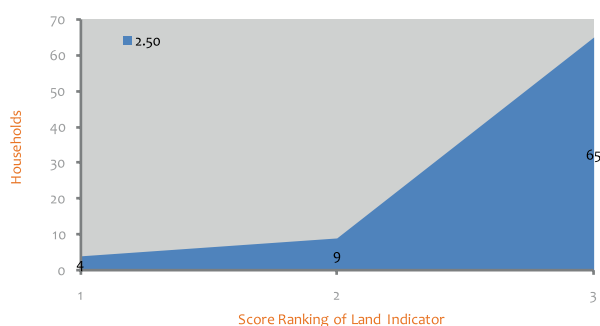
are damaged, some land is washed away and irrigation channels are damaged. During flood, hospital was also damaged which is now shifted to another rented building. Community told that we are organized in committees and we do have the plans for preparedness and mitigation but we lack financial assistance and therefore we are unable to pursue our plans. We need to make protection walls and embankments near river side and most importantly we need land to construct our houses and for this purpose Government should give us land as our land is washed away in food and now we are homeless and living in tents and houses of our relatives. Male community said we need trainings on swimming, firefighting and need to make evacuation route. If we make early warning system then we can be prepared for it and we can make water and food storage. Male community said that we do not allow female participation in committees while on the other hand females are interested in forming committees at village level and do attend first aid trainings and mock drills.



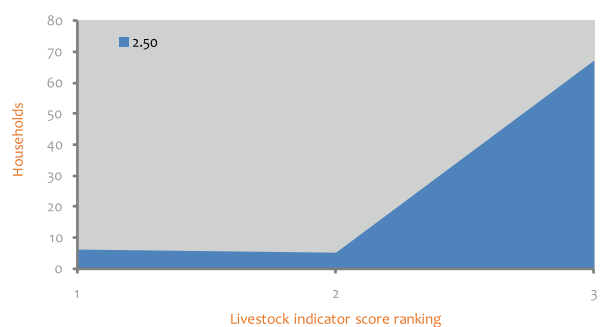
Households under Average Exposure Ranges



Households under Average Fragility



Land Holding Vulnerability



Livestock Vulnerability

Vulnerability Assessment

Summary of household vulnerability

Analysis: In order to set framework for household vulnerability assessment and derive coherent results; there sub indicators including **1) exposure 2) fragility and 3) lack of resilience** were separately analyzed in view of both vulnerability ranking.

However, as hazard scores for the village is cumulatively signified vulnerability ranking would also signify hazard level of 1.5 which is evident in the village.

Exposure Analysis: Exposure analysis signifies property that is exposed to risk, as the overall results shows, 27 out of 78 households stated their vulnerability is in rank of 3 which is the highest rating. It can be reflective of mud houses in the area and their perception regarding flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent hence their property would be under continuous risk.

Fragility Analysis: According to fragility indicator, 53 out of 78 households surveyed fall under range of 2.0-2.5 which is almost touching the higher end. Further analysis would show that under fragility sub indicators majority of basic amenities constituting land, livestock and income which signify monetary wellbeing fall under rank of 3.

Indicative of Socio economic wellbeing of communities in context to Hazard

Land Holding: The village has hazard score of 2.5 and 65 houses out of 78 falls

under rank of 3 for land indicator .As only risk prone households are chosen this indicator would bound to have a higher proportion considering already vulnerable land holding patterns in the village.

Livestock: The village has hazard score of 2.5 and 67 out of 78 HHs fall under rank of 3 for livestock indicator.

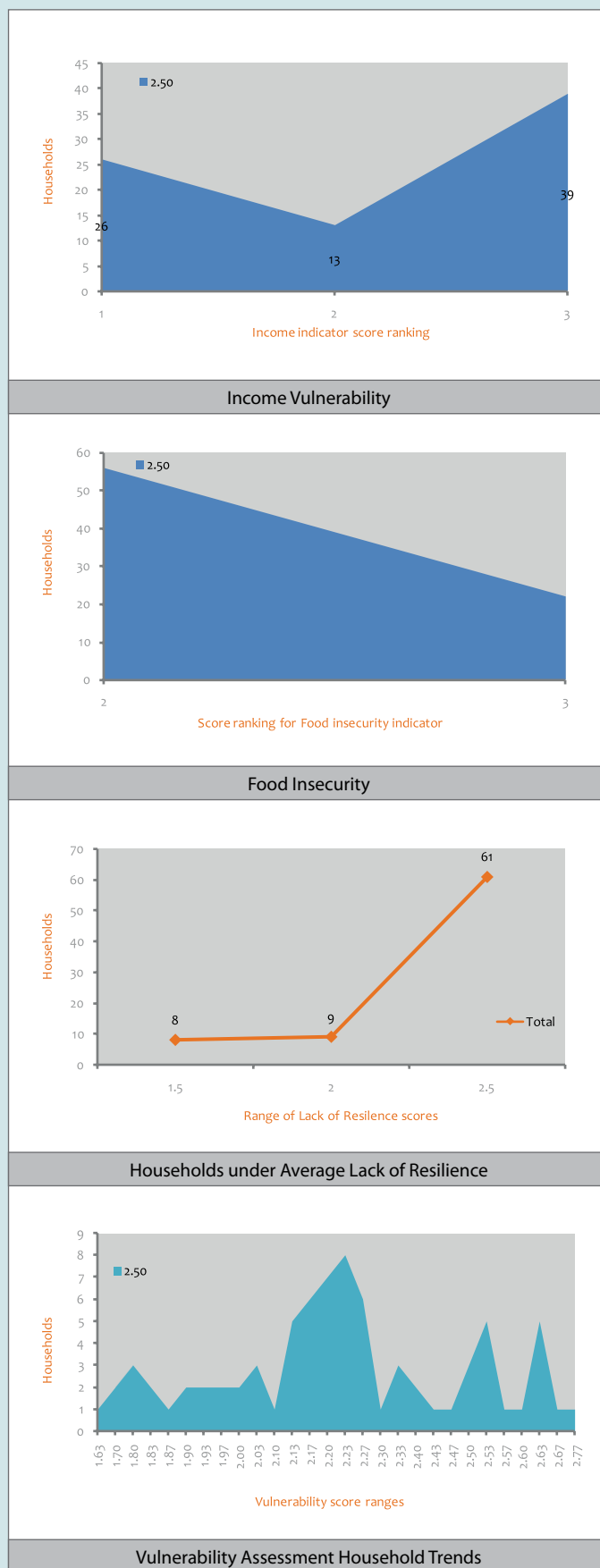
Income: The village has hazard score of 2.5 and 39 houses out of 78 surveyed falls under rank of 3 for income indicator which shows that majority of the houses surveyed have meager source of income. However in comparison to other villages it seems comparatively low.

Food Insecurity: The village has hazard score of 2.50 and 56 out of 78 HHs fall under rank of 2 for food insecurity indicator.

Summary of socio economic indicators: In the village there is a strong correlation between land holding and livestock with both falling under same proportion under rank of 3. However other two indicators such as food insecurity and income are comparatively low, in fact most of the households fall in score of 2 for it which goes to show that despite hazard, food insecurity and income levels are not at critically high vulnerability level.

Lack of resilience

Lack of resilience indicator is measured through education and neighborhood patterns in a village, under it 61 out of 78 HHs surveyed fall within scores of 2.5



which almost touches higher boundaries. For further analysis shows 77% of houses have no member who goes to school so literacy trends to respond to hazard is significantly low; lack of neighborhood support in the village is a lower driver of vulnerability in comparison with 77 households falling in rank of 2. Hence it can be conducted that education level is greater driver of vulnerability in the village.



Overall Vulnerability Score Range

The depiction as per the graph shows that majority of households surveyed fall between ranges of 2.13-2.53 and above equating to 48.

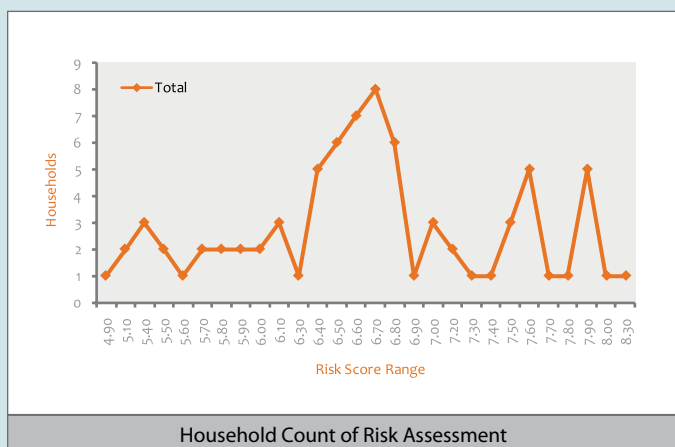
Risk Assessment

In calculating risk assessment number of households in Gharai Village it was revealed that about 63% households equating to 49 falls under risk score of 6-7.5 and above with average hovering around 6.6 considering some of the households which are above the mean average.

Accessibility

Average time that villages are cut off during the winter time: Due to land sliding in winters (Dec – Feb) Village is cut off from Behrain Tehsil.

Availability of public and other transport: Public Transport is available.



Demographic Profile

Name of Village	No. of Total HH	No. of widows & Single Women*	No. of Child-headed HHs*	No. of elderly as head of HHs*	No. of disabled as head of HHs*	Priority Needs	Village Infrastructure
Ghari	800	40	12	8	12	DMC, school safety program to be integrated with plantation, seed banks, energy efficient stoves, food preservation, fortification of safe asylums, sanitation, protection walls at critical points in the village, water harvesting.	4 small markets, one service station, 1 Taxi Stand, 2 Hujra, One Mosque, One graveyard, Crop fields near River, 1 Hospital damaged by flood and now shifted to rented building, Concrete streets, paved roads. Irrigation channels, Hospital building damaged and needs reconstruction

**Estimated figures based on local community knowledge*

Annex 8: Village Report - Shagai Shahgram

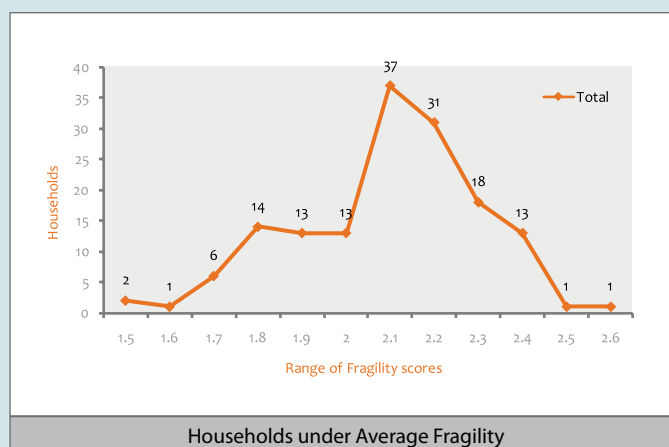
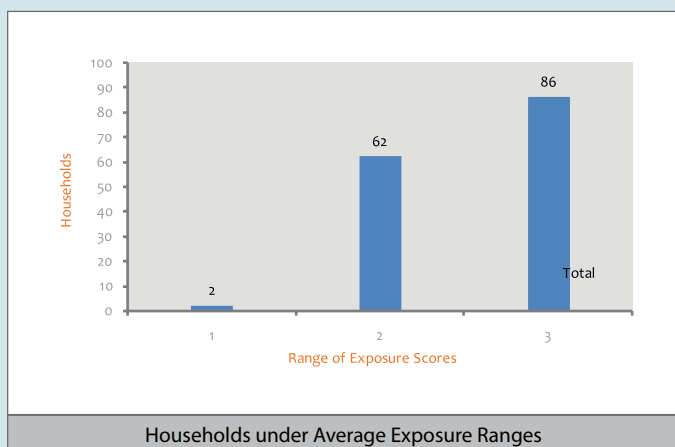
Tehsil: Behrain	Union Council: Madyan	HH Size: 8-10	Hazard Score: 2.25
Hazards in the village: Seasonal flooding, major flooding & soil erosion			
Surveyed household prone to hazards: 150			
NGOs/ Agencies working in the village			
1	NRSP: Livelihoods		

Profile of the Village

Shagai Shahgram is located in Tirat UC and is almost 60 Km away from Mingora, a central city in Swat.

Hazard Assessment

During flood water level in Shahgram was about 20 feet or above, this caused damaged to water channels, water supply, fields and houses. Main roads, bridges, communication and electricity were washed away with flood and there was shortage of food however through mutual cooperation with neighbors they shared food. Major area of Shagai which was effected was drab which is close to the river, presently there are fair chances that river might change course in the future which might hold other communities in close vicinity at risk. The area is also prone to seasonal flooding which causes extensive soil erosion.



Vulnerability Assessment

Summary of household vulnerability Analysis: In order to set framework for household vulnerability assessment and derive coherent results; there sub indicators including **1) exposure 2) fragility and 3) lack of resilience** were separately analyzed in view of vulnerability ranking. However as hazard scores for the villages is cumulatively signified vulnerability ranking would also signify hazard level of 1.5 which is evident in the village.

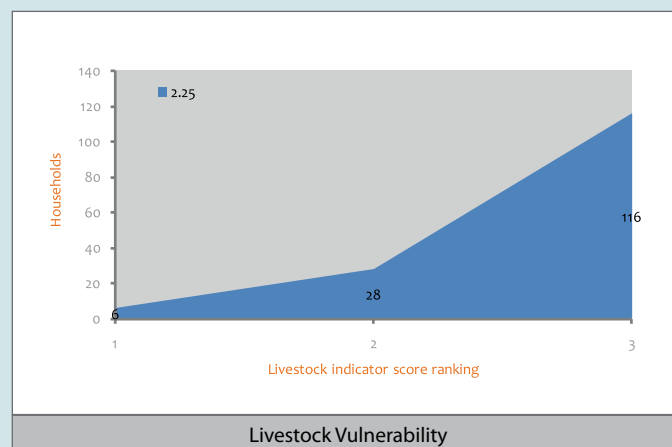
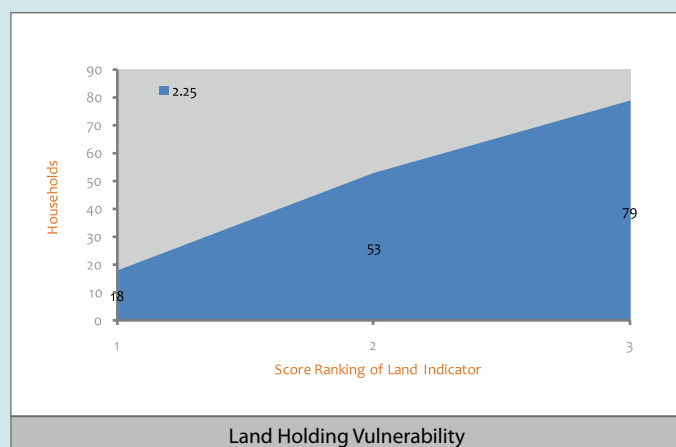
Exposure Analysis: Exposure analysis signifies property that is exposed to risk, as the overall results shows 86 out of 150 households surveyed in the village stated their vulnerability is in rank of 3 which is the highest rating. Possibly it can be reflective of mud houses in the area and their perception regarding flood 2010 and its intensity in respect of which they feel that with changes in climatic conditions natural hazard would become more so prominent hence their property would be continuously under risk.

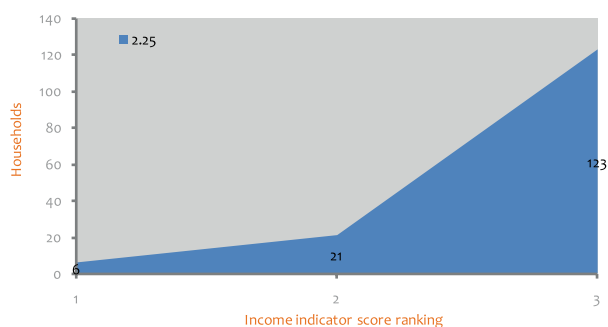
Fragility Analysis: According to fragility indicator, 99 out of 150 households surveyed fall under range of 2.1-2.4 which is almost touching the medium higher end. Further analysis show that under fragility sub indicators majority of basic amenities constituting land, livestock and income which signify monetary wellbeing fall under rank of 3.

Indicators of Socio economic wellbeing of communities in context to Hazard

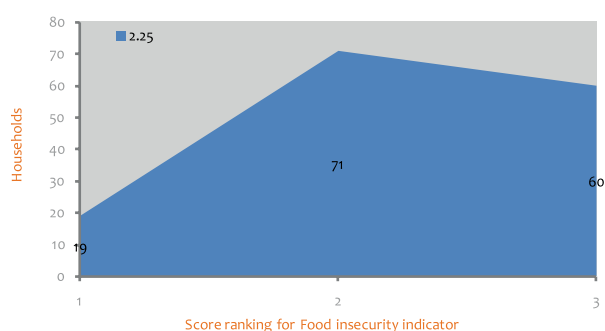
Land Holding: The village has hazard score of 2.25 and 79 out of 150 HHs fall under rank of 3 for land indicator .As only risk prone households are chosen this indicator would bound to have a higher proportion considering already vulnerable land holding patterns in communities but its pattern is almost equally distributed across both score rank of 2 & 3 which attributes to different vulnerability levels of residents in the village.

Livestock: The village has hazard score of 2.25 and 116 out of 150 HHs fall under rank of 3 for livestock indicator. Please note that it livestock vulnerability in this village seems a strong

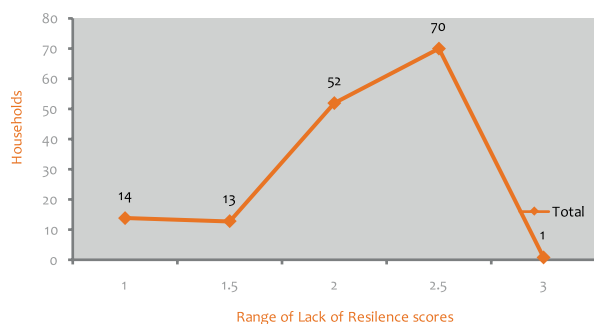




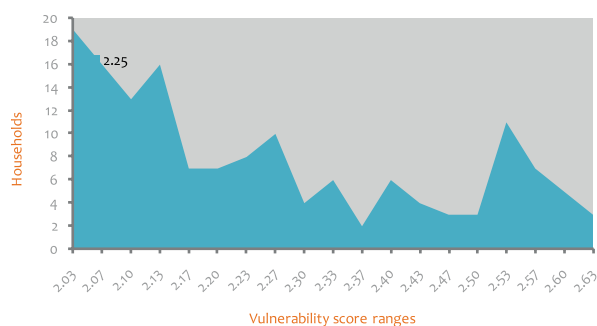
Income Vulnerability



Food Insecurity



Households under Average Lack of Resilience



Vulnerability Assessment Household Trends

driver of vulnerability of households more so than land holding mentioned above.

Income: The village has hazard score of 2.25 and 123 out of 150 HHs surveyed fall under rank of 3 for income indicator which shows that majority of the houses surveyed have meager source of income which feeds into accentuating effects of hazard in the area.

Food Insecurity: The village has hazard score of 2.50 and 71 out of 150 HHs fall under rank of 2 for food insecurity indicator. In comparison to some of other villages its proportion is equally distributed between rank of 2 & 3.

Summary of socio economic

indicators: In the village there is strong correlation between income and livestock with both falling under same proportion under rank of 3. However, in other two indicators such as food insecurity and land, vulnerabilities are spread out between score of 2 & 3 indicating different levels of vulnerabilities in the area, in fact most of the households fall in score of 2 for it which goes to show that despite hazard food insecurity and income levels are not at critically high vulnerability level.

Lack of resilience

Lack of resilience indicator is measured through education and neighborhood patterns in a village. Under it 70 out of 150 HHs surveyed fall within scores of 2.5 which almost touches higher boundaries. For further analysis 65% of houses have no member who goes to school so literacy trends to respond

to hazard is significantly low. Lack of neighborhood support in the village is a lower driver of vulnerability in comparison, with 109 households falling in rank of 2. Hence it can be conducted that education level is greater driver of vulnerability in the village.

Overall Vulnerability Score Range

The graph shows that majority of household surveyed fall between ranges of 2.03-2.30 equating to 100.

Risk Assessment

In calculating risk assessment in Shagai Village it was revealed that about 64% households equating to 96 are clustered in risk scores of 4.58-5.3 which causes average risk score of the village to hover around 5.

Demographic Profile

Name of Village	No. of Total HH	No. of widows & Single Women*	No. of Child-headed HHs*	No. of elderly as head of HHs*	No. of disabled as head of HHs*	Priority Needs	Village Infrastructure
Shagai	300	10	1	3	3	3 school safety programs, DMCs, plantation, seed banks, safe asylum, food preservation, candle making energy efficient stoves	2 schools, 3 mosques,

**Estimated figures based on local community knowledge*



Household count of Risk Assessment



Shagai Shahgram Risk Map

Annex 9: List of Targeted household for Assessment Survey

HH ID	Demographic Profiling				Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Risk Assessment Analysis		
	Name	Sex	Village	UC							Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
1	Unar Hayat	Male	Kokrai	Islampur	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
2	Norin	Female	Kokrai	Islampur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
3	Jamila Bibi	Female	Kokrai	Islampur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
4	Gul Zarin	Female	Kokrai	Islampur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
5	Gul Nazar	Male	Kokrai	Islampur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
6	Arshad Ali	Male	Kokrai	Islampur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
7	Gulab Sher	Male	Kokrai	Islampur	3	2.5	2	2	0	2.5	2.50	2.25	5.63
8	Alam Khan	Male	Kokrai	Islampur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
9	Fateh Khan	Male	Kokrai	Islampur	3	2.5	2	2	0	2.5	2.50	2.25	5.63
10	Khaperar	Male	Kokrai	Islampur	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
11	Abdul Haq	Male	Kokrai	Islampur	3	2	2.5	2	0	2.5	2.50	2.25	5.63
12	Shazia	Female	Kokrai	Islampur	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
13	Naseem	Female	Kokrai	Islampur	3	1.9	3	2	0	2.5	2.63	2.25	5.93
14	M. Fazal	Male	Kokrai	Islampur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
15	Sher Bahadar Khan	Male	Kokrai	Islampur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
16	Khurshida Begum	Female	Kokrai	Islampur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
17	Aziz-ur-Rehman	Male	Kokrai	Islampur	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
18	Mohammad Hayat	Male	Kokrai	Islampur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
19	Shamim	Female	Kokrai	Islampur	3	1.9	3	2	0	2.5	2.63	2.25	5.93
20	Fehmida	Female	Kokrai	Islampur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
21	Husan Zeba	Female	Kokrai	Islampur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
22	Naseem	Female	Kokrai	Islampur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
23	Nazia	Female	Kokrai	Islampur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
24	Yasmeen	Female	Kokrai	Islampur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
25	Sajida	Female	Kokrai	Islampur	3	2	3	2	0	2.5	2.67	2.25	6.00
26	Bibi Maryam	Female	Kokrai	Islampur	3	2.6	2	2	0	2.5	2.53	2.25	5.70
27	Sakina	Female	Kokrai	Islampur	3	2.7	2	2	0	2.5	2.57	2.25	5.78
28	Nargis	Female	Kokrai	Islampur	3	2.6	2	2	0	2.5	2.53	2.25	5.70
29	Gul Zada	Male	Kokrai	Islampur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
30	Unar Rehman	Male	Kokrai	Islampur	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
31	Shanzadai	Female	Kokrai	Islampur	3	2	2.5	2	0	2.5	2.50	2.25	5.63
32	Gulmina	Female	Kokrai	Islampur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
33	Rahamisa	Female	Kokrai	Islampur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
34	Nasib	Female	Kokrai	Islampur	3	2.5	2	2	0	2.5	2.50	2.25	5.63
35	Nigat	Female	Kokrai	Islampur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
36	Reshman	Female	Kokrai	Islampur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
37	Aqal Mina	Female	Kokrai	Islampur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
38	Badraj Begam	Female	Kokrai	Islampur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
39	Iqbal Ahmad	Male	Kokrai	Islampur	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
40	Gul Zada	Male	Kokrai	Islampur	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
41	Habib Khan	Male	Gul Bandai	Islampur	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
42	Bakht Wahid	Male	Gul Bandai	Islampur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
43	Unar Ali	Male	Gul Bandai	Islampur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
44	Sayed Muhammad	Male	Gul Bandai	Islampur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
45	Badshah Anwar	Male	Gul Bandai	Islampur	3	2	2.5	2	0	2.5	2.50	2.25	5.63
46	Muhammad Sher	Male	Gul Bandai	Islampur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
47	Rahim Gul	Male	Gul Bandai	Islampur	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
48	Muhammad Parvez	Male	Gul Bandai	Islampur	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
49	Sardaro Begum	Female	Gul Bandai	Islampur	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
50	Rahim Dad	Male	Gul Bandai	Islampur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
51	Ahmad Shah	Male	Gul Bandai	Islampur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
52	Hazrat Rehman	Male	Gul Bandai	Islampur	3	2	2.5	2	0	2.5	2.50	2.25	5.63
53	Hussan Zada	Male	Gul Bandai	Islampur	3	2.4	2	2	0	2.5	2.47	2.25	5.55

Demographic Profiling					Risk Assessment Analysis								
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
54	Ibni Rehman	Male	Gul Bandai	Islampur	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
55	Sultan Muhammad	male	Gul Bandai	Islampur	3	2.4	2.5	2	2.4	2.5	2.63	2.25	5.93
56	Sultani Gul	Male	Gul Bandai	Islampur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
57	Gul Muhammad	Male	Gul Bandai	Islampur	3	2.3	2.5	2	2.60	2.5	2.60	2.25	5.85
58	Asiya	Female	Gul Bandai	Islampur	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
59	Muhammad Afzal	Male	Gul Bandai	Islampur	3	2.4	2	2	2.4	2.5	2.47	2.25	5.55
60	Muhammad Iqbal	Male	Gul Bandai	Islampur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
61	Begum	Female	Gul Bandai	Islampur	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
62	Bakht Jamala Bibi	Female	Gul Bandai	Islampur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
63	Rajai	Female	Gul Bandai	Islampur	3	2.3	2.5	2	2.60	2.5	2.60	2.25	5.85
64	Anjad Ali	Male	Gul Bandai	Islampur	3	2.5	2	2	0	2.5	2.50	2.25	5.63
65	Gul Faraz	Male	Gul Bandai	Islampur	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
66	Laal Gul	Male	Gul Bandai	Islampur	3	2	2.5	2	0	2.5	2.50	2.25	5.63
67	Asmat	Female	Gul Bandai	Islampur	3	2.4	2	2	0	2.5	2.47	2.25	5.55
68	Shoukat Ali	Male	Gul Bandai	Islampur	3	2.5	2	2	0	2.5	2.50	2.25	5.63
69	Gul Bashir	Male	Gul Bandai	Islampur	3	2.4	2	2	0	2.5	2.47	2.25	5.55
70	Jomae	Female	Gul Bandai	Islampur	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
71	Ezat bibi	Female	Gul Bandai	Islampur	3	2.4	2.5	2	2.63	2.5	2.63	2.25	5.93
72	Asif	Male	Saidu	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
73	Abdul Rehman	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
74	Aqal Gul	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
75	Gul Sher	Male	Saidu	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
76	Parveez Khan	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
77	Fareez Gul	Male	Saidu	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
78	Haseena Bibi	Female	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
79	Sadar Ali	Male	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
80	Sultan Zadgai	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
81	Yasmeen	Female	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
82	Sher Bacha	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
83	Ghaffar	Male	Saidu	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
84	Muhammad Wazir	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
85	Seema	Female	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
86	Noor Wahid	Male	Saidu	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
87	Umar Wahid	Male	Saidu	saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
88	Shah Bakht Rawan	Male	Saidu	Saidu	3	2.3	2	2	0	2.5	2.43	2.25	5.48
89	Muhammad Rawan	Male	Saidu	Saidu	3	2.1	3	2	0	2.5	2.70	2.25	6.08
90	Ibrahim Khan	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
91	Anwar zeb	Male	Saidu	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
92	Shermay bibi	Female	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
93	Jan sila	Female	Saidu	Saidu	3	2.6	2.5	2	2.70	2.5	2.70	2.25	6.08
94	Ibrahim	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
95	Mian Said Ghafar	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
96	Fazal Dad	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
97	Anir Mahmood	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
98	Sar zamin Khan	Male	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
99	Ismail	Male	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
100	Bakht Sultana	Female	Saidu	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
101	2009		Saidu	Saidu	3	2.2	2.5	2	2.78	2.5	2.57	2.25	5.78
102	Saif ul Malik	Male	Saidu	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
103	Zahoor	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
104	Momin Khan	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
105	Saif ul Malook	Male	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
106	Bakht Zaman	Male	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
107	Aftab Ahmad	Male	Saidu	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
108	Bakht Rehman	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
109	Naeem Akhtar	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30

Demographic Profiling				Risk Assessment Analysis									
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
110	Unar Khaliq	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
111	Zia-ur-Rehman	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
112	Ibrahim	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
113	Hameya	Female	Saidu	Saidu	3	2.2	2	2	0	2.5	2.40	2.25	5.40
114	Hakim Khan	Male	Saidu	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
115	Jabran	Male	Saidu	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
116	Baseerat	Female	Saidu	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
117	Fartinay	Female	Saidu	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
118	Hazrat Ali	Male	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
119	Altar	Male	Saidu	Saidu	3	2.6	2	2	0	2.5	2.53	2.25	5.70
120	Muhammad Alam	Male	Saidu	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
121	Amir Zeb	Male	Saidu	Saidu	3	2.6	2	2	0	2.5	2.53	2.25	5.70
122	Mohammad Anwar	Male	Saidu	saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
123	Badshah Khan	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
124	Sadiqeen	Male	Saidu	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
125	Daulat Khan	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
126	Fazal Wahid	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
127	Saba Gul	Female	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
128	Muhammad Aqil	Male	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
129	Iqbal Hussain	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
130	Nihayat Bibi	Female	Saidu	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
131	Anwar Hussain	Male	Saidu	Saidu	3	2.5	2	2	0	2.5	2.50	2.25	5.63
132	Alam Zeb	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
133	Sarfaraz	Male	Saidu	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
134	Shah Zamin	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
135	Said Qahar	Male	Saidu	Saidu	3	2.7	3	2	0	2.5	2.90	2.25	6.53
136	Fazal Khaliq	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
137	Shamroz Khan	Male	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
138	Dawat Khan	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
139	Muhammad Alim	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
140	Muhammad Wakil	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
141	Muhammad Zada	Male	Saidu	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
142	Shaukat Ali	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
143	Begum	Female	Saidu	Saidu	3	2.5	2	2	0	2.5	2.50	2.25	5.63
144	Fazal Subhan	Male	Saidu	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
145	Ghulam Jan	Male	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
146	Waqar Ali Khan	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
147	Farzana	Female	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
148	Fazal Khaliq	Male	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
149	Nisar Ahmad	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
150	Hariz Kamran	Male	Saidu	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
151	Said ur Rehman	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
152	Bakht Rawan	Female	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
153	Akhtar Saba	Female	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
154	Gul Azar	Female	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
155	Fanos Khan	Male	Saidu	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
156	Hussain Shah	Male	Saidu	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
157	Samar	Female	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
158	Muhammad Afzal	Male	Saidu	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
159	Habibullah	Male	Saidu	Saidu	3	2.6	2	2	0	2.5	2.53	2.25	5.70
160	Zainab	Female	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
161	Bacha Hilal	Male	Saidu	saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
162	Mian Said Jalal	Male	Saidu	Saidu	3	2.9	3	2	0	2.5	2.97	2.25	6.68
163	Rustam Khan	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
164	Dilawar Khan	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
165	Zeba	Female	Saidu	Saidu	3	2.7	1.5	2	0	2.5	2.40	2.25	5.40

Demographic Profiling				Risk Assessment Analysis									
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
166	Fazal Hussain	Male	Saidu	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
167	Ghufran	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
168	Muhammad Iqbal	Male	Saidu	Saidu	3	2.7	2	2	0	2.5	2.57	2.25	5.78
169	Muhammad Amin	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
170	Hameed Hussain	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
171	Umaru Khan	Female	Saidu	Saidu	3	2.6	2	2	0	2.5	2.53	2.25	5.70
172	Bakht Raja	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
173	Dunyaya Amin	Male	Saidu	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
174	Akbar Ali	Male	Saidu	Saidu	3	2.4	1.5	2	0	2.5	2.30	2.25	5.18
175	Rehman Ali	Male	Saidu	Saidu	3	2.4	1.5	2	0	2.5	2.30	2.25	5.18
176	Fazal Rahim	Male	Saidu	Saidu	3	2.7	1.5	2	0	2.5	2.40	2.25	5.40
177	Niaz Minm Khan	Male	Saidu	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
178	Anjad Hussain	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
179	Dunyaya Zada	Female	Saidu	Saidu	3	2.7	2	2	0	2.5	2.57	2.25	5.78
180	Khurshed Iqbal	Male	Saidu	Saidu	3	2.7	2	2	0	2.5	2.57	2.25	5.78
181	Isanullah	Male	Saidu	Saidu	3	2.8	1	2	0	2.5	2.27	2.25	5.10
182	Tahir Hussain	Male	Saidu	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
183	Fazal Khatib	Male	Saidu	Saidu	3	2.6	1.5	2	0	2.5	2.37	2.25	5.33
184	Irshad Rehman	Male	Saidu	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
185	Akhtar Ali	Male	Saidu	Saidu	3	2.4	2	2	0	2.5	2.47	2.25	5.55
186	Muhammad Imran	Male	Saidu	Saidu	2	2.3	3	2	0	2.5	2.43	2.25	5.48
187	Jahan Sher	Male	Saidu	Saidu	2	2.2	3	2	0	2.5	2.40	2.25	5.40
188	Akhtar Ali	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
189	Parveena	Female	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
190	Wazir Zada	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
191	Sami ullah	Male	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
192	Pir Muhammad Khan	Male	Saidu	Saidu	2	2.3	3	2	0	2.5	2.43	2.25	5.48
193	Nuzuhit Jabir	Female	Saidu	Saidu	3	2.3	2	2	0	2.5	2.43	2.25	5.48
194	Sardar Hussain	Male	Saidu	Saidu	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
195	Aqal Mand	Male	Saidu	Saidu	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
196	Anjad	Male	Saidu	Saidu	3	2.4	1.5	2	0	2.5	2.30	2.25	5.18
197	Dost Muhammad Khan	Male	Saidu	Saidu	2	2.5	3	2	0	2.5	2.50	2.25	5.63
198	Shabnam Bibi	Female	Saidu	Saidu	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
199	Bakht Naseeba	Female	Saidu	Saidu	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
200	Khan Sher	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
201	Nasrin	Female	Saidu	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
202	Bakht Naseeba	Female	Saidu	Saidu	2	2.3	3	2	0	2.5	2.43	2.25	5.48
203	Gul Sherin	Female	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
204	Talimand	Male	Saidu	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
205	Anir Badshah	Male	Saidu	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
206	Bano	Female	Saidu	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
207	Rasheed Khan	Male	Saidu	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
208	Muhammad Nawaz Khan	Male	Saidu	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
209	Gul Rehman	Male	Saidu	Saidu	3	2.7	2	2	0	2.5	2.57	2.25	5.78
210	Muhammad Ikram	Male	Saidu	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
211	Rahim Taj	Male	Saidu	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
212	Khan Sher	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
213	Muhammad Amin	Male	Saidu	saidu	3	2	2.5	2	0	2.5	2.50	2.25	5.63
214	Akbar Hussain	Male	Saidu	Saidu	3	2	2	2	0	2.5	2.33	2.25	5.25
215	Badshah Khan	Male	Saidu	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
216	Shoukat Iqbal	Male	Saidu	Saidu	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
217	Bakht Sardara	Female	Saidu	Saidu	3	2.2	2	2	0	2.5	2.40	2.25	5.40
218	Bahram Shah	Male	Saidu	Saidu	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
219	Maryam Bibi	Female	Saidu	Saidu	3	2.5	1.5	2	0	2.5	2.33	2.25	5.25
220	Rafaat Bibi	Female	Saidu	Saidu	3	2.6	2	2	0	2.5	2.53	2.25	5.70
221	Hamish Gul	Male	Saidu	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38

Demographic Profiling				Risk Assessment Analysis									
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Assessment Score	Total Risk Assessment Score
2222	Dawood Khan	Male	Saidu	Saidu	3	2.5	2	2	0	2.5	2.50	2.25	5.63
2223	Mudassar Khan	Male	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
2224	Shoukat Iqbal	Male	Saidu	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
2225	Zabarg Jamir	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
2226	Umer Zada	Male	Saidu	Saidu	3	2.4	1.5	2	0	2.5	2.30	2.25	5.18
2227	Hamid Hussain	Male	Saidu	Saidu	3	2.1	1.5	2	0	2.5	2.20	2.25	4.95
2228	Rahim ullah	Male	Saidu	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
2229	Abdul Wahid	Male	Saidu	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
230	Khurshed Ahmed	Male	Saidu	Saidu	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
231	Shahi Muhammad	Male	Saidu	Saidu	3	2.2	2	2	0	2.5	2.40	2.25	5.40
232	Muhammad Rehman	Male	Saidu	Saidu	3	2.2	2	2	0	2.5	2.40	2.25	5.40
233	Zarin Bashar	Female	Saidu	Saidu	3	2	2	2	0	2.5	2.33	2.25	5.25
234	Gul Bashar	Male	Saidu	Saidu	2	2	3	2	0	2.5	2.33	2.25	5.25
235	Sahib Sardar	Male	Saidu	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
236	Gul Sherwan	Male	Saidu	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
237	Bina zira bibi	Female	Shagai	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
238	Said Rehman	Male	Shagai	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
239	M. Zarin	Male	Shagai	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
240	Bacha Khan	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
241	Naimatullah	Male	Shagai	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
242	Bakht Siraja	Female	Shagai	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
243	Ihsanullah	Male	Shagai	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
244	Gulab	Female	Shagai	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
245	Jamshaid	Male	Shagai	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
246	Rukhsana bibi	Female	Shagai	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
247	Fary mirman	Female	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
248	Zebida	Female	Shagai	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
249	Rukhsana bibi	Female	Shagai	Saidu	3	1.9	3	2	0	2.5	2.63	2.25	5.93
250	Juma Khan	Male	Shagai	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
251	Naseem ullah	Male	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
252	Muhammad Sher Ali	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
253	Fazal Rehman	Male	Shagai	Saidu	3	2.7	3	2	0	2.5	2.90	2.25	6.53
254	Najmila bibi	Female	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
255	Naheeda	Female	Shagai	Saidu	3	2.1	3	2	0	2.5	2.70	2.25	6.08
256	Shah Aeeran	Female	Shagai	Saidu	3	2	3	2	0	2.5	2.67	2.25	6.00
257	Maaraj Bibi	Female	Shagai	Saidu	3	1.8	3	2	0	2.5	2.60	2.25	5.85
258	Asia	Female	Shagai	Saidu	3	1.9	3	2	0	2.5	2.63	2.25	5.93
259	Zainab	Female	Shagai	Saidu	3	2.7	3	2	0	2.5	2.90	2.25	6.53
260	Hassan Khan	Male	Saidu	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
261	Fazal Khan	Male	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
262	Umara Khan	Female	Shagai	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
263	Asghar Khan	Male	Shagai	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
264	Habib Khan	Male	Shagai	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
265	Nazia	Female	Shagai	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
266	Bibi Nek Zada	Female	Shagai	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
267	Umer Ali	Male	Shagai	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
268	Salah ud din	Male	Shagai	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
269	Bakht Zeba	Female	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
270	Noshi rawan	Female	Shagai	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
271	Bakht Rawan	Female	Shagai	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
272	Aurash	Female	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
273	Amina Bibi	Female	Shagai	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
274	Bakht Sherawn	Female	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
275	Shebar	Female	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
276	Mehboob Ali	Male	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
277	Umer Khan	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38

Demographic Profiling				Risk Assessment Analysis									
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
278	Zakia	Female	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
279	Bakht Zada	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
280	Ghulam Rahim	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
281	Nazia Imran	Female	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
282	Sarbala	Female	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
283	Gul Malook	Female	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
284	Saeeda	Female	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
285	Gul Zada	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
286	Umer Nasir	Male	Shagai	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
287	Farman Ali	Male	Shagai	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
288	Sher Zada bibi	Female	Shagai	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
289	Ghulam Haider	Male	Shagai	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
290	Bano	Male	Shagai	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
291	Hazrat Ahmed	Male	Shagai	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
292	Sher Zada	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
293	Ibrahim	Male	Shagai	Saidu	3	2.8	3	2	0	2.5	2.93	2.25	6.60
294	Akbar Ali	Male	Shagai	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
295	Sheren	Female	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
296	Gul Namreez	Male	Shagai	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
297	Khan Zada	Male	Shagai	Saidu	3	2.8	2	2	0	2.5	2.60	2.25	5.85
298	Muhammad Afzal	Male	Shagai	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
299	Ahmad Aleem	Male	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
300	Sardara	Female	Shagai	Saidu	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
301	Taj Muhammad Khan	Male	Shagai	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
302	Said Rehman	Male	Shagai	Saidu	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
303	Alamgir	Male	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
304	Muhammad Khan	Male	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
305	Sakina	Female	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
306	Liaqat Ali	Male	Shagai	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
307	Ajab Khan	Male	Shagai	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
308	Zeenat	Female	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
309	Gul Muhammad Khan	Male	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
310	Shalbar	Male	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
311	Hakim Khan	Male	Shagai	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
312	Rabia	Female	Shagai	Saidu	3	2.1	3	2	0	2.5	2.70	2.25	6.08
313	Fahmia	Female	Shagai	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
314	Arjoon	Female	Shagai	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
315	Shahada	Female	Shagai	Saidu	3	2.1	3	2	0	2.5	2.70	2.25	6.08
316	Javed	Male	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
317	Parveez	Male	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
318	Abdul Kabir	Male	Shagai	Saidu	3	2.6	3	2	0	2.5	2.87	2.25	6.45
319	Abdul Sattar	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38
320	Umer Din	Male	Shagai	Saidu	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
321	Naseem	Female	Shagai	Saidu	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
322	Mashoo	Female	Shagai	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
323	Jamal Khan	Male	Shagai	Saidu	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
324	Muhammad Rahim	Male	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
325	Abdul Malik	Male	Shagai	Saidu	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
326	Rasool Khan	Male	Shagai	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
327	Musharraf Khan	Male	Shagai	Saidu	3	2.2	3	2	0	2.5	2.73	2.25	6.15
328	Muhammad Sherin	Male	Shagai	Saidu	3	2.7	3	2	0	2.5	2.90	2.25	6.53
329	Khalista Khan	Male	Shagai	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
330	Khalista Muhammad	Male	Shagai	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
331	Gul Muhammad	Male	Shagai	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
332	Niaz ud din	Male	Shagai	Saidu	3	2.3	3	2	0	2.5	2.77	2.25	6.23
333	Babu Jan	Male	Shagai	Saidu	3	2.5	3	2	0	2.5	2.83	2.25	6.38

Demographic Profiling				Risk Assessment Analysis									
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Assessment Score	Total Risk Assessment Score
334	Bibi Haroon	Female	Shagai	Saidu	3	2.4	3	2	0	2.5	2.80	2.25	6.30
335	Bakht Zameena	Female	Shagai	Saidu	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
336	Yasir Hussain	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
337	Sultana	Female	Spal Bandai	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
338	Ghazala	Female	Spal Bandai	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
339	Kinkhaba	Female	Spal Bandai	islam pur	3	2.7	3	2	0	2.5	2.90	2.25	6.53
340	Chand Bibi	Female	Spal Bandai	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
341	Zohra	Female	Spal Bandai	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
342	Nazia	Female	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
343	Habib Zar	Male	Spal Bandai	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
344	Gvandai	Female	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
345	Nasib ranva	Female	Spal Bandai	islam pur	3	2	3	2	0	2.5	2.67	2.25	6.00
346	Shreer Muhammad Khan	Male	Spal Bandai	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
347	Iqbal	Female	Spal Bandai	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
348	Yasmin	Female	Spal Bandai	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
349	Gul bibi	Female	Spal Bandai	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
350	Razia	Female	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
351	Norista	Female	Spal Bandai	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
352	Cavara	Female	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
353	Akhtar Begum	Female	Spal Bandai	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
354	Robina	Female	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
355	Shamim	Female	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
356	Shahnaz	Female	Spal Bandai	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
357	Mareena	Female	Spal Bandai	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
358	Husn Zeba	Female	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
359	Sahiba	Female	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
360	Anir Naushad	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
361	Fazal Zada	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
362	Bashmand Khan	Male	Spal Bandai	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
363	kajal Bibi	Female	Spal Bandai	islam pur	3	1.9	3	2	0	2.5	2.63	2.25	5.93
364	Warra Bibi	Female	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
365	Siraja Bibi	Female	Spal Bandai	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
366	Muhammad Hanifa	Male	Spal Bandai	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
367	Liaqat Ali	Male	Spal Bandai	islam pur	3	2	3	2	0	2.5	2.67	2.25	6.00
368	Akhtar Nawab	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
369	Akhtar Ali	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
370	Rozi Khan	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
371	Momin Khan	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
372	Bakhti Rawana	Female	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
373	Muhammad Amin	Male	Spal Bandai	islam pur	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
374	Bakht Waseela	Female	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
375	Sultan Ghani	Male	Spal Bandai	islam pur	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
376	Shah Zaman	Male	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
377	Zain ul Akbar	Male	Spal Bandai	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
378	Jan Saba	Female	Spal Bandai	islam pur	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
379	Jehan Akbar	Male	Spal Bandai	islam pur	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
380	Bakht Ameen	Male	Spal Bandai	islam pur	3	2.8	2	2	0	2.5	2.60	2.25	5.85
381	Abdullah Jan	Male	Spal Bandai	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
382	Taj Meh Khan	Male	Spal Bandai	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
383	Alam Zeb	Male	Spal Bandai	islam pur	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
384	Khaista Gul	Male	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
385	Said Karim	Male	Spal Bandai	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
386	Fazal Rehman	Male	Spal Bandai	islam pur	3	2	3	2	0	2.5	2.67	2.25	6.00
387	Usman Ghamni	Male	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
388	Rag Begum	Female	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
389	Fazal Rehman	Male	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30

Demographic Profiling					Risk Assessment Analysis								
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
390	Bahroz Khan	Male	Spal Bandai	islam pur	3	1.9	3	2	0	2.5	2.63	2.25	5.93
391	Gul Sherin	Male	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
392	Khurshid Ali	Male	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
393	Yahya khan	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
394	Imran	Male	Spal Bandai	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
395	Bakht Haram	Female	Spal Bandai	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
396	Resal Khan	Female	Spal Bandai	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
397	Afsar Zaman	Male	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
398	Laal Gul	Male	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
399	Painda Gul	Male	Spal Bandai	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
400	Umer Gul	Male	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
401	Akbar Jan	Male	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
402	Luqman Ali	Male	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
403	Bakht Nameer	Male	Spal Bandai	islam pur	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
404	Farooq Khan	Male	Spal Bandai	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
405	Iqbal Hussain	Male	Spal Bandai	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
406	Akhtar Hussain	Male	Spal Bandai	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
407	Gulshan Zada	Male	Spal Bandai	islam pur	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
408	Umer Zaman	Male	Spal Bandai	islam pur	3	2.5	2.5	2	0	2.5	2.67	2.25	6.00
409	Umer Ali Khan	Male	Spal Bandai	islam pur	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
410	Muhammad Akbar	Male	Spal Bandai	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
411	Imran Zada	Male	Spal Bandai	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
412	Umer Deer	Male	Spal Bandai	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
413	Mehnaz	Female	Spal Bandai	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
414	Naimat Khan	Male	Spal Bandai	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
415	Luqman Nakim	Male	Spal Bandai	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
416	Soltanat Khan	Male	Spal Bandai	islam pur	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
417	Muhammad Irshad	Male	Spal Bandai	islam pur	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
418	Sher Wali Khan	Male	Spal Bandai	islam pur	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
419	Shahi Sultana	Female	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
420	Zahida	Female	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
421	Shabana	Female	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
422	Muhammad Asha	Male	Murguzar	islam pur	3	2.7	2.5	2	0	2.5	2.73	2.25	6.15
423	Dawar Khan	Male	Murguzar	islam pur	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
424	Said Ghawas	Male	Murguzar	islam pur	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
425	Abdul Rauf	Male	Murguzar	islam pur	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
426	Bahadar Khan	Male	Murguzar	islam pur	3	2.8	2.5	2	0	2.5	2.77	2.25	6.23
427	Shaeeda	Female	Spal Bandai	islam pur	3	2	3	2	0	2.5	2.67	2.25	6.00
428	Rasool Khan	Male	Kokrai	islam pur	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
429	Hamida Bibi	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
430	Aziz ur Rehman	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
431	Hazrat Zaidgal	Male	Murguzar	islam pur	3	2.7	3	2	0	2.5	2.90	2.25	6.53
432	Hazrat Akbar	Male	Murguzar	islam pur	3	2.2	1.5	2	0	2.5	2.23	2.25	5.03
433	khan sherin	Male	Murguzar	islam pur	3	1.8	2.5	2	0	2.5	2.43	2.25	5.48
434	Amira	Female	Murguzar	islam pur	3	2.8	3	2	0	2.5	2.93	2.25	6.60
435	Bakht Masala	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
436	Zuhra Begum	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
437	Iqbal Jahan	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
438	Daulat Begum	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
439	Subhania	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
440	Sulana	Female	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
441	Shaheen	Female	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
442	Bakht Zamin	Female	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
443	Umer Ali	Male	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
444	Khairati	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
445	Babuzai	Male	Murguzar	islam pur	3	2.8	3	2	0	2.5	2.93	2.25	6.60

Demographic Profiling				Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Risk Assessment Analysis			
HH ID	Name	Sex	Village							UC	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
446	Taj Bibi	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
447	Shahzadgai	Male	Murguzar	islam pur	3	2.7	3	2	0	2.5	2.90	2.25	6.53
448	Gul Farosha	Female	Murguzar	islam pur	3	2.7	3	2	0	2.5	2.90	2.25	6.53
449	Parveez	Male	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
450	Fazal Rabbi	Male	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
451	Jamila	Female	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
452	Bala Nista	Female	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
453	Firzadgai	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
454	Mian Bar Khan	Male	Murguzar	islam pur	3	2.8	3	2	0	2.5	2.93	2.25	6.60
455	Free Zada	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
456	Bakht Alam	Male	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
457	Abdul Kabir	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
458	Misri Khan	Male	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
459	Abdul Khaliq	Male	Murguzar	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
460	Mam Rahim	Male	Murguzar	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
461	Speen	Male	Murguzar	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
462	Fazal Khalid	Male	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
463	Sher Zada Khan	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
464	Mian Said Jan	Male	Murguzar	islam pur	3	2.5	2	2	0	2.5	2.50	2.25	5.63
465	Aziz ur Rashid	Male	Murguzar	islam pur	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
466	Barkat Ali	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
467	Tobi Khan	Male	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
468	Jomaraz	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
469	Akbar Khan	Male	Murguzar	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
470	Farteht Khan	Male	Murguzar	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
471	Bakht Amin	Female	Murguzar	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
472	Akhter Munir	Male	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
473	Sadaqat	Male	Murguzar	islam pur	3	2	3	2	0	2.5	2.67	2.25	6.00
474	Khanay	Female	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
475	Inayat Khan	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
476	Bakht Zada	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
477	Umer Khan	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
478	Anwar Sherin	Male	Murguzar	islam pur	3	1.9	2.5	2	0	2.5	2.47	2.25	5.55
479	Zahir Khan	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
480	Asalm Khan	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
481	Baeha Sherin	Female	Murguzar	islam pur	3	2	2	2	0	2.5	2.33	2.25	5.25
482	Umar Baeha	Male	Murguzar	islam pur	2	2.1	2	2	0	2.5	2.03	2.25	4.58
483	Zafar Ali	Male	Murguzar	islam pur	3	2.1	2	2	0	2.5	2.37	2.25	5.33
484	Said Umer	Male	Murguzar	islam pur	3	2.5	1.5	2	0	2.5	2.33	2.25	5.25
485	Fazal muhammad	Male	Murguzar	islam pur	2	2.1	2.5	2	0	2.5	2.20	2.25	4.95
486	Abdul Rehman	Male	Murguzar	islam pur	3	2	3	2	0	2.5	2.67	2.25	6.00
487	Fazal Ghami	Male	Murguzar	islam pur	2	2.2	2.5	2	0	2.5	2.23	2.25	5.03
488	Muhammad Khan	Male	Murguzar	islam pur	2	1.8	2.5	2	0	2.5	2.10	2.25	4.73
489	Akbar Khan	Male	Murguzar	islam pur	2	2	3	2	0	2.5	2.33	2.25	5.25
490	Momin Khan	Male	Murguzar	islam pur	2	1.9	2.5	2	0	2.5	2.13	2.25	4.80
491	Kaki	Female	Murguzar	islam pur	2	2	2.5	2	0	2.5	2.17	2.25	4.88
492	Meema Bibi	Female	Murguzar	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
493	Bakht Bibi	Female	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	6.23
494	Sher	Male	Murguzar	islam pur	3	2	3	2	0	2.5	2.67	2.25	6.00
495	Qudrat Ali	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
496	Abdul Mateen	Male	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
497	Waleezat	Female	Murguzar	islam pur	3	2.8	3	2	0	2.5	2.93	2.25	6.60
498	Satrin	Female	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
499	Bakht Unas	Female	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
500	Azim Khan	Male	Murguzar	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
501	Ali Rehman	Male	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	2.00

Demographic Profiling					Risk Assessment Analysis								
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
502	Shah Muhammad	Male	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	2.00
503	Fazal Rahim	Male	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	2.00
504	Rozi Khan	Male	Murguzar	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	2.00
505	Gul muhammad Khan	Male	Murguzar	islam pur	3	2.3	2	2	0	2.5	2.43	2.25	2.00
506	Mohabbat Khan	Male	Murguzar	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
507	Khalista Rehman	Male	Murguzar	islam pur	3	2	3	2	0	2.5	2.67	2.25	2.00
508	Wazir	Male	Murguzar	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
509	Baseer	Male	Murguzar	islam pur	3	2	2	2	0	2.5	2.33	2.25	5.25
510	Poshad	Male	Murguzar	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
511	Saltanat Khan	Male	Murguzar	islam pur	3	2.2	3	2	0	2.5	2.73	2.25	6.15
512	Gul Rashid	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
513	Abdul Qayyum	Male	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
514	Hakam Khan	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
515	Muslim	Male	Murguzar	islam pur	3	2.4	2	2	0	2.5	2.47	2.25	5.55
516	Ismael	Male	Murguzar	islam pur	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
517	Mustafa	Male	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
518	Habib Ullah	Male	Murguzar	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
519	Barooz Khan	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
520	Sahib Zara	Male	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
521	Rahim Khan	Male	Murguzar	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
522	Akhter Ali	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
523	Gul Bacha	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
524	Ruqla	Female	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
525	Sher Ali Khan	Male	Murguzar	islam pur	3	2.3	3	2	0	2.5	2.77	2.25	2.00
526	Sher Malik	Male	Murguzar	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
527	Juma Mohammad	Male	Murguzar	islam pur	3	2.6	2.5	2	0	2.5	2.70	2.25	6.08
528	Bastan	Male	Murguzar	islam pur	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
529	Muhammad Ali	Male	Murguzar	islam pur	3	2.6	3	2	0	2.5	2.87	2.25	6.45
530	Amir Jan	Male	Murguzar	islam pur	3	2.5	3	2	0	2.5	2.83	2.25	6.38
531	Kher Faqir	Male	Murguzar	islam pur	3	2.1	3	2	0	2.5	2.70	2.25	6.08
532	Umer Khatib	Male	Murguzar	islam pur	3	2.9	2.5	2	0	2.5	2.80	2.25	6.30
533	Aziz Raheem	Male	Murguzar	islam pur	2	2.2	2.5	2	0	2.5	2.23	2.25	5.03
534	Shah Wazir Khan	Male	Murguzar	islam pur	2	2.2	3	2	0	2.5	2.40	2.25	5.40
535	Muhammad Rauf	Male	Murguzar	islam pur	2	2.1	3	2	0	2.5	2.37	2.25	5.33
536	Khan Zada	Male	Murguzar	islam pur	2	2	3	2	0	2.5	2.33	2.25	5.25
537	Ihsan Ali	Male	Murguzar	islam pur	3	2.4	3	2	0	2.5	2.80	2.25	6.30
538	Muzaffar Shah	Male	Shagae Shahgram	Tirat	2	2.4	2	2	0	2.5	2.13	2.25	4.80
539	Ali Zar	Male	Shagae Shahgram	Tirat	2	2.3	2.5	2	0	2.5	2.27	2.25	5.10
540	Juma Gul	Male	Shagae Shahgram	Tirat	2	2.4	2.5	2	0	2.5	2.30	2.25	5.18
541	Sayed	Male	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
542	Qalandar shah	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
543	Sayed Alam Shah	Male	Shagae Shahgram	Tirat	3	2	2	2	0	2.5	2.33	2.25	5.25
544	Husn Zeba Bibi	Female	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
545	Muhammad Saeed	Male	Shagae Shahgram	Tirat	2	2.1	2.5	2	0	2.5	2.20	2.25	4.95
546	Bakht Afsar	Female	Shagae Shahgram	Tirat	1	2.6	2.5	2	0	2.5	2.03	2.25	4.58
547	Sami ud Din	Male	Shagae Shahgram	Tirat	3	1.8	2	2	0	2.5	2.27	2.25	5.10
548	Muhammad Sadiq Mian	Male	Shagae Shahgram	Tirat	2	1.9	2.5	2	0	2.5	2.13	2.25	4.80
549	Syed Rehman	Male	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
550	Syed Masoom Shah	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
551	Alam Zeb	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
552	Rafiq ul Hadi	Female	Shagae Shahgram	Tirat	3	1.9	2	2	0	2.5	2.30	2.25	5.18
553	Henif ul Hadi	Male	Shagae Shahgram	Tirat	3	1.8	2	2	0	2.5	2.27	2.25	5.10
554	Gul Tahar ud din	Male	Shagae Shahgram	Tirat	3	2.1	1	2	0	2.5	2.03	2.25	4.58
555	Haji Naqab	Male	Shagae Shahgram	Tirat	3	2.1	2	2	0	2.5	2.37	2.25	5.33
556	Nasir Khan	Male	Shagae Shahgram	Tirat	2	2.3	2.5	2	0	2.5	2.27	2.25	5.10
557	Akbar Jehan	Male	Shagae Shahgram	Tirat	3	2	2	2	0	2.5	2.33	2.25	5.25

Demographic Profiling				Risk Assessment Analysis									
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Assessment Score	Total Risk Assessment Score
558	Hasan Pari	Female	Shagae Shahgram	Tirat	2	2.2	2.5	2	0	2.5	2.23	2.25	5.03
559	Sharoon Mian	Male	Shagae Shahgram	Tirat	3	1.6	2	2	0	2.5	2.20	2.25	4.95
560	Iqbal ud din	Male	Shagae Shahgram	Tirat	3	1.9	2	2	0	2.5	2.30	2.25	5.18
561	Sayed Rahim Shah	male	Shagae Shahgram	Tirat	3	2.2	1	2	0	2.5	2.07	2.25	4.65
562	Mian Rahim Jan	male	Shagae Shahgram	Tirat	3	2.1	2	2	0	2.5	2.37	2.25	5.33
563	Shahred Khan	Male	Shagae Shahgram	Tirat	3	2.1	1	2	0	2.5	2.03	2.25	4.58
564	Sar Zameen Mian	Male	Shagae Shahgram	Tirat	3	2.1	1	2	0	2.5	2.03	2.25	4.58
565	Wazeera Bibi	Female	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
566	Zaim ul Huda	Female	Shagae Shahgram	Tirat	2	2.3	2	2	0	2.5	2.10	2.25	4.73
567	Shetrad Gai	Male	Shagae Shahgram	Tirat	2	2.3	2	2	0	2.5	2.10	2.25	4.73
568	Mian Sayed Hayat	Male	Shagae Shahgram	Tirat	3	2	2	2	0	2.5	2.33	2.25	5.25
569	Muhammad Gul	Male	Shagae Shahgram	Tirat	2	2.4	2	2	0	2.5	2.13	2.25	4.80
570	Maifrat Shah	Male	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
571	M Ashraf Khan	Male	Gharri	Madyan	2	1.7	2	2	3	0	1.90	2.50	5.70
572	Asaf Khan	Male	Gharri	Madyan	3	2.2	2	2	3	0	2.40	2.50	7.20
573	Kishwar Yahan Mian	Male	Gharri	Madyan	2	1.6	1.5	2	3	0	1.70	2.50	5.10
574	Japar Yahan Miam	Male	Gharri	Madyan	2	1.8	2	2	3	0	1.93	2.50	5.80
575	Azghar Khan	Male	Gharri	Madyan	3	1.8	2.5	2	3	0	2.43	2.50	7.30
576	Zoor Talab Khan	Male	Gharri	Madyan	3	1.5	2.5	2	3	0	2.33	2.50	7.00
577	Muhammad Aslam	Male	Gharri	Madyan	1	2	2.5	2	3	0	1.83	2.50	5.50
578	Afzal Khan	Male	Gharri	Madyan	1	1.9	2	2	3	0	1.63	2.50	4.90
579	Fazal Rahim	Male	Gharri	Madyan	1	2.1	2	2	3	0	1.70	2.50	5.10
580	Aamir Mashal	Male	Gharri	Madyan	3	1.9	2	2	3	0	2.30	2.50	6.90
581	Aftab Ali	Male	Gharri	Madyan	2	1.8	2	2	3	0	1.93	2.50	5.80
582	Nourul Mashal	Male	Gharri	Madyan	2	2	2.5	2	3	0	2.17	2.50	6.50
583	Fazal ullah	Male	Gharri	Madyan	2	2.2	2.5	2	3	0	2.23	2.50	6.70
584	Haseena	Female	Gharri	Madyan	3	2.8	2.5	2	3	0	2.77	2.50	8.30
585	Azeem Khan	Male	Gharri	Madyan	3	2.4	2.5	2	3	0	2.63	2.50	7.90
586	Muftah uddin	Male	Gharri	Madyan	3	2	2.5	2	3	0	2.50	2.50	7.50
587	Hamid ullah	Male	Gharri	Madyan	2	2.3	2.5	2	3	0	2.27	2.50	6.80
588	Abdul Rashid	Male	Gharri	Madyan	3	2.4	2.5	2	3	0	2.63	2.50	7.90
589	Main Muhammad	Male	Gharri	Madyan	2	2.3	2	2	3	0	2.10	2.50	6.30
590	Rahmat Wahab	Male	Gharri	Madyan	2	2.5	2.5	2	3	0	2.33	2.50	7.00
591	Sheraj wahab	Male	Gharri	Madyan	1	2.5	2.5	2	3	0	2.00	2.50	6.00
592	Amir Hamza	Male	Gharri	Madyan	3	2.4	2.5	2	3	0	2.63	2.50	7.90
593	Muhammad Nawab	Male	Gharri	Madyan	1	1.9	2.5	2	3	0	1.80	2.50	5.40
594	Sher Muhammad	Male	Gharri	Madyan	1	2.1	2.5	2	3	0	1.87	2.50	5.60
595	Sher Malik	Male	Gharri	Madyan	2	2.3	2.5	2	3	0	2.27	2.50	6.80
596	Umar Shah	Male	Gharri	Madyan	2	2.3	2.5	2	3	0	2.27	2.50	6.80
597	Ihsanullah	Male	Gharri	Madyan	3	2.3	2.5	2	3	0	2.60	2.50	7.80
598	Rahim Zada	Male	Gharri	Madyan	3	2.1	2.5	2	3	0	2.53	2.50	7.60
599	Ahmed Zada Mian	Male	Gharri	Madyan	3	1.7	2.5	2	3	0	2.40	2.50	7.20
600	Imran Sher Zada	Male	Gharri	Madyan	3	1.6	1.5	2	3	0	2.03	2.50	6.10
601	M. Ishaq	Male	Gharri	Madyan	2	1.5	2	2	3	0	1.83	2.50	5.50
602	Noor ul Hakim Mian	Male	Gharri	Madyan	2	1.9	2.5	2	3	0	2.13	2.50	6.40
603	Azhar Yayha Mian	Male	Gharri	Madyan	2	1.4	2.5	2	3	0	1.97	2.50	5.90
604	Mian Dost Mian	Male	Gharri	Madyan	2	1.6	2.5	2	3	0	2.03	2.50	6.10
605	Muhammad ishaq Khan	Male	Gharri	Madyan	2	1.6	2.5	2	3	0	2.03	2.50	6.10
606	Saeed Ullah	Male	Gharri	Madyan	2	2.5	1.5	2	3	0	2.00	2.50	6.00
607	Muhammad Haroon	Male	Gharri	Madyan	2	2.5	2.5	2	3	0	2.33	2.50	7.00
608	Sardar Ali	Male	Gharri	Madyan	2	1.9	2.5	2	3	0	2.13	2.50	6.40
609	Fazal Subhan	Male	Gharri	Madyan	2	1.9	2.5	2	3	0	2.13	2.50	6.40
610	Nazar Ahmed	Male	Gharri	Madyan	2	2.2	2.5	2	3	0	2.23	2.50	6.70
611	Rashid Ahmed	Male	Gharri	Madyan	2	1.9	2.5	2	3	0	2.13	2.50	6.40
612	Fazal Ghani	Male	Gharri	Madyan	2	1.4	2.5	2	3	0	1.97	2.50	5.90
613	Babar Zaman	Male	Gharri	Madyan	3	2.1	2.5	2	3	0	2.53	2.50	7.60

Demographic Profiling				Risk Assessment Analysis									
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
614	Gul Nazaka Bibi	Female	Gharri	Madyan	2	2.1	2.5	2	3	0	2.20	2.50	6.60
615	Muhammad Iran	Male	Gharri	Madyan	3	2.1	1.5	2	3	0	2.20	2.50	6.60
616	Azeem khan	Male	Gharri	Madyan	3	2.2	2.5	2	3	0	2.57	2.50	7.70
617	Adalat Khan	Male	Gharri	Madyan	3	2.1	2.5	2	3	0	2.53	2.50	7.60
618	Razi Khan	Male	Gharri	Madyan	3	2.1	1.5	2	3	0	2.20	2.50	6.60
619	Muhammad Iqbal	Male	Gharri	Madyan	2	1.9	2.5	2	3	0	2.13	2.50	6.40
620	Bachawas Khan	Male	Gharri	Madyan	2	2	2.5	2	3	0	2.17	2.50	6.50
621	Sarzanman Khan	Male	Gharri	Madyan	2	2.3	2.5	2	3	0	2.27	2.50	6.80
622	Liaquat Ali	Male	Gharri	Madyan	3	1.9	2.5	2	3	0	2.47	2.50	7.40
623	Habib ullah	Male	Gharri	Madyan	3	2.1	2.5	2	3	0	2.53	2.50	7.60
624	Syed Akbar	Male	Gharri	Madyan	2	2.1	2.5	2	3	0	2.20	2.50	6.60
625	Nargis Begum	Female	Gharri	Madyan	3	2.1	2.5	2	3	0	2.53	2.50	7.60
626	Anwar ullah	Male	Gharri	Madyan	3	2.4	2.5	2	3	0	2.63	2.50	7.90
627	Khalid Shah mian	Male	Gharri	Madyan	3	2.4	2.5	2	3	0	2.63	2.50	7.90
628	Bakhtyar	Male	Gharri	Madyan	2	2.2	2.5	2	3	0	2.23	2.50	6.70
629	Insanullah	Male	Gharri	Madyan	2	2	2.5	2	3	0	2.17	2.50	6.50
630	Fazal Rabi	Male	Gharri	Madyan	2	2	2.5	2	3	0	2.17	2.50	6.50
631	Sultanat Khan	Male	Gharri	Madyan	3	2	2.5	2	3	0	2.50	2.50	7.50
632	Allah Baksh Mullah	Male	Gharri	Madyan	3	2.2	1.5	2	3	0	2.23	2.50	6.70
633	Sujat Ali	Male	Gharri	Madyan	2	2	2.5	2	3	0	2.17	2.50	6.50
634	Anwar zeb	Male	Gharri	Madyan	2	2.3	2.5	2	3	0	2.27	2.50	6.80
635	Ahmed Zeb	Male	Gharri	Madyan	2	2.2	1.5	2	3	0	1.90	2.50	5.70
636	Amir gazam	Male	Gharri	Madyan	2	2.2	2.5	2	3	0	2.23	2.50	6.70
637	Tote Muhammad	Male	Gharri	Madyan	2	2.1	2.5	2	3	0	2.20	2.50	6.60
638	Kheshta Muhammad	Male	Gharri	Madyan	2	2.3	2.5	2	3	0	2.27	2.50	6.80
639	Aamir Muhammad	Male	Gharri	Madyan	2	2.2	2.5	2	3	0	2.23	2.50	6.70
640	Sher Hassan	Male	Gharri	Madyan	2	1.9	1.5	2	3	0	1.80	2.50	5.40
641	Sardar Yahya Mian	Male	Gharri	Madyan	2	2.1	2.5	2	3	0	2.20	2.50	6.60
642	Ahmed Zada	Male	Gharri	Madyan	2	2.1	2.5	2	3	0	2.20	2.50	6.60
643	Iqbal Hassan	Male	Gharri	Madyan	2	2.2	2.5	2	3	0	2.23	2.50	6.70
644	Mian Sher Zada	Male	Gharri	Madyan	3	2	2.5	2	3	0	2.50	2.50	7.50
645	Insan Ali	Male	Gharri	Madyan	2	2.2	2.5	2	3	0	2.23	2.50	6.70
646	Onar Zada	Male	Gharri	Madyan	3	2.5	2.5	2	3	0	2.67	2.50	8.00
647	Insanullah	Male	Gharri	Madyan	2	2	2.5	2	3	0	2.17	2.50	6.50
648	Usaf Ali Shah	Male	Gharri	Madyan	1	1.9	2.5	2	3	0	1.80	2.50	5.40
649	M. Ali Jan	Male	Shagae Shahgram	Tirat	3	2.2	2	2	0	2.5	2.40	2.25	5.40
650	Anwar Badshah	Male	Shagae Shahgram	Tirat	2	2.4	2	2	0	2.5	2.13	2.25	4.80
651	M. Zahid Iqbal	Male	Shagae Shahgram	Tirat	3	2.1	1	2	0	2.5	2.03	2.25	4.58
652	Parvez Iqbal	Male	Shagae Shahgram	Tirat	3	2.1	1	2	0	2.5	2.03	2.25	4.58
653	Saba Khan	Female	Shagae Shahgram	Tirat	2	2.3	2	2	0	2.5	2.10	2.25	4.73
654	M. Sadiq Main	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
655	Alamzeb Khan	Male	Shagae Shahgram	Tirat	3	2.3	1	2	0	2.5	2.10	2.25	4.73
656	Nadir Khan	Male	Shagae Shahgram	Tirat	3	2.2	2	2	0	2.5	2.40	2.25	5.40
657	Aqim ud din	Male	Shagae Shahgram	Tirat	3	2.2	1	2	0	2.5	2.07	2.25	4.65
658	Sayed Ali Shah	Male	Shagae Shahgram	Tirat	3	2.4	1	2	0	2.5	2.13	2.25	4.80
659	Pir Madar Shah	Male	Shagae Shahgram	Tirat	3	2.2	1	2	0	2.5	2.07	2.25	4.65
660	Rubina	Male	Shagae Shahgram	Tirat	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
661	Feroz Khan	Male	Shagae Shahgram	Tirat	2	2.4	2	2	0	2.5	2.13	2.25	4.80
662	Sofi Jan	Male	Shagae Shahgram	Tirat	3	2.4	1	2	0	2.5	2.13	2.25	4.80
663	Said Nawab	Male	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
664	Aziz Khan	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
665	Bakhtiyar Khan	Male	Shagae Shahgram	Tirat	3	2.2	2	2	0	2.5	2.40	2.25	5.40
666	Said Fazal Khan	Male	Shagae Shahgram	Tirat	3	2	1.5	2	0	2.5	2.17	2.25	4.88
667	Khurshid Alam	Male	Shagae Shahgram	Tirat	3	2.3	1.5	2	0	2.5	2.27	2.25	5.10
668	Rehmat Shah	Male	Shagae Shahgram	Tirat	2	2.2	2.5	2	0	2.5	2.23	2.25	5.03
669	Shah Miroz Jan	Male	Shagae Shahgram	Tirat	2	2.2	2.5	2	0	2.5	2.23	2.25	5.03

Demographic Profiling				Risk Assessment Analysis									
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Assessment Score	Total Risk Assessment Score
670	Said Abdul Hayat	Male	Shagae Shahgram	Tirat	2	2.2	2.5	2	0	2.5	2.23	2.25	5.03
671	Jafar Khan	Male	Shagae Shahgram	Tirat	2	2.1	2.5	2	0	2.5	2.20	2.25	4.95
672	Amir Halder	Male	Shagae Shahgram	Tirat	3	2.1	1.5	2	0	2.5	2.20	2.25	4.95
673	Haider Shah	Male	Shagae Shahgram	Tirat	3	2.3	1.5	2	0	2.5	2.27	2.25	5.10
674	Burhan ud din	Male	Shagae Shahgram	Tirat	3	1.7	2	2	0	2.5	2.23	2.25	5.03
675	Mian Saeed Rahim	Male	Shagae Shahgram	Tirat	3	2.4	1	2	0	2.5	2.13	2.25	4.80
676	Aziz ud din	male	Shagae Shahgram	Tirat	2	2	2.5	2	0	2.5	2.17	2.25	4.88
677	Amir Iqbal	Male	Shagae Shahgram	Tirat	3	1.9	2.5	2	0	2.5	2.47	2.25	5.55
678	Bashir Ahmed	Male	Shagae Shahgram	Tirat	3	1.5	2	2	0	2.5	2.17	2.25	4.88
679	Abkar Khan	Male	Shagae Shahgram	Tirat	2	2.3	2.5	2	0	2.5	2.27	2.25	5.10
680	Badrai Haram	Female	Shagae Shahgram	Tirat	1	2.2	3	2	0	2.5	2.07	2.25	4.65
681	Kamin Khan	Male	Shagae Shahgram	Tirat	2	1.8	2.5	2	0	2.5	2.10	2.25	4.73
682	Mian Said	Male	Shagae Shahgram	Tirat	2	2.3	2.5	2	0	2.5	2.27	2.25	5.10
683	Tamadar Mian	Male	Shagae Shahgram	Tirat	2	1.8	2.5	2	0	2.5	2.10	2.25	4.73
684	Said Manzoor Mian	Male	Shagae Shahgram	Tirat	3	1.9	1.5	2	0	2.5	2.13	2.25	4.80
685	Ali Akbar	Male	Shagae Shahgram	Tirat	3	2	2.5	2	0	2.5	2.50	2.25	5.63
686	Fazal Ahad	Male	Shagae Shahgram	Tirat	3	2.1	1.5	2	0	2.5	2.20	2.25	4.95
687	Arzomand Mian	Male	Shagae Shahgram	Tirat	3	1.7	2.5	2	0	2.5	2.40	2.25	5.40
688	Bladar Mian	Male	Shagae Shahgram	Tirat	3	1.7	2	2	0	2.5	2.23	2.25	5.03
689	Sajid Ali	Male	Shagae Shahgram	Tirat	3	1.8	1.5	2	0	2.5	2.10	2.25	4.73
690	Imran Khan	Male	Shagae Shahgram	Tirat	2	1.8	2.5	2	0	2.5	2.10	2.25	4.73
691	Nasid Rana	Male	Shagae Shahgram	Tirat	3	1.9	2.5	2	0	2.5	2.47	2.25	5.55
692	Muhammad Qasim	Male	Shagae Shahgram	Tirat	3	1.7	2.5	2	0	2.5	2.40	2.25	5.40
693	Sadar Jahan	Male	Shagae Shahgram	Tirat	3	1.9	2.5	2	0	2.5	2.47	2.25	5.55
694	Taj Mahal Bibi	Female	Shagae Shahgram	Tirat	2	1.8	2.5	2	0	2.5	2.10	2.25	4.73
695	Said Wakeel Shah	Male	Shagae Shahgram	Tirat	3	2	2.5	2	0	2.5	2.50	2.25	5.63
696	Zeena Begum	Female	Shagae Shahgram	Tirat	2	2.1	2.5	2	0	2.5	2.20	2.25	4.95
697	Sana ullah Khan	Male	Shagae Shahgram	Tirat	3	1.5	2.5	2	0	2.5	2.33	2.25	5.25
698	Anan ud din	Male	Shagae Shahgram	Tirat	3	1.8	2.5	2	0	2.5	2.43	2.25	5.48
699	Mian Gul Farosh	Male	Shagae Shahgram	Tirat	3	1.7	1.5	2	0	2.5	2.07	2.25	4.65
700	Mafan ud din	Male	Shagae Shahgram	Tirat	3	1.8	1.5	2	0	2.5	2.10	2.25	4.73
701	Muhammad Sayed	Male	Shagae Shahgram	Tirat	2	1.9	2.5	2	0	2.5	2.13	2.25	4.80
702	Said Salah ud din	male	Shagae Shahgram	Tirat	3	1.8	1.5	2	0	2.5	2.10	2.25	4.73
703	Muhammad Ali Shah	Male	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
704	Shams ud din	Male	Shagae Shahgram	Tirat	3	1.9	2	2	0	2.5	2.30	2.25	5.18
705	Khawar ud din	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
706	Ali Akbar Mian	Male	Shagae Shahgram	Tirat	3	1.9	1.5	2	0	2.5	2.13	2.25	4.80
707	Ajeeb ud din	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
708	Bahar ud din	Male	Shagae Shahgram	Tirat	3	2	2	2	0	2.5	2.33	2.25	5.25
709	Rekhangea Bibi	Male	Shagae Shahgram	Tirat	2	2.3	2	2	0	2.5	2.10	2.25	4.73
710	Muhammad Nabi	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
711	Saltanat Shah	Male	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
712	Hamaish Gul	Male	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
713	Muzaffar Saeed	Male	Shagae Shahgram	Tirat	2	1.9	2.5	2	0	2.5	2.13	2.25	4.80
714	Shah Bibi	Female	Shagae Shahgram	Tirat	2	2.5	2.5	2	0	2.5	2.33	2.25	5.25
715	Muhammad Parvez Khan	Male	Shagae Shahgram	Tirat	2	2.1	2.5	2	0	2.5	2.20	2.25	4.95
716	Haji Nawab	Male	Shagae Shahgram	Tirat	2	2.2	2.5	2	0	2.5	2.23	2.25	5.03
717	Sayed Sadiq Shah	Male	Shagae Shahgram	Tirat	2	2.2	2	2	0	2.5	2.07	2.25	4.65
718	Shah Qayyum	Male	Shagae Shahgram	Tirat	2	1.9	2.5	2	0	2.5	2.13	2.25	4.80
719	Bacha Khila	Male	Shagae Shahgram	Tirat	2	2.3	2.5	2	0	2.5	2.27	2.25	5.10
720	Noor Jahan	Female	Shagae Shahgram	Tirat	2	2.2	2.5	2	0	2.5	2.23	2.25	5.03
721	Shahbaz Khan	Male	Shagae Shahgram	Tirat	2	2	2.5	2	0	2.5	2.17	2.25	4.88
722	Mian Khan ud din	Male	Shagae Shahgram	Tirat	3	2.4	1	2	0	2.5	2.13	2.25	4.80
723	Zakir ud din	Male	Shagae Shahgram	Tirat	3	2.2	1	2	0	2.5	2.07	2.25	4.65
724	Sayed Yousaf Shah	Male	Shagae Shahgram	Tirat	3	1.8	1.5	2	0	2.5	2.10	2.25	4.73
725	Sayed Ismail	Male	Shagae Shahgram	Tirat	3	2	1.5	2	0	2.5	2.17	2.25	4.88

Demographic Profiling					Risk Assessment Analysis								
HH ID	Name	Sex	Village	UC	Exposure Score	Fragility Score	Lack of resilience Score	Hazard Analysis for Flood 2010 Score	Hazard Analysis for Seasonal Flooding Score	Soil Erosion Score	Total Vulnerability Score	Total Hazard Score	Total Risk Assessment Score
726	Syed Ayub Shah	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
727	Said Ali Shah	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
728	Said Abdul Ghafar	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
729	Ali Muhammad	Male	Shagae Shahgram	Tirat	2	2	2.5	2	0	2.5	2.17	2.25	4.88
730	Tahir Shah Mian	Male	Shagae Shahgram	Tirat	2	1.7	2.5	2	0	2.5	2.07	2.25	4.65
731	Gul Bahar Khan	Male	Shagae Shahgram	Tirat	3	1.8	2	2	0	2.5	2.27	2.25	5.10
732	Saeed Ahmed Khan	Male	Shagae Shahgram	Tirat	2	1.9	2.5	2	0	2.5	2.13	2.25	4.80
733	Zaibah Shah	Male	Shagae Shahgram	Tirat	2	2.1	2	2	0	2.5	2.03	2.25	4.58
734	Misbah ud din	Male	Shagae Shahgram	Tirat	3	2.2	2	2	0	2.5	2.40	2.25	5.40
735	Mirab ud din	Male	Shagae Shahgram	Tirat	2	2	2.5	2	0	2.5	2.17	2.25	4.88
736	Sher Muhammad	Male	Shagae Shahgram	Tirat	2	2.4	2	2	0	2.5	2.13	2.25	4.80
737	Waleyat Bibi	Male	Shagae Shahgram	Tirat	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
738	Aliq Tabar	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
739	Goga	Male	Shagae Shahgram	Tirat	3	2.3	2	2	0	2.5	2.43	2.25	5.48
740	Mohammad Saleem	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
741	Muhammad Atzal Khan	Male	Shagae Shahgram	Tirat	3	2	2.5	2	0	2.5	2.50	2.25	5.63
742	Abdullah Shah	Male	Shagae Shahgram	Tirat	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
743	Said Luqman Shah	Male	Shagae Shahgram	Tirat	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
744	Muzafar Said	Male	Shagae Shahgram	Tirat	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
745	Shah Nasim	Male	Shagae Shahgram	Tirat	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
746	Said Anwar Shah	Male	Shagae Shahgram	Tirat	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
747	Muhammad Khan	Male	Shagae Shahgram	Tirat	3	2.4	2.5	2	0	2.5	2.63	2.25	5.93
748	Muhammad Alam Khan	Male	Shagae Shahgram	Tirat	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
749	Syed Karim Shah	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
750	Arshad Ali	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
751	Mian Sayeed Karim	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
752	Bakht Shaheen	Female	Shagae Shahgram	Tirat	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
753	Miraj ud din Mian	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
754	Bakht Zamin Shah	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
755	Zahid Shah Mian	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
756	Habib Akbar	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
757	Makro Yar Bibi	Female	Shagae Shahgram	Tirat	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
758	Monjara Khan	Male	Shagae Shahgram	Tirat	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
759	Nawad ud din	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
760	Jahan ud dini	Male	Shagae Shahgram	Tirat	3	2.2	2.5	2	0	2.5	2.57	2.25	5.78
761	Najeeb ullah	Male	Shagae Shahgram	Tirat	3	1.8	2.5	2	0	2.5	2.43	2.25	5.48
762	Mohambar	Male	Shagae Shahgram	Tirat	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85
763	Jahan Saba	Male	Shagae Shahgram	Tirat	3	2.1	2.5	2	0	2.5	2.53	2.25	5.70
764	Muhammad Sherin	Male	Shagae Shahgram	Tirat	3	1.8	2.5	2	0	2.5	2.43	2.25	5.48
765	Said Alam	Male	Shagae Shahgram	Tirat	3	2.3	2.5	2	0	2.5	2.60	2.25	5.85

Annex 10: Questionnaire for conducting Vulnerability Assessment

Parameters	Indicators	Scale	Score
Exposure (A)	High (all property has exposure to hazard)	3	
	Medium (more than half of property has exposure to hazard)	2	
	Low (less than half of property has exposure to hazard)	1	
	Score for section A		
Fragility (B)	1 - Land		
	No land	3	
	Less than 3 Acres	2	
	3 or more Acres	1	
	2 - Livestocks		
	No animals	3	
	1 large animal OR less than 5 small animals	2	
	More than 1 large animal OR more than 5 small animals	1	
	3 - Vulnerable (Elderly, disabled & long term sick people not able to work)		
	More than 3	3	
	2	2	
	0-1	1	
	4 - Children (less than 12 years old)		
	6 or more	3	
	3 to 5 children	2	
	Less than 3	1	
	5 – Women / Men Headed Household		
	Women head of household with children	3	
	Women with no children	2	
	Men headed household (includes Man with Women)	1	
	6 - Assets		
	No assets	3	
	Small assets; Radio	2	
	Large assests; TV, bicycle, vehicle	1	
	7 - Income		
	Farming only or irregular labour work	3	
	Labour OR Farming plus other income	2	
	Regular income from more than one source	1	
	8- Remittance		
	No family member outside of family home giving assistance	3	
	Family member from outside sending money irregularly	2	
	Family member from outside sending money regularly	1	
	9- Habitat		
	6 or more people sleeping per room	3	
	3 to 5 people sleeping per room	2	
	2 or less people sleeping per room	1	
	10- Housing		
	Mud house in poor condition	4	
	Mud house in good condition	2	
	Brick or concrete house	1	
	11- Food		
	Limited food available for cooking more than 1 month in the year	3	
	Limited food available for cooking less than 1 month in the year	2	
	No food restriction all year	1	
	Total for section B divided by number of indicators in section B		
Lack of resilience (C)	1- Education level		
	No member of family who attended and completed school	3	
	At least one person attended and completed primary school	2	
	At least one person attended and completed high school or above	1	
	2 - Neighbourhood/ Support systems		
	Families live isolated	3	
	Families occasionally help each other (but unable during disaster situation)	2	
	Families always help each other	1	
	Total for section C divided by number of indicators in section C		
Vulnerability	(A + B + C) / 3		
Any other significant factor influencing vulnerability:			

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